Time Series Analysis

Lecture notes, Chapters 1–8

Based on Sangyeol Lee, *Time Series Analysis: Theory and Practice with R*

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Chapter 1. Time Series Data

1.1 Time series data with dependence

Motivation. Introductory statistics rests on the i.i.d. assumption, but observations recorded over time (temperatures, stock prices, economic indicators) are typically correlated with their own past. Time series analysis studies such dependent data; this section shows the simplest descriptive devices (smoothing) and explains why a model for the dependence is needed.

1.1.1 Trend and moving-average smoothing

Let $x_1, \dots, x_n$ be observations, e.g. monthly closing values of a stock index. Raw prices fluctuate so much that the underlying movement is hard to see. The *moving average* of span m,

$$y_t = \frac{x_{t-m+1} + \cdots + x_t}{m},$$

replaces each value by the mean of the last m observations. Larger m gives a smoother curve; a short-span average describes short-run movement, a long-span one the long-run trend. Producing such a curve is called *smoothing*.

To see why it works, suppose the data follow the signal-plus-noise model

$$x_t = f(t) + z_t, \qquad t = 1, \dots, n, \tag{1.1}$$

where f is an unknown deterministic function and z_t are i.i.d. errors with mean 0 and variance σ^2. Then

$$y_t = \frac{f(t-m+1) + \cdots + f(t)}{m} + \frac{z_{t-m+1} + \cdots + z_t}{m}. \tag{1.2}$$

For moderately large m the second term is close to $Ez_1 = 0$ by the law of large numbers (its variance is σ^2/m), so $y_t \approx f_m(t) := m^{-1} \sum_{j=0}^{m-1} f(t-j)$, a local average of the trend.

1.1.2 Exponential smoothing

An alternative to equal weights is to discount the past geometrically. For $0 < w < 1$ put

$$S_t = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j}.$$

Given an initial value S_0^*, define recursively

$$S_t^* = w x_t + (1-w) S_{t-1}^*. \tag{1.3}$$

Iterating (1.3) gives

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^* = S_t + (1-w)^t S_0^*, \tag{1.4}$$

so $S_t^* \approx S_t$ for large t, and the cheap recursion may be used instead of S_t. Common choices of S_0^* are x_1 or the sample mean; w can be chosen to minimize the one-step prediction error

$$SSE(w) = \sum_{t=2}^{n} (x_t - S_{t-1}^*)^2. \tag{1.5}$$

1.1.3 Trend, seasonality and noise

Writing T_t for the trend $f(t)$, one also uses the decomposition $x_t = T_t + S_t + z_t$, where the *seasonal component* satisfies $S_t = S_{t+s}$ for a period s ($s = 4$ for quarterly, $s = 12$ for monthly data). The i.i.d. assumption on z_t is only a convenience: real errors are usually strongly correlated. The next section introduces stationarity, the basic framework for modelling such dependence.

1.2 Stationary time series

Motivation. To do inference from a single realization we need the probabilistic structure to be stable over time. Stationarity formalizes this, and the autocovariance function becomes the central object describing the dependence.

Definition 1.2.1 (Time series)

If T is the set of integers $\mathbb{Z}$ or of positive integers $\mathbb{Z}^+ = \{1, 2, \dots\}$, a collection of random variables $\{X_t : t \in T\}$ is called a *time series*.

Definition 1.2.2 (Strictly stationary time series)

$\{X_t\}$ is *strictly stationary* if for all $k \geq 1$, all integers $t_1, \dots, t_k$ and h, and all real $x_1, \dots, x_k$,

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k).$$

That is, every finite-dimensional distribution is invariant under time shifts.

Definition 1.2.3 (Weakly stationary time series)

$\{X_t\}$ is *(weakly) stationary* if
(1) $EX_t = \mu$ for all t;
(2) $EX_t^2 < \infty$ for all t;
(3) $\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(X_0, X_h)$ for all t, h.

Condition (1) says the level is constant, (2) that variances exist, and (3), the essential one, that the covariance between two observations depends only on their time distance; in particular the variance is constant. Unless stated otherwise, "stationary" means weakly stationary.

Example 1.2.4 (Strict versus weak stationarity)

(1) An i.i.d. sequence $\{X_t\}$ is strictly stationary.
(2) If moreover $EX_1^2 < \infty$, it is also weakly stationary. An i.i.d. Cauchy sequence is strictly but not weakly stationary, since the variance does not exist.
(3) If ϵ_t are i.i.d. with mean 0, variance 1, then $X_t = \epsilon_t + \epsilon_{t-1}$ is strictly and weakly stationary.

Neither notion contains the other, but a strictly stationary series with finite variance is weakly stationary: shift invariance of the bivariate law of (X_t, X_{t+h}) gives (1) and (3). Throughout, stationary series are assumed to have finite variance.

Definition 1.2.5 (Autocovariance function)

For a stationary $\{X_t\}$ the real function

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}), \qquad h = 0, \pm 1, \pm 2, \ldots,$$

is called the *autocovariance function* (ACVF).

Theorem 1.2.6 (Basic properties of the ACVF)

The autocovariance function satisfies
(1) $\gamma(0) \geq 0$;
(2) $\gamma(-h) = \gamma(h)$;
(3) $|\gamma(h)| \leq \gamma(0)$.

Proof. [Supplement]

Step 1. $\gamma(0) = \mathrm{Var}(X_t) \geq 0$.

Step 2. By stationarity and symmetry of covariance, $\gamma(-h) = \mathrm{Cov}(X_t, X_{t-h}) = \mathrm{Cov}(X_{t-h}, X_t) = \gamma(h)$, the last equality by putting $s = t - h$.

Step 3. By the Cauchy–Schwarz inequality, $|\mathrm{Cov}(X_t, X_{t+h})| \leq \sqrt{\mathrm{Var}\, X_t \ \mathrm{Var}\, X_{t+h}} = \gamma(0)$. ■

Theorem 1.2.7 (Nonnegative definiteness)

For any reals $x_1, \ldots, x_k$ and integers $t_1, \ldots, t_k$,

$$\sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j \geq 0.$$

Proof. [Supplement] Let $Y = \sum_{i=1}^{k} x_i X_{t_i}$. Since $\mathrm{Cov}(X_{t_i}, X_{t_j}) = \gamma(t_j - t_i) = \gamma(t_i - t_j)$ by Theorem 1.2.6(2),

$$0 \leq \mathrm{Var}(Y) = \sum_{i,j=1}^{k} x_i x_j \,\mathrm{Cov}(X_{t_i}, X_{t_j}) = \sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j.$$

■

In matrix form: the covariance matrix $\Gamma_k = [\gamma(t_i - t_j)]_{i,j=1}^{k}$ of any finite sub-vector is nonnegative definite. (Conversely, every even nonnegative definite function on $\mathbb{Z}$ is the ACVF of some stationary, indeed Gaussian, series.)

Definition 1.2.8 (White noise)

A stationary series $\{X_t\}$ whose ACVF satisfies $\gamma(h) = 0$ for $h \neq 0$ is called *white noise*, written $\{X_t\} \sim WN(0, \sigma^2)$ with $\sigma^2 = \gamma(0)$ (mean zero is understood). If in addition the X_t are i.i.d., we write $\{X_t\} \sim IID(0, \sigma^2)$.

Definition 1.2.9 (Autocorrelation function)

For a stationary $\{X_t\}$,

$$\rho(h) = \gamma(h)/\gamma(0)$$

is the *autocorrelation function* (ACF); for fixed h, $\rho(h)$ is the autocorrelation coefficient at lag h.

By Theorem 1.2.6, $\rho(0) = 1$, $\rho(-h) = \rho(h)$ and $|\rho(h)| \le 1$.

Definition 1.2.10 (Multivariate normal distribution)

Let $\mu = (\mu_1, \dots, \mu_k)^T \in \mathbb{R}^k$ and let $\Sigma = (\sigma_{ij})_{i,j=1}^k$ be symmetric positive definite: $\sigma_{ij} = \sigma_{ji}$ and $a^T \Sigma a > 0$ for all $a \in \mathbb{R}^k \setminus \{0\}$. If the random vector $Z = (Z_1, \dots, Z_k)^T$ has joint density

$$f(z) = \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\Big\{-\tfrac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)\Big\}, \qquad z \in \mathbb{R}^k,$$

then Z follows the *multivariate normal distribution* with mean μ and covariance Σ; here $\mu_i = EZ_i$ and $\sigma_{ij} = \mathrm{Cov}(Z_i, Z_j)$.

Definition 1.2.11 (Gaussian time series)

$\{X_t\}$ is a *Gaussian time series* if $(X_{t_1}, \dots, X_{t_k})^T$ is multivariate normal for every k and all $t_1, \dots, t_k$.

Since a normal law is determined by its mean and covariance, a weakly stationary Gaussian series is strictly stationary. Normality rarely holds exactly, but it makes the theory tractable and is the natural starting point for extensions. A key consequence concerns prediction: for a zero-mean stationary Gaussian series,

$$E(X_{k+1} \mid X_1, \dots, X_k) = \sum_{i=1}^{k} a_i X_i$$

for some reals a_i, i.e. the best predictor is linear. Given data $X_1, \dots, X_n$, such coefficients can be estimated as in regression, by minimizing $\sum_{t=k+1}^{n} (X_t - a_1 X_{t-1} - \cdots - a_k X_{t-k})^2$.

[Supplement] Proof of linearity of the conditional mean. Let $\mathbf{X} = (X_1, \dots, X_k)^T$, $\Gamma_k = \mathrm{Cov}(\mathbf{X})$ (assume invertible) and $\gamma_k = \mathrm{Cov}(\mathbf{X}, X_{k+1})$. Put $a = \Gamma_k^{-1}\gamma_k$ and $U = X_{k+1} - a^T\mathbf{X}$. Then $\mathrm{Cov}(\mathbf{X}, U) = \gamma_k - \Gamma_k a = 0$. Since $(\mathbf{X}, U)$ is jointly normal (a linear image of a normal vector), zero covariance implies U is independent of $\mathbf{X}$, so $E(U \mid \mathbf{X}) = EU = 0$. Hence $E(X_{k+1} \mid \mathbf{X}) = a^T\mathbf{X} + E(U \mid \mathbf{X}) = a^T\mathbf{X}$. ■

Stationarity is the backbone of time series analysis. Real data are often nonstationary, but transformations (e.g. differencing, leading to the ARIMA models of Chapter 5) can frequently reduce them to stationary ones.

1.3 Selected exercises

Motivation. These textbook exercises practise the basic definitions: the exponential-smoothing recursion, the gap between strict and weak stationarity, Gaussian conditional means, and how autocovariances combine under addition.

Exercise 1.1 (Exponential smoothing recursion)

Let $0 < w < 1$ and let S_t^* be defined by (1.3) from an initial value S_0^*. Show (1.4), i.e.

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^*, \qquad t \ge 1,$$

and deduce that $S_t^* - S_t \to 0$ as $t \to \infty$.

Solution. We argue by induction on t.

Step 1. For $t = 1$, (1.3) gives $S_1^* = wx_1 + (1-w)S_0^*$, which is the claimed formula (the sum has the single term $j = 0$).

Step 2. Let $t \geq 2$ and assume the formula for $t-1$. Substituting it into (1.3),

$$\begin{aligned} S_t^* &= wx_t + (1-w)\Big\{w\sum_{j=0}^{t-2}(1-w)^j x_{t-1-j} + (1-w)^{t-1}S_0^*\Big\} \\ &= wx_t + w\sum_{k=1}^{t-1}(1-w)^k x_{t-k} + (1-w)^t S_0^*, \end{aligned}$$

where $k = j+1$. The term wx_t is the $k = 0$ term of the sum, so $S_t^* = w\sum_{k=0}^{t-1}(1-w)^k x_{t-k} + (1-w)^t S_0^*$.

Step 3. By the definition of S_t, $S_t^* - S_t = (1-w)^t S_0^*$. Since $0 < 1-w < 1$, $(1-w)^t \to 0$, so $S_t^* - S_t \to 0$ for every fixed S_0^*. ■

Exercise 1.3 (Strict versus weak stationarity without independence)

Using examples that are *not* i.i.d., show that
(a) a strictly stationary series need not be weakly stationary;
(b) a weakly stationary series need not be strictly stationary.

Solution.

(a) Let Z be standard Cauchy and $X_t = Z$ for all t. For any $t_1, \ldots, t_k$, h and $x_1, \ldots, x_k$,

$$P(X_{t_1} \leq x_1, \ldots, X_{t_k} \leq x_k) = P(Z \leq \min_i x_i) = P(X_{t_1+h} \leq x_1, \ldots, X_{t_k+h} \leq x_k),$$

so $\{X_t\}$ is strictly stationary. It is not i.i.d., since $X_1 = X_2$ and X_1 is non-degenerate, so X_1, X_2 are dependent. It is not weakly stationary, because $EX_t^2 = EZ^2 = \infty$ violates condition (2) of Definition 1.2.3.

(b) Let $\{X_t\}$ be independent with $X_t \sim N(0,1)$ for odd t, and $P(X_t = 1) = P(X_t = -1) = \frac{1}{2}$ for even t. Every X_t has mean 0 and variance 1, and by independence $\mathrm{Cov}(X_t, X_{t+h}) = \mathbb{1}\{h = 0\}$ for all t. Hence $\{X_t\}$ is weakly stationary (indeed $WN(0,1)$). It is not i.i.d., since the marginal laws differ. It is not strictly stationary: taking $k = 1$, $t_1 = 1$, $h = 1$, $x_1 = \frac{1}{2}$,

$$P(X_1 \leq \tfrac{1}{2}) = \Phi(\tfrac{1}{2}) \approx 0.691 \neq \tfrac{1}{2} = P(X_2 \leq \tfrac{1}{2}),$$

where Φ is the standard normal distribution function. ■

Exercise 1.6 (Gaussian one-step conditional mean)

(a) (Exercise 1.5.) Let (X, Y) be bivariate normal with $\mathrm{Var}(Y) > 0$. Show that $E(X \mid Y) = aY + b$ with $a = \mathrm{Cov}(X,Y)/\mathrm{Var}(Y)$ and $b = EX - aEY$.
(b) Let $\{X_t\}$ be a stationary Gaussian series with mean μ and ACVF γ, $\gamma(0) > 0$. Express $E(X_t \mid X_{t-1})$ and its mean squared error through μ and γ.

Solution.

(a) Put $U = X - aY - b$.

Step 1. (U, Y) is a linear image of the normal vector (X, Y), so every linear combination $sU + tY$ is a linear combination of X, Y and hence univariate normal (possibly degenerate). Therefore (U, Y) has characteristic function $Ee^{i(sU+tY)} = \exp\{i(sEU + tEY) - \frac{1}{2}\mathrm{Var}(sU + tY)\}$.

Step 2. $EU = EX - aEY - b = 0$ and $\operatorname{Cov}(U, Y) = \operatorname{Cov}(X, Y) - a \operatorname{Var}(Y) = 0$. Hence $\operatorname{Var}(sU + tY) = s^2 \operatorname{Var} U + t^2 \operatorname{Var} Y$, and the characteristic function factorizes into a function of s times a function of t. Thus U and Y are independent.

Step 3. Consequently $E(U \mid Y) = EU = 0$, and $E(X \mid Y) = aY + b + E(U \mid Y) = aY + b$.

(b) By Definition 1.2.11, (X_t, X_{t-1}) is bivariate normal with $\operatorname{Var} X_{t-1} = \gamma(0) > 0$ and $\operatorname{Cov}(X_t, X_{t-1}) = \gamma(1)$. Part (a) with $X = X_t$, $Y = X_{t-1}$ gives $a = \gamma(1)/\gamma(0) = \rho(1)$ and $b = \mu - \rho(1)\mu$, so

$$E(X_t \mid X_{t-1}) = \mu + \frac{\gamma(1)}{\gamma(0)} (X_{t-1} - \mu).$$

The prediction error is $U = X_t - \mu - \rho(1)(X_{t-1} - \mu)$, whose mean square is

$$EU^2 = \gamma(0) - 2\rho(1)\gamma(1) + \rho(1)^2\gamma(0) = \gamma(0)(1 - \rho(1)^2) = \gamma(0) - \frac{\gamma(1)^2}{\gamma(0)}.$$

■

Exercise 1.7 (Sum of independent stationary series)

Let $\{X_t\}$ and $\{Y_t\}$ be independent stationary series with ACVFs γ_X, γ_Y. Show that $\{X_t + Y_t\}$ is stationary and find its ACVF.

Solution. Let $Z_t = X_t + Y_t$, $\mu_X = EX_t$, $\mu_Y = EY_t$. We check Definition 1.2.3.

Step 1. $EZ_t = \mu_X + \mu_Y$ for all t.

Step 2. $EZ_t^2 \le 2EX_t^2 + 2EY_t^2 < \infty$.

Step 3. By bilinearity of covariance,

$$\operatorname{Cov}(Z_t, Z_{t+h}) = \operatorname{Cov}(X_t, X_{t+h}) + \operatorname{Cov}(X_t, Y_{t+h}) + \operatorname{Cov}(Y_t, X_{t+h}) + \operatorname{Cov}(Y_t, Y_{t+h}).$$

For any s, u, X_s and Y_u are independent with finite second moments, so $EX_sY_u = EX_s\,EY_u$ and $\operatorname{Cov}(X_s, Y_u) = 0$. Hence

$$\gamma_Z(h) = \operatorname{Cov}(Z_t, Z_{t+h}) = \gamma_X(h) + \gamma_Y(h),$$

which does not depend on t. Thus $\{Z_t\}$ is stationary with ACVF $\gamma_X + \gamma_Y$.

Only $\operatorname{Cov}(X_s, Y_u) = 0$ was used, so uncorrelated series suffice. If $\gamma_X(0) + \gamma_Y(0) > 0$, the ACF is $\rho_Z(h) = \{\gamma_X(0)\rho_X(h) + \gamma_Y(0)\rho_Y(h)\}/\{\gamma_X(0) + \gamma_Y(0)\}$, a variance-weighted average of the two ACFs (with the convention that a term with zero variance is omitted). ■

Chapter 2. ARMA Models

2.1 Introduction

Motivation. The autoregressive moving average (ARMA) models are the workhorse of stationary time series modelling. They arise from two natural ways of building a dependent stationary series out of i.i.d. noise: adding up neighbouring noise terms, and regressing the series on its own past.

Let $\epsilon_1, \epsilon_2, \dots$ be i.i.d. with mean 0 and variance σ^2.

2.1.1 Summing noise: MA(1)

For a real θ put

$$X_t = \epsilon_t + \theta\epsilon_{t-1}. \tag{2.1}$$

Then $EX_t = 0$, $\mathrm{Var}(X_t) = (1+\theta^2)\sigma^2$, and since $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$,

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1},\, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \geq 2, \end{cases}$$

which does not depend on t. Thus $\{X_t\}$ is stationary; it is the *moving average model of order* 1, MA(1), the simplest stationary model.

2.1.2 Regressing on the past: AR(1)

The simplest regression of the series on itself is

$$X_t = \phi X_{t-1} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2). \tag{2.2}$$

Does a stationary series satisfy (2.2)?

Case $\phi = 1$ (random walk). Iterating gives $X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t$, so $\mathrm{Var}(X_t - X_0) = t\sigma^2$. If $\{X_t\}$ were stationary with ACVF γ, then

$$\mathrm{Var}(X_t - X_0) = 2\gamma(0) - 2\gamma(t), \tag{2.3}$$

so $2\gamma(0) - 2\gamma(t) = t\sigma^2$. The left side is at most $4\gamma(0)$ for all t by Theorem 1.2.6(3), while the right side diverges, a contradiction. Hence no stationary solution exists; the process with $\phi = 1$ is the *random walk*, whose paths wander with no fixed level. The case $\phi = -1$ is similar: then $X_{2t} - X_0 = \sum_{j=1}^{2t} (-1)^j \epsilon_j$ has variance $2t\sigma^2$, again contradicting stationarity.

Case $|\phi| < 1$. Suppose a stationary solution exists. Iterating (2.2),

$$X_t = \phi X_{t-1} + \epsilon_t = \phi(\phi X_{t-2} + \epsilon_{t-1}) + \epsilon_t = \cdots = \sum_{j=0}^{n-1} \phi^j \epsilon_{t-j} + \phi^n X_{t-n}. \tag{2.4}$$

Since $E(\phi^n X_{t-n})^2 = \phi^{2n} EX_0^2 \to 0$, the only candidate is

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-i} \tag{2.5}$$

(the series converges in mean square and almost surely since $\sum |\phi|^i < \infty$). It is indeed a solution, because

$$\phi X_{t-1} + \epsilon_t = \sum_{i=0}^{\infty} \phi^{i+1} \epsilon_{t-1-i} + \epsilon_t = \sum_{j=1}^{\infty} \phi^j \epsilon_{t-j} + \epsilon_t = X_t,$$

and it is stationary: $EX_t = 0$ and

$$\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}\Big(\sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}, \sum_{k=0}^{\infty} \phi^k \epsilon_{t+h-k}\Big) = \sum_{j=0}^{\infty} \phi^j \phi^{j+h} \sigma^2 = \phi^h \frac{\sigma^2}{1-\phi^2}, \quad h \geq 0.$$

The model (2.2) is the *AR(1) model*. These two constructions are the building blocks of the general models below.

2.2 Autoregressive models

Motivation. An AR(p) model regresses X_t on p of its own lags. Whether it has a stationary solution is decided by where the roots of a polynomial lie, and the backshift operator turns the model into polynomial algebra.

Definition 2.2.1 (Autoregressive model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.6}$$

then $\{X_t\}$ follows an *autoregressive model of order* p, written $\{X_t\} \sim AR(p)$.

Define the *characteristic polynomial* $\phi : \mathbb{C} \to \mathbb{C}$,

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p. \tag{2.7}$$

If every root of $\phi(z) = 0$ satisfies $|z| > 1$, then $1/\phi(z)$ has a power series expansion valid on $|z| \leq 1$,

$$\frac{1}{\phi(z)} = \sum_{i=0}^{\infty} a_i z^i, \tag{2.8}$$

and the stationary solution of (2.6) is

$$X_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i}. \tag{2.9}$$

Thus X_t is a function of ϵ_s, $s \leq t$, only, so X_t is uncorrelated with (for i.i.d. noise, independent of) ϵ_s, $s > t$. Such a representation is called *causal*. These facts are proved in Proposition 2.2.B below.

2.2.1 The backshift operator

The *backshift operator* B acts by $BX_t = X_{t-1}$, and on functions by $Bf(t) = f(t-1)$; thus $B^j X_t = X_{t-j}$. For power series $a(z) = \sum_{i \geq 0} a_i z^i$, $b(z) = \sum_{j \geq 0} b_j z^j$ with $\sum |a_i| < \infty$, $\sum |b_j| < \infty$, put

$$c(z) = (ab)(z) = a(z)b(z) = \sum_{k=0}^{\infty} c_k z^k, \qquad c_k = \sum_{i=0}^{k} a_i b_{k-i}.$$

For a stationary $\{X_t\}$ one then has

$$a(B)\big(b(B)X_t\big) = (ab)(B)X_t = b(B)\big(a(B)X_t\big), \tag{2.10}$$

and all these series are stationary. With this notation (2.6) reads $\phi(B)X_t = \epsilon_t$, and (2.9) reads

$$X_t = \frac{1}{\phi(B)}\epsilon_t, \qquad \frac{1}{\phi(B)} = \sum_{i=0}^{\infty} a_i B^i. \tag{2.11}$$

Lemma 2.2.A (Operator calculus; supplement)

Let $\{X_t\}$ satisfy $\sup_t E|X_t| < \infty$ (e.g. stationary), and let $\sum|a_i| < \infty$, $\sum|b_j| < \infty$. Then $b(B)X_t = \sum_j b_j X_{t-j}$ converges absolutely a.s. and in L^1, (2.10) holds a.s., and if $\{X_t\}$ is stationary so is $\{b(B)X_t\}$, with ACVF $\sum_{j,k} b_j b_k \gamma_X(h+j-k)$.

Proof. [Supplement]

Step 1. Let $M = \sup_t E|X_t|$. By monotone convergence $E\sum_j |b_j||X_{t-j}| \le M\sum_j |b_j| < \infty$, so the series converges absolutely a.s. and in L^1. If $\{X_t\}$ is stationary, then $E|X_t| \le (EX_t^2)^{1/2}$ is bounded, and the same argument in L^2 (Minkowski: $\|\sum_j |b_j||X_{t-j}|\|_2 \le \sum_j |b_j|\,\|X_0\|_2$) gives mean-square convergence.

Step 2. Since $\sum_i\sum_j |a_i||b_j|E|X_{t-i-j}| \le M\sum_i |a_i|\sum_j |b_j| < \infty$, the double series $\sum_{i,j} a_i b_j X_{t-i-j}$ converges absolutely a.s., so it may be summed in any order (Fubini). Summing over j first gives $a(B)(b(B)X_t)$; grouping by $k = i+j$ gives $\sum_k c_k X_{t-k} = (ab)(B)X_t$. Since $ab = ba$, the third expression in (2.10) follows by symmetry.

Step 3. $E\,b(B)X_t = \mu\sum b_j$, and by L^2 convergence covariances can be taken termwise: $\mathrm{Cov}(b(B)X_t, b(B)X_{t+h}) = \sum_{j,k} b_j b_k \gamma_X(h+j-k)$, which is free of t and absolutely convergent. ■

Proposition 2.2.B (Causal stationary solution; supplement)

Suppose $\phi(z) \neq 0$ for all $|z| \le 1$. Then:
(1) there are $r > 1$ and $C < \infty$ with $1/\phi(z) = \sum_{i\ge 0} a_i z^i$ for $|z| < r$ and $|a_i| \le C r_0^{-i}$ for any fixed $1 < r_0 < r$;
(2) $X_t = \sum_i a_i \epsilon_{t-i}$ is a stationary solution of (2.6);
(3) it is the only stationary solution.

Proof. [Supplement]

Step 1. The roots of the polynomial ϕ are finitely many and all have modulus > 1; let $r > 1$ be the smallest modulus. Then $1/\phi$ is analytic on the disc $|z| < r$ and has a Taylor series $\sum a_i z^i$ there. For $1 < r_0 < r$, Cauchy's estimate on the circle $|z| = r_0$ gives $|a_i| \le r_0^{-i}\max_{|z|=r_0}|1/\phi(z)| = C r_0^{-i}$. In particular $\sum|a_i| < \infty$.

Step 2. By the Lemma, $X_t = a(B)\epsilon_t$ is well defined and stationary, and $\phi(B)X_t = (\phi a)(B)\epsilon_t = \epsilon_t$ because $\phi(z)a(z) = 1$ for $|z| \le 1$, so the coefficient sequence of ϕa is $(1, 0, 0, \dots)$.

Step 3. Let $\{Y_t\}$ be any stationary solution, $\phi(B)Y_t = \epsilon_t$. Applying $a(B)$ and using the Lemma,

$$a(B)\epsilon_t = a(B)(\phi(B)Y_t) = (a\phi)(B)Y_t = Y_t,$$

so $Y_t = X_t$ a.s. ■

For $p = 1$, $\phi(z) = 1 - \phi z$ has root $1/\phi$, which lies outside the unit circle iff $|\phi| < 1$, and $a_i = \phi^i$, recovering (2.5).

Remark 2.2.2 (Nonzero mean)

A stationary AR(p) model with mean μ is written

$$X_t - \mu - \phi_1(X_{t-1} - \mu) - \phi_2(X_{t-2} - \mu) - \cdots - \phi_p(X_{t-p} - \mu) = \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2).$$

Example 2.2.3 (Variance of a stationary AR(1))

Let $X_t = \phi X_{t-1} + \epsilon_t$ with $|\phi| < 1$. Since X_{t-1} depends only on ϵ_s, $s \le t-1$, $\mathrm{Cov}(X_{t-1}, \epsilon_t) = 0$, so

$$\gamma(0) = \mathrm{Var}(\phi X_{t-1} + \epsilon_t) = \phi^2 \gamma(0) + \sigma^2, \qquad \text{hence} \qquad \gamma(0) = \frac{\sigma^2}{1-\phi^2}.$$

2.3 Moving average models

Motivation. MA(q) models are finite linear filters of white noise. They are always stationary, and their autocorrelation vanishes beyond lag q, a signature that identifies them in data.

Definition 2.3.1 (Moving average model)

If $\{X_t\}$ satisfies

$$X_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.12}$$

then $\{X_t\}$ follows a *moving average model of order* q, written $\{X_t\} \sim MA(q)$.

With the backshift operator,

$$X_t = \theta(B)\epsilon_t, \qquad \theta(z) = \sum_{i=0}^{q} \theta_i z^i, \quad \theta_0 = 1. \tag{2.13}$$

For MA(1), $X_t = \epsilon_t + \theta \epsilon_{t-1}$ with $\{\epsilon_t\} \sim WN(0, \sigma^2)$, we found in §2.1

$$\gamma(h) = \mathrm{Cov}(\epsilon_t + \theta\epsilon_{t-1}, \epsilon_{t+h} + \theta\epsilon_{t+h-1}) = \begin{cases} (1+\theta^2)\sigma^2, & h = 0, \\ \theta\sigma^2, & |h| = 1, \\ 0, & |h| \ge 2. \end{cases} \tag{2.14}$$

In general, using $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$, for $h \ge 0$

$$\gamma(h) = \sum_{i=0}^{q} \sum_{j=0}^{q} \theta_i \theta_j \,\mathrm{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_{i=0}^{q-h} \theta_i \theta_{i+h},$$

the sum being empty when $h > q$. Hence, by symmetry in h,

$$\gamma(h) = \begin{cases} 0, & |h| \ge q+1, \\ (\theta_{|h|} + \theta_1 \theta_{1+|h|} + \cdots + \theta_{q-|h|}\theta_q)\sigma^2, & |h| \le q. \end{cases} \tag{2.15}$$

Since γ depends only on h, an MA(q) is always stationary, with mean 0 and variance $(1 + \theta_1^2 + \cdots + \theta_q^2)\sigma^2$. Its ACF is zero beyond lag q; conversely, a sample ACF that vanishes from some lag on suggests an MA model.

2.4 Autoregressive moving average models

Motivation. Combining an AR and an MA part gives models that describe rich dependence with few parameters. Two root conditions matter: causality (roots of ϕ) for stationarity, and invertibility (roots of θ) for recovering the noise from the data.

Definition 2.4.1 (ARMA model)

If $\{X_t\}$ satisfies

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \tag{2.16}$$

then $\{X_t\}$ follows an *autoregressive moving average model of orders p and q*, written $\{X_t\} \sim ARMA(p, q)$. With $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$, (2.16) reads

$$\phi(B) X_t = \theta(B) \epsilon_t. \tag{2.17}$$

If $\{X_t\}$ is stationary with mean $\mu \neq 0$, X_t is replaced by $X_t - \mu$ in (2.17).

ARMA models came out of the experience that pure AR fits often need a large order. The *principle of parsimony*, prefer the model with fewer parameters among models explaining the data equally well, underlies all model selection, and parsimonious ARMA models are known to fit real data well.

As for AR(p), if $\phi(z) \neq 0$ for all $|z| \leq 1$ then $\{X_t\}$ is stationary and

$$X_t = \sum_{i=0}^{\infty} a_i \epsilon_{t-i}, \qquad \sum_{i=0}^{\infty} a_i z^i = \frac{\theta(z)}{\phi(z)}, \tag{2.18}$$

that is, by (2.10),

$$X_t = \frac{\theta(B)}{\phi(B)} \epsilon_t = \left(\frac{\theta}{\phi}\right)(B) \epsilon_t. \tag{2.19}$$

The ACVF of (2.18) decays to 0 geometrically as h grows; for this reason ARMA processes are called *short-memory processes*.

Proposition 2.4.A (Causal ARMA and geometric decay; supplement)

If $\phi(z) \neq 0$ for $|z| \leq 1$, then (2.18) is the unique stationary solution of (2.17), with $|a_i| \leq C r_0^{-i}$ for some $C < \infty$, $r_0 > 1$, and

$$\gamma(h) = \sigma^2 \sum_{i=0}^{\infty} a_i a_{i+|h|}, \qquad |\gamma(h)| \leq C' r_0^{-|h|}.$$

Proof. [Supplement]

Step 1. θ/ϕ is analytic on $|z| < r$ ($r > 1$ the smallest modulus of a root of ϕ), so the Cauchy estimate gives $|a_i| \leq C r_0^{-i}$ exactly as in Proposition 2.2.B.

Step 2. By the Lemma, $X_t = a(B)\epsilon_t$ is stationary and $\phi(B)X_t = (\phi a)(B)\epsilon_t = \theta(B)\epsilon_t$, because $\phi a = \theta$ as power series. If Y is another stationary solution, then $Y_t = (\phi^{-1}\phi)(B) Y_t = \phi^{-1}(B)\theta(B)\epsilon_t = a(B)\epsilon_t$, where $\phi^{-1}(z) = 1/\phi(z)$ has absolutely summable coefficients.

Step 3. For $h \geq 0$, $\gamma(h) = \sum_{i,j} a_i a_j \operatorname{Cov}(\epsilon_{t-i}, \epsilon_{t+h-j}) = \sigma^2 \sum_i a_i a_{i+h}$ (take $j = i + h$), and

$$|\gamma(h)| \leq \sigma^2 C^2 \sum_{i \geq 0} r_0^{-i} r_0^{-i-h} = \frac{\sigma^2 C^2}{1 - r_0^{-2}} r_0^{-h}.$$

■

2.4.1 Invertibility

If $\{X_t\}$ is a stationary ARMA process and $\theta(z) \neq 0$ for all $|z| \leq 1$, then

$$\sum_{j=0}^{\infty} \pi_j X_{t-j} = \epsilon_t, \qquad \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \quad \pi_0 = 1. \tag{2.20}$$

In this case $\{X_t\}$ is called *invertible*: the current noise, and hence the present value, is a linear combination of infinitely many past observations.

Proposition 2.4.B (Invertibility; supplement)

Let $\{X_t\}$ be a stationary solution of $\phi(B)X_t = \theta(B)\epsilon_t$ and suppose $\theta(z) \neq 0$ for $|z| \leq 1$. Then (2.20) holds a.s., with $|\pi_j|$ decaying geometrically.

Proof. [Supplement] As in Proposition 2.2.B, $1/\theta(z) = \sum_j \eta_j z^j$ with geometrically decaying η_j, and $\pi(z) = \phi(z)/\theta(z) = \phi(z)\eta(z)$ also has geometrically decaying coefficients (ϕ is a polynomial). By the Lemma applied to the stationary series $\{X_t\}$ and $\{\epsilon_t\}$,

$$\pi(B)X_t = \eta(B)(\phi(B)X_t) = \eta(B)(\theta(B)\epsilon_t) = (\eta\theta)(B)\epsilon_t = \epsilon_t.$$

■

Example 2.4.2 (One ACF, two MA(1) models)

The models

$$X_t = \epsilon_t + 2\epsilon_{t-1} \quad \text{(not invertible)}, \qquad X_t = \epsilon_t + \tfrac{1}{2}\epsilon_{t-1} \quad \text{(invertible)} \tag{2.21}$$

have the same ACF: $\rho(1) = \theta/(1+\theta^2) = 2/5$ for $\theta = 2$ and for $\theta = 1/2$, and $\rho(h) = 0$ for $|h| \geq 2$. (Generally θ and $1/\theta$ give the same MA(1) ACF; the noise variances differ so that the ACVFs also agree: σ^2 for $\theta = 2$ corresponds to $4\sigma^2$ for $\theta = 1/2$.)

Thus a single ACF corresponds to both an invertible and a non-invertible model, and the invertible one is preferred. The reason is prediction: given $X_1, \ldots, X_n$ with n large, (2.20) gives the easily computed predictor

$$\hat{X}_{n+1} = -\pi_1 X_n - \pi_2 X_{n-1} - \cdots - \pi_n X_1,$$

whereas no such representation is available in the non-invertible case.

The orders p, q must be determined from data. Too many parameters violate parsimony, too few cause *model bias*; criteria such as AIC and BIC are used (Chapter 4). The choice matters for inference: e.g. for an AR(p) with i.i.d. noise and $n > p$,

$$E(X_{n+1} \mid X_1, \ldots, X_n) = E(\phi_1 X_n + \cdots + \phi_p X_{n-p+1} + \epsilon_{n+1} \mid X_1, \ldots, X_n) = \phi_1 X_n + \cdots + \phi_p X_{n-p+1},$$

since ϵ_{n+1} is independent of $X_1, \ldots, X_n$ by causality; a wrong p yields a wrong predictor.

Example 2.4.3 (ACVF of a causal ARMA(1,1))

Let $X_t = \phi X_{t-1} + \epsilon_t + \theta\epsilon_{t-1}$, $|\phi| < 1$. Since X_{t-h} is uncorrelated with $\epsilon_t, \epsilon_{t-1}$ for $h \geq 2$,

$$\gamma(h) = \mathrm{Cov}(\phi X_{t-1} + \epsilon_t + \theta\epsilon_{t-1}, X_{t-h}) = \phi\gamma(h-1), \quad \text{so} \quad \gamma(h) = \phi^{h-1}\gamma(1), \ h \geq 1.$$

Using $\mathrm{Cov}(\epsilon_{t-1}, X_{t-1}) = \sigma^2$ and $\mathrm{Cov}(\epsilon_t, X_t) = \sigma^2$, $\mathrm{Cov}(\epsilon_{t-1}, X_t) = (\phi+\theta)\sigma^2$:

$$\begin{aligned} \gamma(1) &= \phi\gamma(0) + \theta\sigma^2, \\ \gamma(0) &= \phi^2\gamma(0) + \sigma^2 + \theta^2\sigma^2 + 2\phi\theta\sigma^2. \end{aligned}$$

Hence

$$\gamma(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1-\phi^2}, \qquad \gamma(1) = \frac{\sigma^2(\phi+\theta)(1+\phi\theta)}{1-\phi^2}.$$

2.5 The partial autocorrelation function

Motivation. The ACF identifies MA orders but not AR orders. The partial autocorrelation removes the linear effect of the intermediate observations and cuts off after lag p for an AR(p), giving the complementary identification tool.

Let $\{X_t\}$ be stationary with mean 0. To measure the correlation of X_1 and X_{1+k} after removing the effect of $X_2, \dots, X_k$, let

$$F = \Big\{ \sum_{i=2}^{k} a_i X_i : a_i \in \mathbb{R} \Big\},$$

and let $P_F Y$ denote the *projection* of Y onto F: the element of F with $E\big[(Y - P_F Y) X_i\big] = 0$ for all $i = 2, \dots, k$. ($P_F Y$ is the best linear predictor of Y from $X_2, \dots, X_k$; for $k = 1$, $F = \{0\}$.)

Definition 2.5.1 (Partial autocorrelation function)

For $k \geq 1$,

$$\phi_{kk} = \mathrm{Corr}(X_{k+1} - P_F X_{k+1},\ X_1 - P_F X_1)$$

is the *partial autocorrelation function* (PACF); for fixed k, ϕ_{kk} is the partial autocorrelation coefficient at lag k.

2.5.1 Computing the PACF

Let ρ be the ACF. Solve

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(h-1) \\ \rho(1) & \rho(0) & \cdots & \rho(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(h-1) & \rho(h-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_{h1} \\ \phi_{h2} \\ \vdots \\ \phi_{hh} \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(h) \end{pmatrix}. \tag{2.22}$$

The last component ϕ_{hh} of the solution is the PACF at lag h. By Cramer's rule, writing $R_h = [\rho(i-j)]_{i,j=1}^{h}$ and R_h^* for R_h with its last column replaced by $(\rho(1), \dots, \rho(h))^T$,

$$\phi_{hh} = \frac{\det R_h^*}{\det R_h}.$$

Proposition 2.5.A (PACF via the prediction equations; supplement)

Let $\{X_t\}$ be stationary, mean 0, with $\Gamma_h = [\gamma(i-j)]_{i,j=1}^{h}$ nonsingular. Then the best linear predictor of X_{h+1} from $X_1, \dots, X_h$ is $\sum_{j=1}^{h} \phi_{hj} X_{h+1-j}$ with coefficients solving (2.22), and its coefficient ϕ_{hh} on X_1 equals the PACF of Definition 2.5.1.

Proof. [Supplement]

Step 1. A linear combination $\hat{X} = \sum_j c_j X_{h+1-j}$ is the projection of X_{h+1} onto $\mathrm{span}(X_1, \dots, X_h)$ iff $E[(X_{h+1} - \hat{X}) X_{h+1-i}] = 0$ for $i = 1, \dots, h$, i.e. $\sum_j c_j \gamma(i-j) = \gamma(i)$. Dividing by $\gamma(0)$ gives (2.22), uniquely solvable since Γ_h is nonsingular.

Step 2. Let $F = \mathrm{span}(X_2, \dots, X_h)$, $U = X_1 - P_F X_1 \perp F$ (here $\perp$ means uncorrelated). Then $\mathrm{span}(X_1, \dots, X_h) = F \oplus \mathrm{span}(U)$ orthogonally, so the projection of X_{h+1} is

$$P_F X_{h+1} + \frac{E(X_{h+1} U)}{E U^2}\, U = P_F X_{h+1} + \frac{E(X_{h+1} U)}{E U^2}\, (X_1 - P_F X_1).$$

Since $P_F X_{h+1}, P_F X_1 \in F$ do not involve X_1, the coefficient of X_1 is $\phi_{hh} = E(X_{h+1}U)/EU^2$ (coefficients are unique as Γ_h is nonsingular).

Step 3. Let $V = X_{h+1} - P_F X_{h+1}$. As $U \perp F$, $E(X_{h+1}U) = E(VU)$, so $\phi_{hh} = E(VU)/EU^2$. By stationarity the covariance matrix of $(X_2, \dots, X_h, X_{h+1})$ equals that of $(X_h, \dots, X_2, X_1)$ (both are $[\gamma(i-j)]$, as γ is even), so predicting X_{h+1} from $X_2, \dots, X_h$ and X_1 from $X_h, \dots, X_2$ have the same error variance: $EV^2 = EU^2$. Hence $\phi_{hh} = E(VU)/\sqrt{EV^2 EU^2} = \mathrm{Corr}(V, U)$. ■

Example 2.5.2 (PACF of an AR(1))

For a stationary AR(1), $\rho(h) = \phi^h$. At $h = 1$: $\phi_{11} = \rho(1) = \phi$. At $h = 2$:

$$\begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \phi_{21} \\ \phi_{22} \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix} \implies \begin{pmatrix} \phi_{21} \\ \phi_{22} \end{pmatrix} = \frac{1}{1-\phi^2} \begin{pmatrix} 1 & -\phi \\ -\phi & 1 \end{pmatrix} \begin{pmatrix} \phi \\ \phi^2 \end{pmatrix} = \begin{pmatrix} \phi \\ 0 \end{pmatrix},$$

so $\phi_{22} = 0$; likewise $\phi_{hh} = 0$ for all $h \geq 2$.

Proposition 2.5.B (PACF of a causal AR(p); supplement)

If $\{X_t\}$ is a causal AR(p) with $\phi_p \neq 0$ and $\sigma^2 > 0$, then $\phi_{pp} = \phi_p$ and $\phi_{hh} = 0$ for all $h > p$.

Proof. [Supplement]

Step 1. Γ_h is nonsingular for every h. Let $a \neq 0$ and let i be the largest index with $a_i \neq 0$. By causality $X_i = \epsilon_i + \sum_{k \geq 1} a'_k \epsilon_{i-k}$ (the leading coefficient of $1/\phi$ is 1), and $X_1, \dots, X_{i-1}$ are functions of ϵ_u, $u < i$. Hence $Y = \sum_{j \leq i} a_j X_j = a_i \epsilon_i + W$ with W uncorrelated with ϵ_i, so $a^T \Gamma_h a = \mathrm{Var}(Y) \geq a_i^2 \sigma^2 > 0$.

Step 2. Let $h \geq p$. The candidate $\hat{X} = \phi_1 X_h + \cdots + \phi_p X_{h+1-p} \in \mathrm{span}(X_1, \dots, X_h)$ has error $X_{h+1} - \hat{X} = \epsilon_{h+1}$, which is uncorrelated with $X_1, \dots, X_h$ by causality (each X_s is a function of ϵ_u, $u \leq s$). Hence $\hat{X}$ is the projection, and by uniqueness its coefficients are $(\phi_1, \dots, \phi_p, 0, \dots, 0)$.

Step 3. By Proposition 2.5.A, ϕ_{hh} is the coefficient on $X_1 = X_{h+1-h}$: it equals ϕ_p when $h = p$ and 0 when $h > p$. ■

Thus the PACF of an AR(p) vanishes after lag p, while the ACF of an MA(q) vanishes after lag q; the opposite functions decay geometrically. Figure 1 illustrates this duality, which is the basis for choosing p and q from data (Chapters 3 and 4).

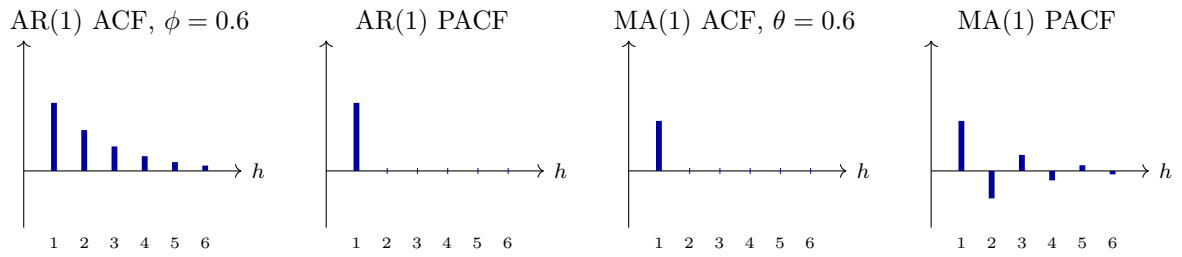


Figure 1: ACF and PACF of an AR(1) and an MA(1). The AR(1) PACF and the MA(1) ACF cut off after lag 1; the other two decay geometrically.

2.6 Selected exercises

Motivation. These textbook exercises go beyond the propositions above: sums of MA processes, models that share an ACF, and an exact invertibility criterion for ARMA(1,1). Covariance formulas

already proved in the text (the ACVF of a linear process, the operator calculus, the MA(q) ACVF) are not repeated.

Exercise 2.8 (Sum of independent MA(1) and MA(2))

Let $X_t = \epsilon_t + a\epsilon_{t-1}$ with $\{\epsilon_t\} \sim IID(0, \sigma_1^2)$ and $Y_t = \eta_t + b_1\eta_{t-1} + b_2\eta_{t-2}$ with $\{\eta_t\} \sim IID(0, \sigma_2^2)$, the two noise sequences being independent, and $\sigma_1^2 + \sigma_2^2 > 0$. Show that $Z_t = X_t + Y_t$ can be regarded as an MA(2) model: there are a white noise $\{u_t\} \sim WN(0, v)$ with $v > 0$ and constants c_1, c_2 such that

$$Z_t = u_t + c_1 u_{t-1} + c_2 u_{t-2} \qquad \text{for all } t.$$

Solution. All random variables below lie in L^2 with the inner product $E(UV)$; "$\perp$" means uncorrelated. Let H_t be the closed linear span of $\{Z_s : s \le t\}$ and P_t the orthogonal projection onto H_t.

Step 1. (Z is stationary with ACVF vanishing beyond lag 2.) By (2.15), $\gamma_X(h) = 0$ for $|h| \ge 2$ and $\gamma_Y(h) = 0$ for $|h| \ge 3$. Since $\mathrm{Cov}(X_s, Y_u) = 0$ for all s, u, Exercise 1.7 gives that Z is stationary with mean 0 and $\gamma_Z = \gamma_X + \gamma_Y$; hence $\gamma_Z(h) = 0$ for $|h| \ge 3$ and $\gamma_Z(0) = (1+a^2)\sigma_1^2 + (1+b_1^2+b_2^2)\sigma_2^2 > 0$. In particular $Z_t \perp Z_s$ for $s \le t-3$, so $Z_t \perp H_{t-3}$.

Step 2. (Innovations.) Put $u_t = Z_t - P_{t-1}Z_t$. Then $u_t \in H_t$ and $u_t \perp H_{t-1}$. For $s < t$, $u_s \in H_s \subseteq H_{t-1}$, so $E(u_s u_t) = 0$.

Step 3. (Shift invariance.) For $m \ge 1$ let $\alpha_m = (\alpha_{m,1}, \dots, \alpha_{m,m})^T$ be the minimum-norm solution of $\Gamma_m \alpha_m = (\gamma_Z(1), \dots, \gamma_Z(m))^T$, $\Gamma_m = [\gamma_Z(i-j)]_{i,j=1}^m$. As in Step 1 of the proof of Proposition 2.5.A, $\sum_i \alpha_{m,i} Z_{t-i}$ is the projection of Z_t onto $\mathrm{span}(Z_{t-1}, \dots, Z_{t-m})$, and α_m depends on γ_Z only, not on t. These subspaces increase and their union is dense in H_{t-1}, so the projections converge in L^2 to $P_{t-1}Z_t$. By continuity of the inner product, every covariance $E(u_t u_s)$, $E(u_t Z_s)$ is the limit of a covariance of finite linear combinations of the Z's with t-free coefficients, hence depends on $t - s$ only. In particular $v = Eu_t^2$ does not depend on t, and $\{u_t\} \sim WN(0, v)$ by Step 2.

Step 4. ($v > 0$.) If $v = 0$, then $Z_s = P_{s-1}Z_s \in H_{s-1}$ for every s, so $H_s = H_{s-1}$ for all s and $H_t = H_{t-3}$. By Step 1, $Z_t \perp H_{t-3} = H_t \ni Z_t$, so $EZ_t^2 = 0$, contradicting $\gamma_Z(0) > 0$.

Step 5. (Orthogonal decomposition.) Since $Z_s = u_s + P_{s-1}Z_s$ with $P_{s-1}Z_s \in H_{s-1}$, we have $H_s = H_{s-1} \oplus \mathrm{span}(u_s)$ (the sum of a closed subspace and an orthogonal one-dimensional subspace is closed). Applying this for $s = t-1$ and $s = t-2$,

$$H_{t-1} = H_{t-3} \oplus \mathrm{span}(u_{t-2}) \oplus \mathrm{span}(u_{t-1}).$$

Step 6. (Conclusion.) Projecting Z_t onto the three orthogonal pieces,

$$Z_t = u_t + P_{t-1}Z_t = u_t + \frac{E(Z_t u_{t-1})}{v}\, u_{t-1} + \frac{E(Z_t u_{t-2})}{v}\, u_{t-2} + P_{H_{t-3}} Z_t.$$

The last term is 0 because $Z_t \perp H_{t-3}$, and by Step 3 the coefficients $c_j = E(Z_t u_{t-j})/v$ do not depend on t. This is the claimed MA(2) representation. Moreover $u_{t-2} - Z_{t-2} \in H_{t-3} \perp Z_t$ gives $c_2 = \gamma_Z(2)/v = b_2\sigma_2^2/v$, so the order is exactly 2 when $b_2\sigma_2^2 \neq 0$.

The driving noise $\{u_t\}$ is white noise, in general not i.i.d.; "MA(2)" is meant at the level of second-order properties. ■

Exercise 2.9 (Two filters with the same ACF)

Let $\{Y_t\}$ be stationary with mean 0 and ACVF γ_Y, not a.s. zero, and let

$$X_t = Y_t - 0.4Y_{t-1}, \qquad W_t = Y_t - 2.5Y_{t-1}.$$

(a) Express the ACVFs of $\{X_t\}$ and $\{W_t\}$ through γ_Y.
(b) Show that $\{X_t\}$ and $\{W_t\}$ have the same ACF.

Solution.

(a) For $c \in \mathbb{R}$ and $V_t = Y_t - cY_{t-1}$, bilinearity and stationarity of Y give

$$\begin{aligned}\mathrm{Cov}(V_t, V_{t+h}) &= \gamma_Y(h) - c\,\gamma_Y(h-1) - c\,\gamma_Y(h+1) + c^2\gamma_Y(h) \\ &= (1+c^2)\gamma_Y(h) - c\{\gamma_Y(h-1) + \gamma_Y(h+1)\},\end{aligned}$$

free of t. With $c = 0.4$ and $c = 2.5$,

$$\begin{aligned}\gamma_X(h) &= 1.16\,\gamma_Y(h) - 0.4\{\gamma_Y(h-1) + \gamma_Y(h+1)\}, \\ \gamma_W(h) &= 7.25\,\gamma_Y(h) - 2.5\{\gamma_Y(h-1) + \gamma_Y(h+1)\}.\end{aligned}$$

(b)

Step 1. Since $2.5 = 1/0.4$, we have $7.25 = 6.25 \times 1.16$ and $2.5 = 6.25 \times 0.4$, so $\gamma_W(h) = 6.25\,\gamma_X(h)$ for all h. (In general $(1 + c^{-2}) = c^{-2}(1 + c^2)$ and $c^{-1} = c^{-2}c$.)

Step 2. The ACF is defined if $\gamma_X(0) > 0$. Suppose $\gamma_X(0) = 0$, i.e. $Y_t = 0.4Y_{t-1}$ a.s. for all t. Iterating, $Y_t = 0.4^n Y_{t-n}$, so $\gamma_Y(0) = 0.16^n\gamma_Y(0)$ for every n, forcing $\gamma_Y(0) = 0$, i.e. $Y_t = 0$ a.s., which is excluded.

Step 3. Hence $\rho_W(h) = \gamma_W(h)/\gamma_W(0) = 6.25\gamma_X(h)/(6.25\gamma_X(0)) = \rho_X(h)$.

This generalizes Example 2.4.2: the filters $1 - cB$ and $1 - c^{-1}B$ give the same ACF. ■

Exercise 2.10 (Invertibility of ARMA(1,1))

Let $\{X_t\}$ be the stationary solution of $X_t - \phi X_{t-1} = \epsilon_t + \theta\epsilon_{t-1}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, $\sigma^2 > 0$, $|\phi| < 1$. Find the condition for $\{X_t\}$ to be invertible, i.e. for $\epsilon_t = \sum_{j\geq 0}\pi_j X_{t-j}$ a.s. for some $\sum_j |\pi_j| < \infty$.

Solution. The condition is $|\theta| < 1$.

(a) Sufficiency.

Step 1. If $|\theta| < 1$, the root $-1/\theta$ of $\theta(z) = 1 + \theta z$ lies outside the unit disc, so Proposition 2.4.B applies and $\{X_t\}$ is invertible.

Step 2. Explicitly, for $|z| < 1/|\theta|$,

$$\frac{\phi(z)}{\theta(z)} = (1 - \phi z)\sum_{k\geq 0}(-\theta)^k z^k = 1 + \sum_{j\geq 1}\{(-\theta)^j - \phi(-\theta)^{j-1}\}z^j,$$

so $\pi_0 = 1$ and $\pi_j = -(\phi + \theta)(-\theta)^{j-1}$ for $j \geq 1$, which is absolutely summable.

(b) Necessity. Suppose $\epsilon_t = \pi(B)X_t$ with $\sum|\pi_j| < \infty$ and $|\theta| \geq 1$.

Step 1. By Proposition 2.4.A, $X_t = a(B)\epsilon_t$ with $\sum|a_i| < \infty$ and $a(z) = \theta(z)/\phi(z)$ for $|z| \leq 1$. By Lemma 2.2.A, $\epsilon_t = \pi(B)a(B)\epsilon_t = c(B)\epsilon_t$ with $c = \pi a$ (Cauchy product), $\sum|c_k| < \infty$.

Step 2. Taking the covariance of $\epsilon_t = \sum_k c_k\epsilon_{t-k}$ with ϵ_{t-k} (termwise, by L^2 convergence) gives $\sigma^2\mathbb{1}\{k = 0\} = c_k\sigma^2$, so $c_0 = 1$ and $c_k = 0$ for $k \geq 1$. Since both series converge absolutely on $|z| \leq 1$, $\pi(z)a(z) = c(z) = 1$ there, i.e.

$$\pi(z)\,\theta(z) = \phi(z), \qquad |z| \leq 1.$$

Step 3. The point $z_0 = -1/\theta$ satisfies $|z_0| \leq 1$ and $\theta(z_0) = 0$, so $\phi(z_0) = 1 + \phi/\theta = 0$, i.e. $\phi = -\theta$. But then $|\phi| = |\theta| \geq 1$, contradicting $|\phi| < 1$. Hence $|\theta| < 1$ is necessary.

If $\phi = -\theta$ (common factor), then $|\theta| = |\phi| < 1$ and $a(z) = 1$, so $X_t = \epsilon_t$ is trivially invertible, consistent with the criterion. ■

Chapter 3. Inference for Stationary ARMA Models

3.1 Estimation of the mean

Motivation. For i.i.d. data the central limit theorem gives $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$, the basis of tests and confidence intervals. For a stationary ARMA series the same holds, but the asymptotic variance is a function of the model parameters and can differ greatly from $\mathrm{Var}(X_t)$.

Throughout, the subscript n on statistics is dropped ($\bar{X} = \bar{X}_n$, $\hat{\theta} = \hat{\theta}_n$), and limits are as $n \to \infty$. Consider the AR(1) model

$$X_t - \mu = \phi(X_{t-1} - \mu) + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad |\phi| < 1.$$

By (2.5),

$$X_t = \mu + \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}. \tag{3.1}$$

Theorem 3.1.1 (CLT for the sample mean of an AR(1))

If $\{X_t\}$ is the stationary series (3.1) and $\hat{\mu} = \bar{X} = n^{-1}\sum_{t=1}^{n} X_t$, then

$$\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N\Big(0, \frac{\sigma^2}{(1-\phi)^2}\Big).$$

Proof.

Step 1. (Decomposition.) Put $b_j = \sum_{k \geq j+1} \phi^k = \phi^{j+1}/(1-\phi)$ and $y_t = \sum_{j \geq 0} b_j \epsilon_{t-j}$. Since $\sum |b_j| < \infty$, $\{y_t\}$ is stationary (Lemma 2.2.A). As $b_{j-1} - b_j = \phi^j$ and $b_0 = \phi/(1-\phi)$,

$$\begin{aligned} X_t - \mu &= \epsilon_t + \sum_{j \geq 1} (b_{j-1} - b_j)\epsilon_{t-j} = \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - b_0\epsilon_t + \sum_{j \geq 0} b_j \epsilon_{t-1-j} - \sum_{j \geq 1} b_j \epsilon_{t-j} \\ &= \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - y_t + y_{t-1}. \end{aligned}$$

Step 2. (Telescoping.) Summing over t,

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} (X_t - \mu) = \underbrace{\frac{1}{1-\phi} \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \epsilon_t}_{\text{(I)}} \underbrace{- \frac{1}{\sqrt{n}} y_n + \frac{1}{\sqrt{n}} y_0}_{\text{(II)}}.$$

Step 3. By the i.i.d. CLT, (I) $\xrightarrow{d} N(0, \sigma^2/(1-\phi)^2)$, since $\sum_j \phi^j = 1/(1-\phi)$.

Step 4. $E(\text{II})^2 \leq \frac{2}{n}(Ey_n^2 + Ey_0^2) = \frac{4}{n} Ey_0^2 \to 0$, so (II) $\xrightarrow{p} 0$ by Chebyshev's inequality. Slutsky's theorem gives the claim. ∎

Remark 3.1.2 (General linear processes, ARMA)

The same argument applies to any $X_t = \mu + \sum_{j \geq 0} a_j \epsilon_{t-j}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, with $|a_j| \leq c\rho^j$ ($c > 0$, $0 < \rho < 1$); by Proposition 2.4.A this covers every causal ARMA(p, q). Put $b_j = \sum_{k \geq j+1} a_k$; then $|b_j| \leq c\rho^{j+1}/(1-\rho)$, so $\sum_j |b_j| < \infty$ and $y_t = \sum_j b_j \epsilon_{t-j}$ is stationary.

Exactly as above, $X_t - \mu = (\sum_j a_j)\epsilon_t - y_t + y_{t-1}$, hence

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}}\Big(\sum_{j=0}^{\infty} a_j\Big)\sum_{t=1}^{n}\epsilon_t - \frac{1}{\sqrt{n}}y_n + \frac{1}{\sqrt{n}}y_0 \xrightarrow{d} N\Big(0, \Big(\sum_{j=0}^{\infty} a_j\Big)^2\sigma^2\Big).$$

So the CLT survives without independence, but with a different variance. For testing $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ at level α: under i.i.d. data reject when $|\sqrt{n}\bar{X}| \geq z_{\alpha/2}\sigma$; for a linear process reject when $|\sqrt{n}\bar{X}| \geq z_{\alpha/2}\,|\sum_j a_j|\,\sigma$. Here $z_{\alpha/2}$ satisfies $P(|Z| \leq z_{\alpha/2}) = 1 - \alpha$ for $Z \sim N(0,1)$. For AR(1), $\sum_j a_j = 1/(1-\phi)$, which is huge when ϕ is near 1, so getting the variance right is essential. In particular, the $100(1-\alpha)\%$ confidence interval for μ in an AR(1) model is

$$\bar{X} - \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)} \leq \mu \leq \bar{X} + \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)}.$$

3.2 Sample autocovariance, autocorrelation and partial autocorrelation

Motivation. The ACF and PACF identify ARMA orders, but in practice they must be estimated. This section defines the sample versions and gives Bartlett's theorem, the asymptotic distribution of the sample ACF, which tells us how far a sample autocorrelation can stray from zero by chance.

For an i.i.d. sample $(X_1, Y_1), \ldots, (X_n, Y_n)$ the covariance is estimated by $\widehat{\mathrm{Cov}}(X, Y) = n^{-1}\sum_{j=1}^{n}(X_j - \bar{X})(Y_j - \bar{Y})$. Accordingly:

Definition 3.2.1 (Sample autocovariance function)

For $|h| < n$, the *sample autocovariance function* is

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-|h|}(X_t - \bar{X})(X_{t+|h|} - \bar{X}).$$

For fixed h, $\hat{\gamma}(h)$ is the sample autocovariance coefficient. For stationary ARMA series $P(\hat{\gamma}(h) \to \gamma(h)) = 1$ (by the ergodic theorem), so $\hat{\gamma}(h)$ is consistent. For large h few pairs enter $\hat{\gamma}(h)$, so it is reliable only for small lags; a common rule of thumb is $|h| \leq \sqrt{n}$. The divisor n (rather than $n - |h|$) makes the matrix $[\hat{\gamma}(i-j)]$ nonnegative definite.

Definition 3.2.2 (Sample autocorrelation function)

The *sample autocorrelation function* (SACF), the estimator of $\rho(h)$, is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Like ρ, it satisfies $|\hat{\rho}(h)| \leq 1$.

Definition 3.2.3 (Sample partial autocorrelation)

For fixed h, the last component $\hat{\phi}_{hh}$ of the solution of

$$\begin{pmatrix} \hat{\rho}(0) & \hat{\rho}(1) & \cdots & \hat{\rho}(h-1) \\ \hat{\rho}(1) & \hat{\rho}(0) & \cdots & \hat{\rho}(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\rho}(h-1) & \hat{\rho}(h-2) & \cdots & \hat{\rho}(0) \end{pmatrix} \begin{pmatrix} \hat{\phi}_{h1} \\ \hat{\phi}_{h2} \\ \vdots \\ \hat{\phi}_{hh} \end{pmatrix} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \\ \vdots \\ \hat{\rho}(h) \end{pmatrix}$$

is the *sample partial autocorrelation coefficient*; as a function of h it is the *sample partial autocorrelation function* (SPACF).

Theorem 3.2.4 (Bartlett)

Let $\{X_t\}$ be stationary with

$$X_t = \mu + \sum_{j=0}^{\infty} a_j \epsilon_{t-j}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad \sum_{j=0}^{\infty} |a_j| < \infty, \quad E\epsilon_t^4 < \infty.$$

Then for each fixed $h \geq 1$,

$$\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0, W), \qquad \hat{\boldsymbol{\rho}}(h) = (\hat{\rho}(1), \ldots, \hat{\rho}(h))^T, \quad \boldsymbol{\rho}(h) = (\rho(1), \ldots, \rho(h))^T,$$

where $W = (w_{ij})_{i,j=1}^{h}$ is given by *Bartlett's formula*

$$\begin{aligned} w_{ij} &= \sum_{k=-\infty}^{\infty} \Big\{ \rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j) + 2\rho(i)\rho(j)\rho^2(k) \\ &\qquad - 2\rho(i)\rho(k)\rho(k+j) - 2\rho(j)\rho(k)\rho(k+i) \Big\} \\ &= \sum_{k=1}^{\infty} \{\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k)\}\{\rho(k+j) + \rho(k-j) - 2\rho(j)\rho(k)\}. \end{aligned}$$

Proof. [Supplement] We give the argument at the level of the stated assumptions; two standard limit theorems are quoted. Write $\eta = E\epsilon_t^4/\sigma^4$ and, without loss of generality, $\mu = 0$. Let $\gamma^*(h) = n^{-1}\sum_{t=1}^{n} X_t X_{t+h}$.

Step 1. (Fourth moments.) Since $E\epsilon_i\epsilon_j\epsilon_k\epsilon_l = \sigma^4(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + (\eta - 3)\sigma^4\delta_{ijkl}$, expanding the four series ($a_j = 0$ for $j < 0$) gives

$$\begin{aligned} E[X_t X_{t+p} X_s X_{s+q}] &= \gamma(p)\gamma(q) + \gamma(s-t)\gamma(s-t+q-p) + \gamma(s-t+q)\gamma(s-t-p) \\ &\quad + (\eta - 3)\sigma^4 \sum_i a_i a_{i+p} a_{i+s-t} a_{i+s-t+q}. \end{aligned}$$

Step 2. (Covariances.) Summing over $s, t = 1, \ldots, n$ and grouping by $k = s - t$,

$$\begin{aligned} n\,\mathrm{Cov}(\gamma^*(p), \gamma^*(q)) &= \sum_{|k|<n} \Big(1 - \frac{|k|}{n}\Big) \Big[\gamma(k)\gamma(k+q-p) + \gamma(k+q)\gamma(k-p) \\ &\qquad + (\eta - 3)\sigma^4 \sum_i a_i a_{i+p} a_{i+k} a_{i+k+q}\Big]. \end{aligned}$$

All terms are absolutely summable in k (as $\sum|\gamma(k)| \leq \sigma^2(\sum|a_j|)^2$), so by dominated convergence the limit is

$$V_{pq} = (\eta - 3)\gamma(p)\gamma(q) + \sum_{k=-\infty}^{\infty} [\gamma(k)\gamma(k+q-p) + \gamma(k+q)\gamma(k-p)],$$

using $\sum_k \sum_i a_i a_{i+p} a_{i+k} a_{i+k+q} = \sigma^{-4}\gamma(p)\gamma(q)$.

Step 3. (Joint CLT for γ^*.) For the truncated process $X_t^{(m)} = \sum_{j=0}^{m} a_j \epsilon_{t-j}$, the vectors $(X_t^{(m)} X_{t+l}^{(m)})_{l=0}^{h}$ form a strictly stationary $(m+h)$-dependent sequence, so the CLT for m-dependent sequences (Hoeffding–Robbins) together with the Cramér–Wold device gives $\sqrt{n}(\boldsymbol{\gamma}_m^* - \boldsymbol{\gamma}_m) \xrightarrow{d} N(0, V^{(m)})$, where $V^{(m)} \to V$ as $m \to \infty$ by Step 2 applied to $a^{(m)}$. Applying the computation of Step 2 to the difference $X_t - X_t^{(m)}$ gives $\limsup_n n \operatorname{Var}(\gamma^*(l) - \gamma_m^*(l)) \le C \sum_{j>m} |a_j| \to 0$ as $m \to \infty$. By the standard approximation lemma (Billingsley, Thm. 3.2), $\sqrt{n}(\boldsymbol{\gamma}^* - \boldsymbol{\gamma}) \xrightarrow{d} N(0, V)$, with $\boldsymbol{\gamma}^* = (\gamma^*(0), \dots, \gamma^*(h))^T$.

Step 4. (From γ^* to $\hat{\gamma}$.) Expanding the product in Definition 3.2.1,

$$\hat{\gamma}(l) - \gamma^*(l) = -\frac{1}{n} \sum_{t=n-l+1}^{n} X_t X_{t+l} - \bar{X} \cdot \frac{1}{n} \sum_{t=1}^{n-l} (X_t + X_{t+l}) + \Big(1 - \frac{l}{n}\Big) \bar{X}^2.$$

The first term has L^1 norm $\le l\gamma(0)/n$. The other two are bounded by $C\bar{X}^2 + O_p(n^{-1})|\bar{X}|$, and $\bar{X} = O_p(n^{-1/2})$ by Remark 3.1.2. Hence $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}^*) \xrightarrow{p} 0$ and $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) \xrightarrow{d} N(0, V)$.

Step 5. (Delta method.) $\hat{\rho}(i) = g_i(\hat{\boldsymbol{\gamma}})$ with $g_i(x) = x_i/x_0$, whose gradient at $\boldsymbol{\gamma}$ has entries $-\rho(i)/\gamma(0)$ (in x_0) and $1/\gamma(0)$ (in x_i). Hence $\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0, W)$ with

$$w_{ij} = \frac{V_{ij} - \rho(i)V_{0j} - \rho(j)V_{i0} + \rho(i)\rho(j)V_{00}}{\gamma(0)^2}.$$

The $(\eta - 3)$ terms contribute $(\eta - 3)[\rho(i)\rho(j) - \rho(i)\rho(j) - \rho(j)\rho(i) + \rho(i)\rho(j)] = 0$, so W does not depend on η. With $T_{pq} = V_{pq}/\gamma(0)^2$ (without the η term) and the shifts $k \mapsto k + i$, one gets $T_{ij} = \sum_k [\rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j)]$, $T_{0j} = 2\sum_k \rho(k)\rho(k+j)$, $T_{i0} = 2\sum_k \rho(k)\rho(k+i)$ and $T_{00} = 2\sum_k \rho^2(k)$, which yields the first form of w_{ij}.

Step 6. (Second form.) Expand the product in the second form, sum over $k \ge 1$, and use $\rho(-k) = \rho(k)$ to rewrite each one-sided sum as half of a two-sided sum (the $k = 0$ terms cancel); this reproduces the first form. ∎

Corollary 3.2.A (White-noise and MA(q) bands; supplement)

(1) If $\{X_t\} \sim IID(0, \sigma^2)$ with $EX_t^4 < \infty$, then $\sqrt{n}\,\hat{\boldsymbol{\rho}}(h) \xrightarrow{d} N(0, I_h)$.
(2) If $\{X_t\}$ is MA(q), then for $i > q$, $\sqrt{n}\,\hat{\rho}(i) \xrightarrow{d} N\big(0,\, 1 + 2\sum_{k=1}^{q} \rho^2(k)\big)$.

Proof. [Supplement]

Step 1. For i.i.d. data $\rho(k) = 0$ for $k \ne 0$. In the second form of Bartlett's formula, the factor $\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k)$ with $k \ge 1$, $i \ge 1$ is nonzero only for $k = i$, where it equals 1. Thus $w_{ij} = \mathbb{1}\{i = j\}$.

Step 2. If $i > q$ then $\rho(i) = 0$ and $\rho(k+i) = 0$ for $k \ge 1$, so $w_{ii} = \sum_{k\ge1} \rho^2(k-i) = \sum_{m=1-i}^{\infty} \rho^2(m) = \rho^2(0) + 2\sum_{k=1}^{q} \rho^2(k)$, since $\rho(m) = 0$ for $|m| > q$. ∎

Hence, for white noise, about 95% of the $\hat{\rho}(h)$ lie within $\pm 1.96/\sqrt{n}$; sample autocorrelations outside these bands indicate genuine dependence, and for an MA(q) the bands at lags $> q$ widen by the factor $(1 + 2\sum_{k\le q} \rho^2(k))^{1/2}$.

3.3 The method of moments

Motivation. The simplest estimator equates population and sample autocorrelations. For AR models this leads to a linear system (the Yule–Walker equations); it also supplies starting values for maximum likelihood.

For MA and ARMA models the moment equations are nonlinear and awkward to solve, so we treat AR models only. Consider the stationary (causal) AR(p) model

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2).$$

By causality X_{t-j} is uncorrelated with ϵ_t for $j \geq 1$, so

$$E\big[(X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p}) X_{t-j}\big] = E\epsilon_t X_{t-j} = 0, \qquad j = 1, \ldots, p,$$

which gives the *Yule–Walker equations* $\gamma(j) = \phi_1\gamma(j-1) + \cdots + \phi_p\gamma(j-p)$, $j = 1, \ldots, p$; in matrix form

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(p-1) \\ \rho(1) & \rho(0) & \cdots & \rho(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p) \end{pmatrix}. \tag{3.2}$$

Replacing every $\rho(j)$ by $\hat\rho(j)$ and solving gives the moment estimator $\hat{\boldsymbol\phi}_{YW}$ of $\boldsymbol\phi = (\phi_1, \ldots, \phi_p)^T$, the *Yule–Walker (Y-W) estimator*. Comparing with Definition 3.2.3, the last component of the order-h Y-W estimator is exactly $\hat\phi_{hh}$.

Theorem 3.3.1 (Asymptotic normality of the Y-W estimator)

Let $\hat{\boldsymbol\phi}_{YW} = (\hat\phi_1, \ldots, \hat\phi_p)^T$ be the Y-W estimator of $\boldsymbol\phi$ in a causal AR(p) model. Then

$$\sqrt{n}(\hat{\boldsymbol\phi}_{YW} - \boldsymbol\phi) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2),$$

where Γ is the $p \times p$ matrix with $\Gamma_{ij} = \gamma(i-j)$.

The proof uses the following lemma, which is also the key to Section 3.4.

Lemma 3.3.A (Basic limits for causal AR(p); supplement)

Let $\{X_t\}$ be a causal AR(p) with $\{\epsilon_t\} \sim IID(0, \sigma^2)$, and $\mathbf{X}_{t-1} = (X_{t-1}, \ldots, X_{t-p})^T$. Then
(a) $\frac{1}{n}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s., and $\hat\gamma(h) \to \gamma(h)$ a.s. for each h;
(b) $\frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2\Gamma)$;
(c) Γ is nonsingular.

Proof. [Supplement]

Step 1. (a) $X_t = \sum_j a_j \epsilon_{t-j}$ is a fixed measurable function of the i.i.d. sequence shifted to time t, so $\{X_t\}$ (and any finite-lag function such as $X_{t-i}X_{t-j}$) is strictly stationary and ergodic. Birkhoff's ergodic theorem gives $n^{-1}\sum X_{t-i}X_{t-j} \to EX_{t-i}X_{t-j} = \gamma(i-j)$ and $\bar X \to 0$ a.s.; deleting finitely many terms does not affect the limits. The same applies to $\hat\gamma(h)$.

Step 2. (b) Let $\mathcal{F}_t = \sigma(\epsilon_s : s \leq t)$ and fix $c \in \mathbb{R}^p$. Then $\xi_t = c^T\mathbf{X}_{t-1}\epsilon_t$ is stationary, ergodic, $\mathcal{F}_t$-measurable, and $E(\xi_t \mid \mathcal{F}_{t-1}) = c^T\mathbf{X}_{t-1}E\epsilon_t = 0$, since $\mathbf{X}_{t-1} \in \mathcal{F}_{t-1}$ and ϵ_t is independent of $\mathcal{F}_{t-1}$. Also $E\xi_t^2 = E(c^T\mathbf{X}_{t-1})^2 E\epsilon_t^2 = \sigma^2 c^T\Gamma c < \infty$. By the CLT for stationary ergodic martingale differences (Billingsley–Ibragimov), $n^{-1/2}\sum \xi_t \xrightarrow{d} N(0, \sigma^2 c^T\Gamma c)$. The Cramér–Wold device gives (b).

Step 3. (c) This is Step 1 of the proof of Proposition 2.5.B: $c^T\Gamma c \geq c_i^2\sigma^2 > 0$ for $c \neq 0$. ∎

Proof of Theorem 3.3.1. [Supplement] Let $\hat\Gamma = [\hat\gamma(i-j)]_{i,j=1}^p$ and $\hat{\boldsymbol\gamma}_p = (\hat\gamma(1), \ldots, \hat\gamma(p))^T$, so $\hat\Gamma\hat{\boldsymbol\phi}_{YW} = \hat{\boldsymbol\gamma}_p$ (multiply (3.2) with hats by $\hat\gamma(0)$).

Step 1. Then $\hat\Gamma(\hat{\boldsymbol\phi}_{YW} - \boldsymbol\phi) = \hat{\boldsymbol\gamma}_p - \hat\Gamma\boldsymbol\phi$, whose j-th entry is $\hat\gamma(j) - \sum_k \phi_k\hat\gamma(j-k)$.

Step 2. Replace $\hat{\gamma}(l)$ by $n^{-1}\sum_t X_t X_{t+l}$ (sums over the available indices). The error is a sum of $O(p)$ boundary terms of size $O_p(1/n)$ plus terms of the form $\bar{X}\cdot n^{-1}\sum X_t$, i.e. $O_p(1/n)$ by Remark 3.1.2; after multiplying by $\sqrt{n}$ these vanish in probability. Hence

$$\sqrt{n}\Big(\hat{\gamma}(j)-\sum_k \phi_k\hat{\gamma}(j-k)\Big)=\frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\Big(X_t-\sum_k \phi_k X_{t-k}\Big)+o_p(1)=\frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\epsilon_t+o_p(1).$$

Step 3. By the Lemma, $\sqrt{n}(\hat{\boldsymbol{\gamma}}_p-\hat{\Gamma}\boldsymbol{\phi})\xrightarrow{d}N(0,\sigma^2\Gamma)$ and $\hat{\Gamma}\to\Gamma$ (nonsingular). By Slutsky,

$$\sqrt{n}(\hat{\boldsymbol{\phi}}_{YW}-\boldsymbol{\phi})=\hat{\Gamma}^{-1}\sqrt{n}(\hat{\boldsymbol{\gamma}}_p-\hat{\Gamma}\boldsymbol{\phi})\xrightarrow{d}N(0,\Gamma^{-1}\sigma^2\Gamma\Gamma^{-1})=N(0,\Gamma^{-1}\sigma^2).$$

■

Example 3.3.3 (Y-W estimators for AR(1) and AR(2))

For AR(1), $\rho(1)=\phi$ gives $\hat{\phi}=\hat{\rho}(1)$. For AR(2) the Y-W equations are

$$\rho(1)=\phi_1+\rho(1)\phi_2,\qquad \rho(2)=\rho(1)\phi_1+\phi_2,$$

so

$$\hat{\phi}_1=\frac{\hat{\rho}(1)(1-\hat{\rho}(2))}{1-\hat{\rho}^2(1)},\qquad \hat{\phi}_2=\frac{\hat{\rho}(2)-\hat{\rho}^2(1)}{1-\hat{\rho}^2(1)}.$$

Corollary 3.3.B (Sample PACF of an AR(p); supplement)

For a causal AR(p) and each $h>p$, $\sqrt{n}\,\hat{\phi}_{hh}\xrightarrow{d}N(0,1)$.

Proof. [Supplement]

Step 1. An AR(p) is also an AR(h) with $\boldsymbol{\phi}^{(h)}=(\phi_1,\dots,\phi_p,0,\dots,0)^T$, and $\hat{\phi}_{hh}$ is the last component of the order-h Y-W estimator. By Theorem 3.3.1, $\sqrt{n}\,\hat{\phi}_{hh}\xrightarrow{d}N(0,\sigma^2(\Gamma_h^{-1})_{hh})$.

Step 2. For a positive definite covariance matrix, $1/(\Gamma_h^{-1})_{hh}$ is the residual variance of the last coordinate regressed linearly on the others: here $\mathrm{Var}(X_{t-h}-P(X_{t-h}\mid X_{t-h+1},\dots,X_{t-1}))$. By time reversal (the covariance matrix is unchanged when the order is reversed) this equals the one-step prediction error of X_t from $X_{t-1},\dots,X_{t-h+1}$, which is $\mathrm{Var}(\epsilon_t)=\sigma^2$ because $h-1\geq p$ (Proposition 2.5.B, Step 2). Hence $\sigma^2(\Gamma_h^{-1})_{hh}=1$. ■

Thus $\pm 1.96/\sqrt{n}$ bands on the SPACF beyond lag p serve to choose the AR order.

3.4 Least squares estimators

Motivation. Least squares, the standard tool of regression, applies directly to AR models by regressing X_t on its lags. Despite the dependence of the regressors, it has the same limit distribution as in Theorem 3.3.1.

For the stationary AR(1) model $X_t=\phi X_{t-1}+\epsilon_t$, the *least squares estimator* minimizes $S(\phi)=\sum_{t=2}^n(X_t-\phi X_{t-1})^2$:

$$\hat{\phi}=\frac{\sum_{t=2}^n X_t X_{t-1}}{\sum_{t=2}^n X_{t-1}^2},\qquad \sqrt{n}(\hat{\phi}-\phi)=\frac{n^{-1/2}\sum_{t=2}^n X_{t-1}\epsilon_t}{n^{-1}\sum_{t=2}^n X_{t-1}^2}.$$

In the numerator $EX_{t-1}\epsilon_t = 0$ and $\mathrm{Var}(\sum_{t=2}^n X_{t-1}\epsilon_t) = (n-1)\sigma^2 EX_1^2$ (the cross terms vanish because ϵ_t is independent of $X_{s-1}\epsilon_s X_{t-1}$ for $s < t$). In fact

$$\frac{1}{\sqrt{n}}\sum_{t=2}^n X_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2 EX_1^2) \tag{3.3}$$

(Lemma 3.3.A(b)), while the denominator tends to $\gamma(0) = EX_1^2$.

Theorem 3.4.1 (LS estimator in AR(1))

If $\hat{\phi}$ is the least squares estimator of ϕ in a stationary AR(1) model, then

$$\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N(0, (EX_1^2)^{-1}\sigma^2) = N(0, 1-\phi^2).$$

Proof. [Supplement] By (3.3) the numerator converges in distribution to $N(0, \sigma^2 EX_1^2)$, and by Lemma 3.3.A(a) the denominator converges a.s. to $EX_1^2 > 0$. Slutsky's theorem gives the limit $N(0, \sigma^2 EX_1^2/(EX_1^2)^2)$, and $EX_1^2 = \sigma^2/(1-\phi^2)$ by Example 2.2.3. (This is also the case $p = 1$ of Theorem 3.4.2.) ■

For the stationary AR(p) model $X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t$, write $\mathbf{X}_{t-1} = (X_{t-1}, \ldots, X_{t-p})^T$. The least squares estimator is

$$\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \ldots, \hat{\phi}_p)^T = \Big(\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T\Big)^{-1} \sum_{t=p+1}^n \mathbf{X}_{t-1}X_t,$$

so that, using $X_t = \mathbf{X}_{t-1}^T\boldsymbol{\phi} + \epsilon_t$,

$$\hat{\boldsymbol{\phi}} - \boldsymbol{\phi} = \Big(\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T\Big)^{-1} \sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t.$$

Here $E\mathbf{X}_{t-1}\epsilon_t = 0$ and $\mathrm{Cov}(\sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t) = (n-p)\Gamma\sigma^2$ with $\Gamma_{ij} = \gamma(i-j)$.

Theorem 3.4.2 (LS estimator in AR(p))

If $\hat{\boldsymbol{\phi}}$ is the least squares estimator of $\boldsymbol{\phi}$ in a stationary (causal) AR(p) model, then

$$\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2).$$

Proof. [Supplement]

Step 1. By Lemma 3.3.A(b), $n^{-1/2}\sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0, \Gamma\sigma^2)$.

Step 2. By Lemma 3.3.A(a),(c), $n^{-1}\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s. with Γ nonsingular, so the inverse exists for large n and converges to Γ^{-1}.

Step 3. Slutsky: $\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} \Gamma^{-1}N(0, \Gamma\sigma^2) = N(0, \Gamma^{-1}\sigma^2)$. ■

By Theorems 3.3.1 and 3.4.2, the Y-W and LS estimators have the same asymptotic distribution. Moreover, with $\hat{\Gamma}_{ij} = \hat{\gamma}(i-j)$ and a consistent $\hat{\sigma}^2$,

$$\frac{n(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{\hat{\sigma}^2} \xrightarrow{d} \chi^2(p), \tag{3.4}$$

because $Z_n = \sqrt{n}\,\Gamma^{1/2}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})/\sigma \xrightarrow{d} N(0, I_p)$, the statistic equals $Z_n^T Z_n + o_p(1)$ (Slutsky, as $\hat{\Gamma} \to \Gamma$, $\hat{\sigma}^2 \to \sigma^2$), and $|Z|^2 \sim \chi^2(p)$ for $Z \sim N(0, I_p)$ (continuous mapping). Hence a $100(1-\alpha)\%$ confidence region for $\boldsymbol{\phi}$ is

$$S_\alpha = \Big\{\boldsymbol{\phi} \in \mathbb{R}^p : \frac{n(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})}{\hat{\sigma}^2} \le \chi^2(p, 1-\alpha)\Big\},$$

where $P(\chi^2(p) \ge \chi^2(p, 1-\alpha)) = \alpha$.

3.5 Maximum likelihood estimators

Motivation. Maximum likelihood uses the full Gaussian joint density. For AR models the likelihood factorizes into one-step conditional densities; dropping the first factor gives the conditional MLE, which coincides with least squares.

To write a likelihood we assume $\{X_t\}$ is Gaussian, i.e. $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$; this is also the usual working assumption when the data are not normal. The likelihood equations are rarely solvable by hand, but the large-sample behaviour of the estimators is well known.

3.5.1 Exact likelihood of AR(1)

Let $X_t = \phi X_{t-1} + \epsilon_t$, $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$, $|\phi| < 1$. Then $EX_1 = 0$, $\mathrm{Var}(X_1) = \sigma^2/(1-\phi^2)$, so

$$f_{X_1}(x_1; \phi, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2/(1-\phi^2)}} \exp\Big\{\frac{-x_1^2}{2\sigma^2/(1-\phi^2)}\Big\}.$$

Given $X_1 = x_1$, since ϵ_2 is independent of X_1,

$$P(X_2 \le x \mid X_1 = x_1) = P(\phi X_1 + \epsilon_2 \le x \mid X_1 = x_1) = P(\epsilon_2 \le x - \phi x_1),$$

so $X_2 \mid X_1 = x_1 \sim N(\phi x_1, \sigma^2)$. Likewise, by the Markov structure, $f_{X_t|X_{t-1},\dots,X_1}(x_t \mid x_{t-1}, \dots, x_1) = f_{X_t|X_{t-1}}(x_t \mid x_{t-1})$ is the $N(\phi x_{t-1}, \sigma^2)$ density. Hence

$$f_{X_1,\dots,X_n}(x_1, \dots, x_n; \phi, \sigma^2) = f_{X_1}(x_1; \phi, \sigma^2) \prod_{t=2}^{n} f_{X_t|X_{t-1}}(x_t \mid x_{t-1}; \phi, \sigma^2), \tag{3.5}$$

and the log-likelihood is

$$\begin{aligned} L(\phi, \sigma^2) = &-\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\frac{\sigma^2}{1-\phi^2} - \frac{x_1^2}{2\sigma^2/(1-\phi^2)} \\ &- \frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t - \phi x_{t-1})^2. \end{aligned} \tag{3.6}$$

The MLE maximizes (3.6); the score equations are nonlinear in ϕ, so numerical optimization is used. The same factorization works for AR(p) (the first factor is now the $N(0, \sigma^2 V_p)$ density of $(X_1, \dots, X_p)$, where $\sigma^2 V_p = \Gamma_p$); for ARMA(p, q) the MLE is likewise computed numerically.

3.5.2 Conditional likelihood

The *conditional maximum likelihood estimator* maximizes the density given the first observation(s), i.e. (3.5) without the factor f_{X_1}:

$$L_c(\phi, \sigma^2) = -\frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t - \phi x_{t-1})^2.$$

Its maximizers are

$$\hat\phi = \frac{\sum_{t=2}^{n} X_t X_{t-1}}{\sum_{t=2}^{n} X_{t-1}^2}, \qquad \hat\sigma^2 = \frac{1}{n-1}\sum_{t=2}^{n}(X_t - \hat\phi X_{t-1})^2,$$

which are easy to compute.

Proposition 3.5.A (Conditional MLE = LS for AR(p); supplement)

For a Gaussian causal AR(p), the conditional MLE of $\boldsymbol{\phi}$ given $X_1, \dots, X_p$ equals the LS estimator $\hat{\boldsymbol{\phi}}$, the conditional MLE of σ^2 is $\hat{\sigma}^2 = (n-p)^{-1} \sum_{t=p+1}^{n} (X_t - \mathbf{X}_{t-1}^T \hat{\boldsymbol{\phi}})^2 \to \sigma^2$ a.s., and $\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2)$. The exact MLE has the same limit distribution.

Proof. [Supplement]

Step 1. The conditional log-likelihood is $L_c = -\frac{n-p}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} S(\boldsymbol{\phi})$ with $S(\boldsymbol{\phi}) = \sum_{t=p+1}^{n} (X_t - \mathbf{X}_{t-1}^T \boldsymbol{\phi})^2$. For every σ^2 it is maximized in $\boldsymbol{\phi}$ by minimizing S, i.e. by the LS estimator. Then $\partial L_c / \partial \sigma^2 = -\frac{n-p}{2\sigma^2} + \frac{S(\hat{\boldsymbol{\phi}})}{2\sigma^4} = 0$ gives $\hat{\sigma}^2 = S(\hat{\boldsymbol{\phi}})/(n-p)$.

Step 2. Consistency: $S(\hat{\boldsymbol{\phi}})/(n-p) = n^{-1}\sum \epsilon_t^2 - (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T (n^{-1} \sum \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T)(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) + o(1)$ (expand $X_t - \mathbf{X}_{t-1}^T \hat{\boldsymbol{\phi}} = \epsilon_t - \mathbf{X}_{t-1}^T(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})$ and use the normal equations). The first term $\to \sigma^2$ by the strong law, the second $\to 0$ by Lemma 3.3.A and Theorem 3.4.2. The limit law of $\hat{\boldsymbol{\phi}}$ is Theorem 3.4.2.

Step 3. (Exact MLE, sketch at this level.) The exact log-likelihood is L_c plus the log-density of $(X_1, \dots, X_p)$, which does not grow with n; its derivative is $O_p(1)$, while the Hessian of L_c is of order n ($-n\Gamma/\sigma^2$ asymptotically). A one-step expansion of the score equation then shows that the exact and conditional maximizers differ by $O_p(1/n)$, so $\sqrt{n}$ times their difference tends to 0 and the limits coincide. ■

3.6 Prediction

Motivation. Forecasting is the ultimate goal of time series analysis. The mean-square optimal predictor is a conditional expectation; for ARMA models it can be computed recursively from the model equation.

Given $X_1, \dots, X_n$, the *predictor* of X_{n+h} is

$$f(X_1, \dots, X_n) := E(X_{n+h} \mid X_1, \dots, X_n).$$

Proposition 3.6.A (Optimality of the conditional mean; supplement)

If $EX_{n+h}^2 < \infty$, then $E\big(X_{n+h} - g(X_1, \dots, X_n)\big)^2$ over all measurable g with $Eg^2 < \infty$ is minimized by $g = f$.

Proof. [Supplement] Write $Y = X_{n+h}$, $\mathbf{X} = (X_1, \dots, X_n)$. Then

$$E(Y - g(\mathbf{X}))^2 = E(Y - f(\mathbf{X}))^2 + E(f(\mathbf{X}) - g(\mathbf{X}))^2 + 2E\big[(Y - f(\mathbf{X}))(f(\mathbf{X}) - g(\mathbf{X}))\big].$$

Conditioning on $\mathbf{X}$, the cross term is $2E\big[(f(\mathbf{X}) - g(\mathbf{X}))\, E(Y - f(\mathbf{X}) \mid \mathbf{X})\big] = 0$. Hence $E(Y-g)^2 \geq E(Y-f)^2$, with equality iff $g(\mathbf{X}) = f(\mathbf{X})$ a.s. ■

If $\{X_t\}$ is Gaussian, f is a linear combination $\sum_{i=1}^{n} a_i X_i$ (Section 1.2), and the a_i minimize $E(X_{n+h} - \sum_i a_i X_i)^2$. In general f is not linear, so the best linear predictor is not optimal in the strict sense; but since f is hard to find, linear predictors are used in practice. For stationary series efficient algorithms for the coefficients exist (e.g. the Durbin–Levinson algorithm), and for ARMA models the predictor is a function of the model parameters.

For forecasting from the infinite past write $\hat{X}_t(h) = E(X_{t+h} \mid X_t, X_{t-1}, \dots)$.

Example 3.6.1 (Forecasting an AR(1) with mean)

Let $(1-\phi B)(X_t-\mu)=\epsilon_t$, $\{\epsilon_t\}\sim IID(0,\sigma^2)$, $|\phi|<1$, so $X_t=\mu+\phi(X_{t-1}-\mu)+\epsilon_t$. Since ϵ_{t+1} is independent of $X_t, X_{t-1},\dots,$

$$\hat{X}_t(1)=E(\mu+\phi(X_t-\mu)+\epsilon_{t+1}\mid X_t,X_{t-1},\dots)=\mu+\phi(X_t-\mu).$$

Similarly $\hat{X}_t(h)=\mu+\phi(\hat{X}_t(h-1)-\mu)$, hence

$$\hat{X}_t(h)=\mu+\phi^h(X_t-\mu)\longrightarrow\mu\quad(h\to\infty).$$

In practice μ,ϕ are replaced by $\hat{\mu},\hat{\phi}$.

For a stationary invertible ARMA(1,1), $(1-\phi B)(X_t-\mu)=(1+\theta B)\epsilon_t$ with $|\phi|<1$, $|\theta|<1$, we have $X_{t+1}=\mu(1-\phi)+\phi X_t+\epsilon_{t+1}+\theta\epsilon_t$. By invertibility ϵ_t is a function of $X_t, X_{t-1},\dots,$ so $E(\epsilon_t\mid X_t,X_{t-1},\dots)=\epsilon_t$ and

$$\hat{X}_t(1)=\mu(1-\phi)+\phi X_t+\theta\epsilon_t,\qquad \hat{X}_t(h)=\mu(1-\phi)+\phi\hat{X}_t(h-1),\quad h\geq 2.$$

In practice ϵ_t is unknown and is replaced by the residual $\hat{\epsilon}_t$, and μ,ϕ,θ by their estimates; the forecasts are then computed recursively from $\hat{X}_t(1)$.

3.7 Selected exercises

Motivation. These textbook exercises use the sample statistics of this chapter: a dependence-adjusted interval for the mean, hand computation of the sample ACVF, ACF and PACF, and the proof that sample autocorrelations are bounded by one.

Exercise 3.1 (Inference on the mean of an AR(1))

In an AR(1) model $X_t-\mu=\phi(X_{t-1}-\mu)+\epsilon_t$ with $\phi=0.3$ and $\sigma^2=1$, a sample of size $n=100$ gives $\bar{X}=0.1$.
(a) Find an approximate 95% confidence interval for μ.
(b) Test $H_0:\mu=0$ against $H_1:\mu\neq 0$ at level 5%.

Solution.

(a) By Theorem 3.1.1, $\sqrt{n}(\bar{X}-\mu)(1-\phi)/\sigma\xrightarrow{d}N(0,1)$, so $P(|\bar{X}-\mu|\leq z_{0.025}\sigma/(\sqrt{n}(1-\phi)))\to 0.95$. Here

$$\frac{z_{0.025}\,\sigma}{\sqrt{n}(1-\phi)}=\frac{1.96\times 1}{10\times 0.7}=0.28,$$

so the interval is $0.1\pm 0.28=[-0.18,\,0.38]$.

(b) Reject when $|\sqrt{n}\bar{X}|\geq z_{0.025}\sigma/(1-\phi)=1.96/0.7=2.8$. Since $|\sqrt{n}\bar{X}|=10\times 0.1=1<2.8$, H_0 is not rejected; equivalently, 0 lies in the interval of (a).

For comparison, treating the data as i.i.d. with variance $\gamma(0)=\sigma^2/(1-\phi^2)=1/0.91$ would give the half-width $1.96\times(0.91)^{-1/2}/10\approx 0.205$, too narrow because positive autocorrelation inflates $\mathrm{Var}(\bar{X})$. ■

Exercise 3.2 (Sample ACVF, ACF and PACF by hand)

For the data

$$(x_1,\dots,x_{10})=(16,\ 9,\ 10,\ 3,\ 5,\ 12,\ 14,\ 7,\ 9,\ 11),$$

compute $\hat{\gamma}(h)$ and $\hat{\rho}(h)$ for $h = 0, 1, 2, 3$, and $\hat{\phi}_{hh}$ for $h = 1, 2, 3$.

Solution.

Step 1. $\bar{x} = 96/10 = 9.6$, and the deviations $d_t = x_t - \bar{x}$ are

$$6.4,\ -0.6,\ 0.4,\ -6.6,\ -4.6,\ 2.4,\ 4.4,\ -2.6,\ -0.6,\ 1.4.$$

Step 2. By Definition 3.2.1, $\hat{\gamma}(h) = \frac{1}{10}\sum_{t=1}^{10-h} d_t d_{t+h}$. The lagged cross-product sums are 140.40, 12.44, −43.92, −50.88 for $h = 0, 1, 2, 3$, so

$$\hat{\gamma}(0) = 14.04, \quad \hat{\gamma}(1) = 1.244, \quad \hat{\gamma}(2) = -4.392, \quad \hat{\gamma}(3) = -5.088,$$

and, dividing by $\hat{\gamma}(0)$,

$$\hat{\rho}(0) = 1, \quad \hat{\rho}(1) = 0.0886, \quad \hat{\rho}(2) = -0.3128, \quad \hat{\rho}(3) = -0.3624.$$

Step 3. By Definition 3.2.3, $\hat{\phi}_{11} = \hat{\rho}(1) = 0.0886$, and solving the 2×2 system as in Example 2.5.2,

$$\hat{\phi}_{22} = \frac{\hat{\rho}(2) - \hat{\rho}(1)^2}{1 - \hat{\rho}(1)^2} = \frac{-0.3128 - 0.0079}{1 - 0.0079} = -0.3232.$$

Step 4. For $h = 3$, Cramer's rule gives $\hat{\phi}_{33} = \det \hat{R}_3^* / \det \hat{R}_3$ with $\hat{R}_3 = [\hat{\rho}(i-j)]_{i,j=1}^3$ and $\hat{R}_3^*$ its last column replaced by $(\hat{\rho}(1), \hat{\rho}(2), \hat{\rho}(3))^T$. Numerically $\det \hat{R}_3 = 0.8815$ and $\det \hat{R}_3^* = -0.2947$, so $\hat{\phi}_{33} = -0.3344$.

With $n = 10$ the $\pm 1.96/\sqrt{10} = \pm 0.62$ bands of Corollary 3.2.A contain all these values, so the data show no significant autocorrelation. ■

Exercise 3.4 (Sample autocorrelations are bounded by one)

Let $x_1, \ldots, x_n$ be not all equal.
(a) Show that $|\hat{\rho}(h)| \le 1$ for all $|h| < n$.
(b) Show that the matrix $\hat{\Gamma}_k = [\hat{\gamma}(i-j)]_{i,j=1}^k$ is nonnegative definite for every $k \ge 1$.

Solution. Let $y_t = x_t - \bar{x}$ for $1 \le t \le n$ and $y_t = 0$ for all other integers t. Since $\hat{\gamma}(h) = \hat{\gamma}(-h)$ and the extra terms vanish,

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t \in \mathbb{Z}} y_t y_{t+h}, \qquad h \in \mathbb{Z}.$$

(a)

Step 1. $\hat{\gamma}(0) = n^{-1}\sum_t y_t^2 > 0$ because the x_t are not all equal, so $\hat{\rho}(h)$ is defined.

Step 2. By the Cauchy–Schwarz inequality for finite sums,

$$\Big|\sum_t y_t y_{t+h}\Big| \le \Big(\sum_t y_t^2\Big)^{1/2}\Big(\sum_t y_{t+h}^2\Big)^{1/2} = \sum_t y_t^2,$$

since shifting the index does not change $\sum_t y_t^2$. Dividing by $n\hat{\gamma}(0)$ gives $|\hat{\rho}(h)| \le 1$.

(b) For $a \in \mathbb{R}^k$, substituting $s = t - j$ in the inner sum (all sums are finite),

$$\sum_{i,j=1}^k a_i a_j \hat{\gamma}(i-j) = \frac{1}{n}\sum_{i,j} a_i a_j \sum_s y_{s+j} y_{s+i} = \frac{1}{n}\sum_{s \in \mathbb{Z}}\Big(\sum_{i=1}^k a_i y_{s+i}\Big)^2 \ge 0.$$

Part (b) also implies (a): the 2×2 principal submatrix with rows $1, 1 + |h|$ has nonnegative determinant $\hat{\gamma}(0)^2 - \hat{\gamma}(h)^2$. The zero-padding is exactly why the divisor n, not $n - |h|$, is used in Definition 3.2.1. ■

Chapter 4. Nonstationary Time Series Models and Model Selection

4.1 Autoregressive integrated moving average models

Motivation. Many observed series (prices, output, consumption) wander without returning to a fixed level, so the stationary ARMA theory of the previous chapters does not apply directly. The simplest remedy is to difference the series until it becomes stationary and then fit an ARMA model to the differences; this gives the ARIMA class.

Consider the AR(1) recursion with coefficient one,

$$X_t = X_{t-1} + \epsilon_t, \qquad \text{equivalently} \qquad X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t,$$

the *random walk*. With the backshift operator B it reads $(1-B)X_t = \epsilon_t$, i.e. $\nabla X_t = \epsilon_t$ with $\nabla = 1 - B$.

Example (the random walk is not stationary)

Let $X_0 = 0$ and $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$. Then $EX_t = 0$ but $\text{Var}(X_t) = t\sigma^2$ and $\text{Cov}(X_t, X_{t+h}) = t\sigma^2$ for $h \geq 0$, which depend on t; hence $\{X_t\}$ is not stationary. Its first difference $\nabla X_t = \epsilon_t$ is white noise. Equivalently, the AR polynomial $\phi(z) = 1 - z$ has its root on the unit circle, so the stationarity condition of Chapter 2 fails.

In general a nonstationary series may become stationary after $d \geq 1$ differences: $Z_t = (1-B)^d X_t$ is stationary. If moreover Z_t is an ARMA process, $\phi(B)Z_t = \theta(B)\epsilon_t$, then the model for $\{X_t\}$ is $\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t$.

Definition 4.1.1 (ARIMA(p,d,q) model)

A time series $\{X_t\}$ follows an *autoregressive integrated moving average model* of orders p, d, q, written $\{X_t\} \sim \text{ARIMA}(p,d,q)$, if

$$\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2),$$

where $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$. In practice $d \in \{0, 1, 2\}$.

Throughout, $\phi(z)$ and $\theta(z)$ are assumed to satisfy the causality and invertibility conditions (no zeros in $|z| \leq 1$), so that the d-th difference is a causal invertible ARMA process. The errors are taken IID rather than merely white noise because the inference below (limit distributions, likelihood) needs it.

Remark 4.1.2

The random walk is $\text{ARIMA}(0,1,0)$. If the differenced series has a nonzero mean μ, the model is written

$$\phi(B)\big((1-B)^d X_t - \mu\big) = \theta(B)\epsilon_t, \qquad \mu = E(1-B)^d X_t.$$

Polynomial trends. Not every nonstationary series becomes stationary by differencing, but series with polynomial trend do. For $X_t = t + \epsilon_t$ with $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$,

$$(1-B)X_t = t + \epsilon_t - (t-1) - \epsilon_{t-1} = 1 + \epsilon_t - \epsilon_{t-1},$$

so $(1-B)X_t - 1 = \theta(B)\epsilon_t$ with $\theta(B) = 1 - B$: $\{X_t\}$ is ARIMA$(0,1,1)$ with $\mu = 1$ (with a non-invertible MA part). More generally, for $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$ the series $(1-B)^r X_t$ is stationary: $(1-B)$ maps a polynomial in t of degree $k \geq 1$ to one of degree $k-1$ (the leading term at^k becomes $a\{t^k - (t-1)^k\} = ak\,t^{k-1} + \cdots$), so $(1-B)^r$ turns the trend into the constant $a_r r!$, while $(1-B)^r \epsilon_t$ is a finite moving average of IID noise. Thus polynomial-trend models can be rewritten as ARIMA models. The converse is false, and when a series is known to follow a polynomial trend it is preferable to fit that trend directly.

4.2 Unit root tests

Motivation. Whether to difference is itself a statistical question: does the AR polynomial have a root at $z = 1$? Under a unit root the least-squares estimator converges at rate n rather than $\sqrt{n}$ and its limit is a functional of Brownian motion, not a normal law, so special (Dickey–Fuller) critical values are needed.

4.2.1 The AR(1) case

Consider $X_t = \phi X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$, and the hypotheses

$$H_0 : \phi = 1 \qquad \text{vs} \qquad H_1 : |\phi| < 1.$$

Let $\hat{\phi}_{LS} = \sum_{t=1}^n X_{t-1}X_t / \sum_{t=1}^n X_{t-1}^2$ be the least-squares estimator and

$$T_n = n(\hat{\phi}_{LS} - 1). \tag{4.1}$$

Under H_0, $X_t - X_{t-1} = \epsilon_t$, hence $\hat{\phi}_{LS} - 1 = \sum X_{t-1}\epsilon_t / \sum X_{t-1}^2$ and

$$T_n = \frac{n^{-1}\sum_{t=1}^n X_{t-1}\epsilon_t}{n^{-2}\sum_{t=1}^n X_{t-1}^2}. \tag{4.2}$$

The scaling suggests looking at the partial-sum process

$$W_n(s) = \frac{1}{\sqrt{n}\,\sigma}\sum_{j=1}^{[ns]} \epsilon_j, \qquad 0 \leq s \leq 1, \tag{4.3}$$

where $[\cdot]$ is the integer part. Under H_0, $X_t = \sigma\sqrt{n}\,W_n(t/n)$.

Definition 4.2.1 (Brownian motion)

A stochastic process $\{W(s) : s \geq 0\}$ with continuous sample paths is a (standard) *Brownian motion* if
(1) $W(0) = 0$;
(2) for all $t > s \geq 0$, $W(t) - W(s) \sim N(0, t-s)$;
(3) for all $0 \leq t_1 < \cdots < t_k$, the increments $W(t_1), W(t_2) - W(t_1), \ldots, W(t_k) - W(t_{k-1})$ are independent.

The convergence below is weak convergence of random elements of the space $D[0,1]$ of right-continuous functions with left limits, equipped with the Skorokhod topology: $W_n \xrightarrow{d} W$ means $Ef(W_n) \to Ef(W)$ for every bounded continuous $f : D[0,1] \to \mathbb{R}$. By the continuous mapping theorem, then $h(W_n) \xrightarrow{d} h(W)$ for every measurable h that is continuous at almost every path of W.

Theorem 4.2.2 (Functional central limit theorem)

Let $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$ and define W_n by (4.3). Then, as $n \to \infty$, $W_n \xrightarrow{d} W$, a standard Brownian motion on $[0, 1]$.

Proof. [Supplement] Write $S_k = \epsilon_1 + \cdots + \epsilon_k$, so $W_n(s) = S_{[ns]}/(\sigma\sqrt{n})$. Weak convergence in $D[0, 1]$ follows from (a) convergence of finite-dimensional distributions and (b) tightness of $\{W_n\}$ (Prokhorov's theorem, used as a tool); we verify both.

Step 1 (finite-dimensional laws). Fix $0 = s_0 < s_1 < \cdots < s_k \le 1$ and put $m_j = [ns_j] - [ns_{j-1}]$, so $m_j/n \to s_j - s_{j-1} > 0$. The increment

$$W_n(s_j) - W_n(s_{j-1}) = \sqrt{\frac{m_j}{n}} \cdot \frac{1}{\sigma\sqrt{m_j}} \sum_{i=[ns_{j-1}]+1}^{[ns_j]} \epsilon_i$$

converges in distribution to $N(0, s_j - s_{j-1})$ by the classical CLT and Slutsky's lemma. The k increments are built from disjoint blocks of the ϵ_i, hence independent, so their joint characteristic function factorizes and the vector of increments converges to a vector of independent $N(0, s_j - s_{j-1})$ variables, which is the law of the increments of W. Applying the linear map (increments)$\mapsto$(partial sums) gives $(W_n(s_1), \ldots, W_n(s_k)) \xrightarrow{d} (W(s_1), \ldots, W(s_k))$.

Step 2 (a maximal inequality). For partial-sum processes of IID variables a sufficient condition for tightness (Billingsley, *Convergence of Probability Measures*, Thm. 8.4, used as a tool) is: for every $\varepsilon > 0$ there are $\lambda > 1$ and n_0 with

$$P\Big(\max_{k \le n} |S_k| \ge \lambda\sigma\sqrt{n}\Big) \le \frac{\varepsilon}{\lambda^2} \qquad (n \ge n_0).$$

By Etemadi's inequality, $P(\max_{k\le n} |S_k| \ge 3a) \le 3 \max_{k\le n} P(|S_k| \ge a)$. Take $a = \mu\sigma\sqrt{n}$ with $\mu = \lambda/3$. Let $N \sim N(0, 1)$. By the CLT there is k_μ such that $P(|S_k| \ge \mu\sigma\sqrt{k}) \le 2P(|N| \ge \mu)$ for $k \ge k_\mu$, and $2P(|N| \ge \mu) \le 2EN^4/\mu^4 = 6/\mu^4$. Hence

(i) for $k_\mu \le k \le n$: $P(|S_k| \ge \mu\sigma\sqrt{n}) \le P(|S_k| \ge \mu\sigma\sqrt{k}) \le 6/\mu^4$;

(ii) for $k < k_\mu$: by Chebyshev, $P(|S_k| \ge \mu\sigma\sqrt{n}) \le k/(\mu^2 n) \le k_\mu/(\mu^2 n) \le 1/\mu^4$ once $n \ge \mu^2 k_\mu$.

Therefore $P(\max_{k\le n} |S_k| \ge \lambda\sigma\sqrt{n}) \le 3 \cdot 7/\mu^4 = 1701/\lambda^4$ for large n, which is $\le \varepsilon/\lambda^2$ as soon as $\lambda^2 \ge 1701/\varepsilon$.

Step 3 (conclusion). By Steps 1–2 every subsequence of $\{W_n\}$ has a further subsequence converging weakly, and every limit has the finite-dimensional laws of Brownian motion; since these determine the law on $D[0, 1]$ (and Brownian motion has continuous paths), the whole sequence converges to W. ■

Proposition (Limit of the unit-root statistics)

Under $H_0 : \phi = 1$, with T_n as in (4.1),

$$T_n \xrightarrow{d} T := \frac{\frac{1}{2}(W(1)^2 - 1)}{\int_0^1 W(s)^2\, ds} = \frac{\int_0^1 W(s)\, dW(s)}{\int_0^1 W(s)^2\, ds}.$$

Under $H_1 : |\phi| < 1$, $P(|T_n| \ge c) \to 1$ for every constant c; that is, the test rejecting for large $|T_n|$ is consistent.

Proof. [Supplement]

Step 1 (numerator). Squaring $X_t = X_{t-1} + \epsilon_t$ gives $X_t^2 - X_{t-1}^2 = 2X_{t-1}\epsilon_t + \epsilon_t^2$. Summing over

$t = 1, \dots, n$ with $X_0 = 0$,

$$\frac{1}{n}\sum_{t=1}^{n} X_{t-1}\epsilon_t = \frac{1}{2}\Big(\frac{X_n^2}{n} - \frac{1}{n}\sum_{t=1}^{n}\epsilon_t^2\Big) = \frac{\sigma^2}{2}\Big(W_n(1)^2 - \frac{1}{n\sigma^2}\sum_{t=1}^{n}\epsilon_t^2\Big).$$

Step 2 (denominator). For $s \in [(t-1)/n, t/n)$ we have $[ns] = t-1$, so $W_n(s) = X_{t-1}/(\sigma\sqrt{n})$. Hence exactly

$$\frac{1}{n^2}\sum_{t=1}^{n} X_{t-1}^2 = \sigma^2 \sum_{t=1}^{n}\int_{(t-1)/n}^{t/n} W_n(s)^2\, ds = \sigma^2 \int_0^1 W_n(s)^2\, ds.$$

Step 3 (joint limit). The map $w \mapsto (w(1), \int_0^1 w(s)^2 ds)$ from $D[0,1]$ to $\mathbb{R}^2$ is continuous at every continuous w (Skorokhod convergence to a continuous limit is uniform convergence). By Theorem 4.2.2 and the continuous mapping theorem, $(W_n(1), \int_0^1 W_n^2) \xrightarrow{d} (W(1), \int_0^1 W^2)$; by the law of large numbers $n^{-1}\sum \epsilon_t^2 \xrightarrow{p} \sigma^2$, and Slutsky's lemma gives joint convergence of all three quantities.

Step 4 (ratio). Since $\int_0^1 W(s)^2 ds > 0$ almost surely, the map $(a,b,c) \mapsto \frac{1}{2}(a^2 - c)/b$ is continuous at the limit, and Steps 1–3 yield $T_n \xrightarrow{d} \frac{1}{2}(W(1)^2 - 1)/\int_0^1 W^2$. The second form follows from Itô's formula $W(1)^2 = 2\int_0^1 W\, dW + 1$.

Step 5 (consistency). If $|\phi| < 1$, write $T_n = n(\hat\phi_{LS} - \phi) + n(\phi - 1)$. By the asymptotic normality of the least-squares estimator for a causal AR(1) (Chapter 3), $\sqrt{n}(\hat\phi_{LS} - \phi) \xrightarrow{d} N(0, 1-\phi^2)$, so $n(\hat\phi_{LS} - \phi) = O_p(\sqrt{n})$, whereas $n(\phi - 1) \to -\infty$ linearly. Thus $T_n/n \xrightarrow{p} \phi - 1 < 0$ and $P(|T_n| \geq c) \to 1$. ■

Notice the contrast with the stationary case: the estimator converges at rate n (super-consistency) and the limit T is not normal (it is skewed to the left). The test rejects H_0 at level α when $|T_n| > c_\alpha$, where $P(|T| > c_\alpha) = \alpha$; the quantiles of T are tabulated by simulation.

Dickey–Fuller test. The studentized version is

$$DF_n = \Big(\frac{\sum_{t=1}^n X_{t-1}^2}{\hat\sigma^2}\Big)^{1/2}(\hat\phi_{LS} - 1), \qquad \hat\sigma^2 = \frac{1}{n}\sum_{t=1}^{n}\big(X_t - \hat\phi_{LS}X_{t-1}\big)^2.$$

Under H_0,

$$DF_n = \frac{n^{-1}\sum_{t=1}^n X_{t-1}\epsilon_t}{\hat\sigma\,\big(n^{-2}\sum_{t=1}^n X_{t-1}^2\big)^{1/2}} \xrightarrow{d} DF := \frac{\frac{1}{2}(W(1)^2 - 1)}{\big(\int_0^1 W(s)^2 ds\big)^{1/2}}.$$

Indeed, by Steps 1–3 above the numerator and the bracket in the denominator converge jointly to $\frac{\sigma^2}{2}(W(1)^2 - 1)$ and $\sigma^2\int_0^1 W^2$, and $\hat\sigma^2 \xrightarrow{p} \sigma^2$: expanding $X_t - \hat\phi_{LS}X_{t-1} = \epsilon_t - (\hat\phi_{LS} - 1)X_{t-1}$,

$$\hat\sigma^2 = \frac{1}{n}\sum \epsilon_t^2 - 2(\hat\phi_{LS} - 1)\cdot\frac{1}{n}\sum X_{t-1}\epsilon_t + (\hat\phi_{LS} - 1)^2\cdot\frac{1}{n}\sum X_{t-1}^2,$$

where $\hat\phi_{LS} - 1 = O_p(n^{-1})$, $n^{-1}\sum X_{t-1}\epsilon_t = O_p(1)$ and $n^{-1}\sum X_{t-1}^2 = O_p(n)$, so the last two terms are $O_p(n^{-1})$. One rejects H_0 when $|DF_n| \geq DF_\alpha$, where $P(|DF| \geq DF_\alpha) = \alpha$.

4.2.2 The AR(p) case: augmented Dickey–Fuller test

Analogous tests exist for models with drift or trend; we treat the pure AR(p) model $\phi(B)X_t = \epsilon_t$, $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$, and test whether $\phi(1) = 0$.

Lemma (Error-correction form)

For any $\phi_1, \dots, \phi_p$, with $\Delta X_t = X_t - X_{t-1}$,

$$X_t - \sum_{i=1}^{p} \phi_i X_{t-i} = X_t - \beta X_{t-1} - \sum_{j=1}^{p-1} \psi_j \Delta X_{t-j}, \qquad \beta = \sum_{i=1}^{p} \phi_i = 1 - \phi(1), \quad \psi_j = -\sum_{i=j+1}^{p} \phi_i.$$

In particular $\phi(1) = 0$ iff $\beta = 1$, and then $\phi(B) = \psi(B)(1 - B)$ with $\psi(B) = 1 - \psi_1 B - \cdots - \psi_{p-1} B^{p-1}$.

Proof. [Supplement] Since $X_{t-i} = X_{t-1} - \sum_{j=1}^{i-1} \Delta X_{t-j}$ for $i \geq 1$,

$$\sum_{i=1}^{p} \phi_i X_{t-i} = \Big(\sum_{i=1}^{p} \phi_i\Big) X_{t-1} - \sum_{i=1}^{p} \phi_i \sum_{j=1}^{i-1} \Delta X_{t-j} = \beta X_{t-1} - \sum_{j=1}^{p-1} \Big(\sum_{i=j+1}^{p} \phi_i\Big) \Delta X_{t-j},$$

after exchanging the order of summation; this is the identity. If $\beta = 1$ the right-hand side equals $\Delta X_t - \sum_j \psi_j \Delta X_{t-j} = \psi(B)(1 - B) X_t$. ∎

Hence the AR(p) model can be written as the regression

$$X_t = \beta X_{t-1} + \psi_1 \Delta X_{t-1} + \cdots + \psi_{p-1} \Delta X_{t-p+1} + \eta_t, \qquad \{\eta_t\} \sim \mathrm{IID}(0, \tau^2), \tag{4.4}$$

and under the unit-root hypothesis $\beta = 1$ and $\eta_t = \epsilon_t$. The least-squares estimator $\hat{\beta}$ from (4.4) should then be close to 1, and one rejects for large $\left|(\hat{\beta} - 1)/\operatorname{sd}(\hat{\beta})\right|$. This is the *augmented Dickey–Fuller* (ADF) test; the lagged differences absorb the short-run dependence, and the statistic has the same Dickey–Fuller limit law DF as in the AR(1) case.

4.3 Model selection and diagnostics

Motivation. Good forecasts require the right model. We describe the classical Box–Jenkins identification by sample autocorrelations, then information criteria (AIC, BIC) that select orders automatically, and finally checks of a fitted model by overfitting and residual analysis.

4.3.1 The Box–Jenkins method

The Box–Jenkins procedure for ARMA models has three stages:

(1) *identification* (choose p and q), (2) *estimation* of the parameters (Chapter 3), and (3) *diagnostic checking* (Section 4.3.3). If the model fails the checks one returns to stage 1 (Figure 2).

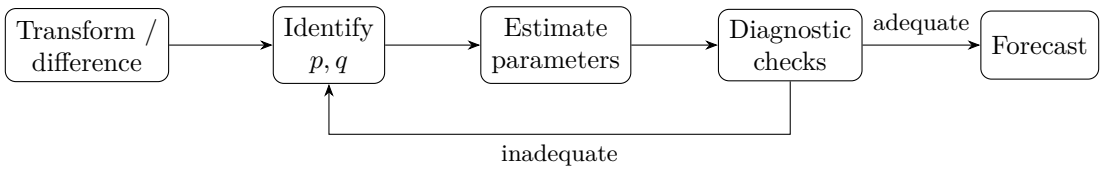


Figure 2: The Box–Jenkins modelling cycle.

Making the series stationary. If the spread of the series is not constant, a variance-stabilizing transformation is applied first: the Box–Cox transform $(y^\lambda - 1)/\lambda$ ($\lambda \neq 0$), or $\sqrt{y}$, $1/y$, $\log y$ (the log is standard for economic data). If the series trends up or down it is differenced. For a stationary series the sample ACF decays to 0 quickly, while for a nonstationary one it decays very slowly; a unit-root test (Section 4.2) settles whether differencing is needed. Usually $d \in \{0, 1, 2\}$.

Identification by ACF and PACF. After transformation, the sample ACF (SACF) and sample PACF (SPACF) are compared with the theoretical patterns of Chapter 2 (Table 1); the SACF mainly guides q and the SPACF mainly guides p. Large p, q increase the number of parameters ($p+q+2$ including mean and variance) and lead to inefficient, complicated models, so parsimonious choices are preferred; usually $p \le 3$, $q \le 3$.

Model	ACF	PACF
AR(p)	decays exponentially or as a damped sine	cuts off after lag p
MA(q)	cuts off after lag q	decays exponentially or as a damped sine
ARMA(p,q)	decays after lag $q-p$	decays after lag $p-q$

Table 1: ACF and PACF of ARMA models (Table 4.3.1).

For an AR(p) model the PACF satisfies $\phi_{kk} = 0$ for $k \ge p+1$, so one tests $H_0 : \phi_{kk} = 0$ vs $H_1 : \phi_{kk} \neq 0$. For $k \ge p+1$, $\hat{\phi}_{kk}$ is approximately normal with $E\hat{\phi}_{kk} \approx 0$ and $\operatorname{Var}\hat{\phi}_{kk} \approx 1/n$; hence H_0 is rejected at level α when $|\hat{\phi}_{kk}| > z_{\alpha/2}/\sqrt{n}$, and $|\hat{\phi}_{kk}| \le z_{\alpha/2}/\sqrt{n}$ is read as $\phi_{kk} = 0$.

For an MA(q) model the ACF cuts off after lag q; one tests $H_0 : \rho(k) = 0$ and rejects when $|\hat{\rho}(k)| > z_{\alpha/2}\,\text{S.E.}(\hat{\rho}(k))$, where by Bartlett's formula

$$\text{S.E.}(\hat{\rho}(k)) \approx \Big(\frac{1 + 2\sum_{j=1}^{q}\hat{\rho}(j)^2}{n}\Big)^{1/2}, \qquad k \ge q+1.$$

If $|\hat{\rho}(k)| \le z_{\alpha/2}\,\text{S.E.}(\hat{\rho}(k))$ for all $k \ge q+1$, an MA(q) model is tentatively chosen. For mixed ARMA models neither function cuts off, and the orders are chosen by the criteria of the next subsection.

4.3.2 Model selection criteria

Choosing a model means choosing a parametric family. For data $X_1, \dots, X_n$ consider, for example, the nested normal models

$$M_1 = \{N(\mu, 1) : -\infty < \mu < \infty\}, \qquad M_2 = \{N(\mu, \sigma^2) : -\infty < \mu < \infty,\ 0 < \sigma^2 < \infty\}.$$

If M_1 is correct, choosing M_2 wastes a parameter (inefficiency); if M_2 is correct, choosing M_1 ignores an essential parameter, which is worse. Akaike's approach: let $f_1(x_1, \dots, x_n; \mu)$ and $f_2(x_1, \dots, x_n; \mu, \sigma^2)$ be the joint densities, compute the maximized likelihoods $MLE_1 = f_1(X_1, \dots, X_n; \hat{\mu})$ and $MLE_2 = f_2(X_1, \dots, X_n; \hat{\mu}, \hat{\sigma}^2)$, and

$$AIC(M_1) = -2\log(MLE_1) + 2 \times 1, \qquad AIC(M_2) = -2\log(MLE_2) + 2 \times 2;$$

the model with the smaller value is selected. In general

$$AIC(\text{model}) = -2\log(MLE) + 2\,(\text{number of parameters in the model}), \tag{4.5}$$

$$BIC(\text{model}) = -2\log(MLE) + \log(n)\,(\text{number of parameters in the model}).$$

The maximized likelihood always increases with the number of parameters, so it alone would favour ever larger models, contradicting parsimony; the second term penalizes the number of parameters and offsets this. The factor 2 in front of the log-likelihood is conventional. Unlike Box–Jenkins identification, information criteria apply directly to ARMA models and rest on a complete theory (AIC estimates the expected Kullback–Leibler discrepancy between the fitted and the true model).

[Supplement] AIC versus BIC

Compare a correct model M with an overfitted model $M' \supset M$ having one more parameter. The likelihood-ratio statistic $\Lambda_n = 2\log(MLE') - 2\log(MLE)$ is asymptotically χ^2_1 under M. AIC selects M' iff $\Lambda_n > 2$, which has limiting probability $P(\chi^2_1 > 2) \approx 0.157 > 0$; BIC selects M' iff $\Lambda_n > \log n$, whose probability tends to 0. If instead M is false and M' true, Λ_n/n converges to a positive constant (a Kullback–Leibler divergence), so both criteria reject M with probability tending to one. Hence BIC is consistent for the true order, while AIC tends to overfit (but is efficient for prediction).

4.3.3 Model diagnostics

After identification and estimation, *diagnostic checking* asks whether the tentative model is adequate. If it is, it becomes the forecasting model; otherwise one returns to identification. Two tools are used: overfitting and residual analysis.

(1) Overfitting. If ARIMA(p, d, q) was chosen, also fit the larger models ARIMA$(p+1, d, q)$ and ARIMA$(p, d, q+1)$ and ask:

(a) are the added parameters judged to be zero?

(b) are the estimates of the original parameters essentially unchanged?

If both answers are yes, the tentative model passes; otherwise another model should be tried. For instance, if MA(1) was chosen, fit ARMA(1,1) and MA(2) and check that ϕ_1 and θ_2 are insignificant and that the three estimates of θ_1 agree. The added estimates are asymptotically normal with mean 0 under H_0: "the tentative model is true", which yields the tests.

(2) Residual analysis. Suppose the selected model is the stationary ARMA(p, q) model

$$\phi(B)(X_t - \mu) = \theta(B)\epsilon_t. \tag{4.6}$$

If it is correct and well estimated, the residuals $\{e_t\}$ should behave like the IID errors $\{\epsilon_t\}$. With estimates $\hat\phi_1, \dots, \hat\phi_p, \hat\theta_1, \dots, \hat\theta_q, \hat\mu$ and $\hat\phi(B) = 1 - \hat\phi_1 B - \cdots - \hat\phi_p B^p$, $\hat\theta(B) = 1 + \hat\theta_1 B + \cdots + \hat\theta_q B^q$, the residuals are

$$e_t = \hat\theta(B)^{-1}\hat\phi(B)(X_t - \hat\mu), \qquad t = 1, \dots, n,$$

computed with $X_t = 0$ for $t \le 0$. One checks:

(a) is the mean zero? (b) is the variance constant? (c) are the residuals independent, i.e. are their autocorrelations negligible? (a) is checked by comparing $\bar e$ with its standard error, (b) by a residual plot. For (c) use the residual sample ACF

$$\rho_e(k) = \frac{\sum_{t=1}^{n-k}(e_t - \bar e)(e_{t+k} - \bar e)}{\sum_{t=1}^{n}(e_t - \bar e)^2}.$$

Under H_0: "(4.6) is the true model" (so $\rho(k) = 0$ for $k \ge 1$), $\mathrm{Var}(\rho_e(k)) \approx 1/n$, and the approximate rejection region at level α is $|\rho_e(k)| \ge z_{\alpha/2}\,\mathrm{S.E.}(\rho_e(k))$. One may also fit ARMA models to the residuals themselves: ARMA(0,0) (white noise) should be identified.

Testing each lag separately at level α ignores the dependence among the $\rho_e(k)$ and inflates the overall type I error, whereas the real hypothesis is the simultaneous one that all residual autocorrelations vanish. The *portmanteau* or *Ljung–Box* statistic

$$Q = n(n+2)\sum_{h=1}^{K}\frac{\rho_e^2(h)}{n-h}$$

is approximately $\chi^2_{K-p-q-1}$ under H_0; the model is rejected at level α if Q exceeds the upper $100\alpha\%$ point of $\chi^2_{K-p-q-1}$.

The *Durbin–Watson* statistic tests no first-order correlation against AR(1) errors:

$$DW = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2} \approx 2(1 - \rho_e(1)),$$

since the numerator equals $\sum_{t\geq 2} e_t^2 + \sum_{t\geq 2} e_{t-1}^2 - 2\sum_{t\geq 2} e_t e_{t-1} \approx 2\sum e_t^2 - 2\sum e_t e_{t-1}$. Thus $0 \leq DW \leq 4$ approximately; $DW \approx 2$ means no lag-one correlation, $DW < 2$ positive and $DW > 2$ negative correlation. Critical values are tabulated by Durbin and Watson (1971).

4.4 Selected exercises

Motivation. These textbook exercises make the differencing argument of Section 4.1 rigorous, treat stochastic trends and over-differencing, and establish the basic properties of the random walk and of Brownian motion used in the unit-root theory. The algebraic identities for T_n already derived in Section 4.2.1 are not repeated.

Exercise 4.1 (Differencing a polynomial trend)

Let $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$ with $r \geq 1$, $a_r \neq 0$, $\{\epsilon_t\} \sim WN(0, \sigma^2)$. Show that
(a) $(1-B)^r t^r = r!$;
(b) $(1-B)^r \epsilon_t = \sum_{j=0}^{r} \binom{r}{j}(-1)^j \epsilon_{t-j}$;
(c) $(1-B)^r X_t$ is stationary, and find its mean and variance.

Solution.

(a)

Step 1. Let p be a polynomial of degree $k \geq 1$ with leading coefficient c. By the binomial theorem,

$$(1-B)t^k = t^k - (t-1)^k = -\sum_{i=0}^{k-1} \binom{k}{i}(-1)^{k-i} t^i = k\, t^{k-1} + (\text{terms of degree} \leq k-2),$$

so by linearity $(1-B)p$ has degree $k-1$ and leading coefficient kc. For a constant c, $(1-B)c = c - c = 0$.

Step 2. By induction on $j = 0, \ldots, r$, $(1-B)^j t^r$ is a polynomial of degree $r-j$ with leading coefficient $r(r-1)\cdots(r-j+1)$. For $j = r$ this is the constant $r!$.

(b) The operators 1 and B commute, so the binomial theorem holds for $(1-B)^r$: $(1-B)^r = \sum_{j=0}^{r} \binom{r}{j}(-1)^j B^j$. Applying it to ϵ_t and using $B^j \epsilon_t = \epsilon_{t-j}$ gives the claim.

(c)

Step 1. For $k < r$, Step 2 of (a) with $j = k$ gives $(1-B)^k t^k = k!$, a constant, so $(1-B)^r t^k = (1-B)^{r-k} k! = 0$ as $r - k \geq 1$. By linearity and (a),(b),

$$Z_t := (1-B)^r X_t = a_r\, r! + \sum_{j=0}^{r} c_j \epsilon_{t-j}, \qquad c_j = \binom{r}{j}(-1)^j.$$

Step 2. $Z_t - a_r r!$ is a finite moving average of white noise. Its ACVF is computed exactly as in (2.15) (only $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$ is used): $\gamma_Z(h) = \sigma^2 \sum_{j=0}^{r-|h|} c_j c_{j+|h|}$ for $|h| \leq r$ and 0 otherwise, free of t. Hence $\{Z_t\}$ is stationary with mean $EZ_t = a_r\, r!$.

Step 3. By Vandermonde's identity $\sum_j \binom{r}{j}\binom{r}{r-j} = \binom{2r}{r}$,

$$\mathrm{Var}(Z_t) = \sigma^2 \sum_{j=0}^{r} \binom{r}{j}^2 = \binom{2r}{r}\sigma^2.$$

■

Exercise 4.2 (Markov property of the random walk)

Let $X_t = X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$ with values in a countable set. Show that, whenever the conditioning events have positive probability,

$$P(X_n = i_n \mid X_0 = i_0, X_1 = i_1, \ldots, X_{n-1} = i_{n-1}) = P(X_n = i_n \mid X_{n-1} = i_{n-1}).$$

Solution.

Step 1. $X_k = \epsilon_1 + \cdots + \epsilon_k$, so $X_0, \ldots, X_{n-1}$ are functions of $\epsilon_1, \ldots, \epsilon_{n-1}$ and are independent of ϵ_n. Let $A = \{X_0 = i_0, \ldots, X_{n-1} = i_{n-1}\}$ and $A' = \{X_{n-1} = i_{n-1}\}$; both belong to $\sigma(\epsilon_1, \ldots, \epsilon_{n-1})$.

Step 2. On A we have $X_n = i_{n-1} + \epsilon_n$, so by independence

$$P(X_n = i_n, A) = P(\epsilon_n = i_n - i_{n-1}, A) = P(\epsilon_n = i_n - i_{n-1})P(A).$$

Dividing by $P(A) > 0$: $P(X_n = i_n \mid A) = P(\epsilon_1 = i_n - i_{n-1})$, using that the ϵ_t are identically distributed.

Step 3. The same computation with A' in place of A gives $P(X_n = i_n \mid A') = P(\epsilon_1 = i_n - i_{n-1})$. The two conditional probabilities coincide.

For general (not necessarily discrete) ϵ_t with distribution function F, the same argument with the independence ("freezing") lemma gives $P(X_n \le x \mid X_0, \ldots, X_{n-1}) = F(x - X_{n-1})$, which is a function of X_{n-1} alone and therefore also equals $P(X_n \le x \mid X_{n-1})$. ■

Exercise 4.3 (Quadratic trend plus a local linear trend)

Let

$$X_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \mu_t + \epsilon_t, \qquad \mu_t = \mu_{t-1} + \beta_t + \eta_t, \qquad \beta_t = \beta_{t-1} + \zeta_t,$$

where $\{\epsilon_t\} \sim WN(0, \sigma_\epsilon^2)$, $\{\eta_t\} \sim WN(0, \sigma_\eta^2)$, $\{\zeta_t\} \sim WN(0, \sigma_\zeta^2)$ are mutually independent. Difference $\{X_t\}$ to obtain a stationary series and find its mean and variance.

Solution.

Step 1. From the recursions, $(1-B)\beta_t = \zeta_t$ and $(1-B)\mu_t = \beta_t + \eta_t$, hence

$$(1-B)^2\mu_t = (1-B)\beta_t + (1-B)\eta_t = \zeta_t + \eta_t - \eta_{t-1}.$$

Step 2. By Exercise 4.1(a) and Step 1 of its part (c), $(1-B)^2(\alpha_0 + \alpha_1 t + \alpha_2 t^2) = 2\alpha_2$, and $(1-B)^2\epsilon_t = \epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2}$. Therefore

$$D_t := (1-B)^2 X_t = 2\alpha_2 + \zeta_t + (\eta_t - \eta_{t-1}) + (\epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2}).$$

Step 3. The three noise terms are uncorrelated with each other, and each is a finite moving average of white noise with ACVF given by (2.15) (with coefficients (1), $(1, -1)$, $(1, -2, 1)$). By Exercise 1.7 (which only uses zero cross-covariances), $\{D_t\}$ is stationary with

$$\gamma_D(0) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2, \quad \gamma_D(\pm 1) = -\sigma_\eta^2 - 4\sigma_\epsilon^2, \quad \gamma_D(\pm 2) = \sigma_\epsilon^2,$$

and $\gamma_D(h) = 0$ for $|h| \geq 3$.

Step 4. Hence $ED_t = 2\alpha_2$ and $\mathrm{Var}(D_t) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2$.

One difference does not suffice in general: $(1-B)X_t = \alpha_1 - \alpha_2 + 2\alpha_2 t + \beta_t + \eta_t + \epsilon_t - \epsilon_{t-1}$ has a mean linear in t when $\alpha_2 \neq 0$, and contains the random walk β_t, whose variance grows linearly in t when $\sigma_\zeta^2 > 0$. ■

Exercise 4.8 (Properties of Brownian motion)

Let $\{W(s) : s \geq 0\}$ be a standard Brownian motion (Definition 4.2.1). Show that
(a) $EW(t)W(s) = t \wedge s$;
(b) $Z(s) = W(\eta s)/\sqrt{\eta}$ is a Brownian motion for every $\eta > 0$;
(c) $G(s) = sW(1/s)$ for $s > 0$, $G(0) = 0$, is a Brownian motion.

Solution.

(a) Let $s \leq t$. By (3) of Definition 4.2.1, $W(s)$ and $W(t) - W(s)$ are independent with mean 0, and by (1),(2), $EW(s)^2 = s$. Hence $EW(t)W(s) = EW(s)^2 + E[W(s)(W(t) - W(s))] = s = s \wedge t$.

A characterization. Before (b),(c) we prove: *a process $\{G(s)\}$ with continuous paths, $G(0) = 0$, whose finite-dimensional laws are jointly normal with mean 0 and $\mathrm{Cov}(G(s), G(t)) = s \wedge t$, is a Brownian motion; and conversely W has these properties.*

Step 1. For $t > s$, $G(t) - G(s)$ is normal with mean 0 and variance $t - 2(s \wedge t) + s = t - s$.

Step 2. For $0 \leq t_0 < t_1 < \cdots < t_k$, the increments $\Delta_i = G(t_i) - G(t_{i-1})$ are a linear image of a normal vector, hence jointly normal. For $i < j$, $t_i \leq t_{j-1}$, so

$$\mathrm{Cov}(\Delta_i, \Delta_j) = t_i \wedge t_j - t_i \wedge t_{j-1} - t_{i-1} \wedge t_j + t_{i-1} \wedge t_{j-1} = t_i - t_i - t_{i-1} + t_{i-1} = 0.$$

Jointly normal and uncorrelated variables are independent (the characteristic function factorizes, as in Exercise 1.6). With $t_0 = 0$ this gives (3) of Definition 4.2.1.

Step 3. Conversely, $(W(t_1), \ldots, W(t_k))$ is a linear image (partial sums) of the independent normal increments, hence jointly normal with mean 0, and its covariance is $s \wedge t$ by (a).

(b) Z has continuous paths and $Z(0) = 0$; its finite-dimensional laws are linear images of those of W, hence jointly normal with mean 0; and $\mathrm{Cov}(Z(s), Z(t)) = (\eta s \wedge \eta t)/\eta = s \wedge t$. The characterization applies.

(c)

Step 1. For $s, t > 0$, $\mathrm{Cov}(G(s), G(t)) = st\,(1/s \wedge 1/t) = st/(s \vee t) = s \wedge t$, and this also holds trivially if s or t is 0. The finite-dimensional laws are linear images of those of W (with the constant $G(0) = 0$), hence jointly normal with mean 0. Paths are continuous on $(0, \infty)$ as compositions of continuous maps.

Step 2. It remains to show continuity at 0, i.e. $W(u)/u \to 0$ a.s. as $u \to \infty$. Since $W(n) = \sum_{k=1}^{n}(W(k) - W(k-1))$ is a sum of i.i.d. $N(0,1)$ variables, the strong law gives $W(n)/n \to 0$ a.s.

Step 3. Let $M_k = \sup_{u \in [k, k+1]} |W(u) - W(k)|$, a supremum over rationals by path continuity. The process $u \mapsto W(k+u) - W(k)$, $u \in [0,1]$, is Gaussian with mean 0 and covariance $u \wedge v$ (by (a)), so it has the same finite-dimensional laws as W on $[0,1]$, and M_k has the law of $M_0 = \sup_{u \leq 1} |W(u)|$. By Doob's L^2 maximal inequality for the martingale W (used as a tool), $EM_0^2 \leq 4EW(1)^2 = 4$, so $EM_0 \leq 2$.

Step 4. For $\varepsilon > 0$, $\sum_{k \geq 1} P(M_k > \varepsilon k) = \sum_{k \geq 1} P(M_0/\varepsilon > k) \leq E(M_0)/\varepsilon < \infty$. By Borel–Cantelli, $M_k \leq \varepsilon k$ eventually a.s.; letting $\varepsilon \downarrow 0$ along a sequence, $M_k/k \to 0$ a.s.

Step 5. For $u \in [k, k+1]$, $|W(u)|/u \leq (|W(k)| + M_k)/k \to 0$ a.s. by Steps 2 and 4. Hence $G(s) = W(1/s)/(1/s) \to 0 = G(0)$ as $s \downarrow 0$, and the characterization shows that G is a Brownian motion. ■

Exercise 4.9 (Over-differencing an MA(2))

Let $X_t = \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}$, $\{\epsilon_t\} \sim IID\, N(0,\sigma^2)$, be stationary, and suppose one fits $(1-B)X_t = Y_t$. Find the ACVF of $\{Y_t\}$.

Solution.

Step 1. $Y_t = X_t - X_{t-1} = \epsilon_t + (\theta_1 - 1)\epsilon_{t-1} + (\theta_2 - \theta_1)\epsilon_{t-2} - \theta_2\epsilon_{t-3}$, an MA(3) with coefficients $c_0 = 1$, $c_1 = \theta_1 - 1$, $c_2 = \theta_2 - \theta_1$, $c_3 = -\theta_2$.

Step 2. By Exercise 2.9(a) with $c = 1$, $\gamma_Y(h) = 2\gamma_X(h) - \gamma_X(h-1) - \gamma_X(h+1)$, where by (2.15) $\gamma_X(0) = (1+\theta_1^2+\theta_2^2)\sigma^2$, $\gamma_X(\pm1) = (\theta_1+\theta_1\theta_2)\sigma^2$, $\gamma_X(\pm2) = \theta_2\sigma^2$ and $\gamma_X(h) = 0$ for $|h| \geq 3$. Hence

$$\begin{aligned}
\gamma_Y(0) &= 2\sigma^2(1+\theta_1^2+\theta_2^2-\theta_1-\theta_1\theta_2),\\
\gamma_Y(\pm1) &= \sigma^2(2\theta_1+2\theta_1\theta_2-\theta_2-1-\theta_1^2-\theta_2^2),\\
\gamma_Y(\pm2) &= \sigma^2(2\theta_2-\theta_1-\theta_1\theta_2), \qquad \gamma_Y(\pm3) = -\theta_2\sigma^2,
\end{aligned}$$

and $\gamma_Y(h) = 0$ for $|h| \geq 4$. (These agree with $\sigma^2\sum_j c_j c_{j+|h|}$ from Step 1.)

Step 3. The MA polynomial of Y is $(1-z)\theta(z)$, which vanishes at $z = 1$: the over-differenced series is not invertible. For example, $\theta_1 = \theta_2 = 0$ gives $\rho_Y(1) = -\frac{1}{2}$; a lag-one sample autocorrelation near -0.5 is a warning sign of unnecessary differencing. ■

Chapter 5. Seasonal Time Series Models

Many series repeat a similar pattern with a known period s: monthly data typically have $s = 12$, quarterly data $s = 4$. Such periodic behaviour is called *seasonality* (a seasonal effect), and the period s is assumed known from the nature of the data. This chapter adapts the ARIMA models of Chapter 4 to seasonal data and introduces the multiplicative seasonal model of Box and Jenkins.

5.1 Time series with a seasonal component

Motivation. A deterministic-looking periodic component makes a series nonstationary in a way that ordinary differencing does not remove; differencing at the seasonal lag s does.

Suppose

$$X_t = S_t + Z_t, \tag{5.1}$$

where $\{S_t\}$ is a *seasonal component* of period s and $\{Z_t\}$ is stationary. Precisely, $S_t = S_{t+s}$ for all t; $S_1, \dots, S_s$ are independent with $ES_t = 0$ and $\mathrm{Var}(S_t) = \sigma_t^2 < \infty$; $EZ_t = 0$ with autocovariance $\gamma_Z(h)$; and $\{S_t\}$ and $\{Z_t\}$ are independent. Then

$$\begin{aligned} EX_{t+h}X_t &= E(S_{t+h} + Z_{t+h})(S_t + Z_t) = ES_{t+h}S_t + \gamma_Z(h) \\ &= \begin{cases} \sigma_t^2 + \gamma_Z(h), & h = 0, \pm s, \pm 2s, \dots, \\ \gamma_Z(h), & \text{otherwise.} \end{cases} \end{aligned}$$

The covariance depends on t through σ_t^2, so $\{X_t\}$ is nonstationary; and even when all σ_t^2 are equal the autocovariance does not decay along the lags $s, 2s, \dots$, so no causal ARMA model can describe $\{X_t\}$.

To remove the seasonal component introduce the *seasonal difference operator*

$$\nabla_s = 1 - B^s.$$

Applying it to (5.1) (*seasonal differencing*) gives

$$Y_t = (1 - B^s)X_t = X_t - X_{t-s} = S_t + Z_t - (S_{t-s} + Z_{t-s}) = Z_t - Z_{t-s},$$

a difference of a stationary series, hence stationary. Fitting an ARMA model $\phi(B)Y_t = \theta(B)\epsilon_t$, $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$, to Y_t we obtain

$$\phi(B)\nabla_s X_t = \theta(B)\epsilon_t. \tag{5.2}$$

Definition 5.1.1 (Seasonal difference model)

The model (5.2) is called the *seasonal difference model*.

5.2 Time series with trend and seasonality

Motivation. When a trend and a seasonal component appear together, one seasonal difference removes the seasonality and lowers the degree of the polynomial trend by one; ordinary differences then remove what is left of the trend.

Suppose

$$X_t = T_t + S_t + Z_t,$$

where T_t is a trend, S_t a seasonal component and Z_t a stationary process, mutually independent, with

$$T_t = \beta_0 + \beta_1 t + \cdots + \beta_r t^r \quad (r \geq 1,\ \beta_r \neq 0), \qquad S_t = S_{t+s} \text{ for all } t.$$

Seasonal differencing gives $\nabla_s X_t = \nabla_s T_t + \nabla_s S_t + \nabla_s Z_t$ with $\nabla_s S_t = 0$.

Lemma (Removing a polynomial trend)

With T_t and Z_t as above,
(1) $(1-B)^{r-1}\nabla_s T_t = \mu$ for the constant $\mu = s\,r!\,\beta_r$;
(2) $(1-B)^{r-1}\nabla_s Z_t$ is stationary with mean 0.
Consequently

$$(1-B)^{r-1}\nabla_s X_t = (1-B)^{r-1}\nabla_s T_t + (1-B)^{r-1}\nabla_s Z_t = \mu + (1-B)^{r-1}\nabla_s Z_t. \tag{5.3}$$

Proof. [Supplement]

Step 1. If $P(t) = at^k + (\text{lower terms})$ is a polynomial of degree $k \geq 1$, then for any integer $m \geq 1$, by the binomial theorem $t^k - (t-m)^k = km\,t^{k-1} + (\text{lower terms})$, so $P(t) - P(t-m)$ is a polynomial of degree $k-1$ with leading coefficient akm. (For $k = 0$ the difference is 0.)

Step 2. Apply Step 1 with $m = s$: $\nabla_s T_t$ has degree $r-1$ and leading coefficient $\beta_r rs$. Apply it $r-1$ more times with $m = 1$: the degree drops to 0 and the leading coefficient becomes $\beta_r rs \cdot (r-1)(r-2)\cdots 1 = s\,r!\,\beta_r$. This proves (1).

Step 3. $(1-B)^{r-1}\nabla_s = \sum_{j=0}^{m} c_j B^j$ is a polynomial in B of finite degree $m = r - 1 + s$. Hence $U_t = \sum_j c_j Z_{t-j}$ has $EU_t = 0$ and $\mathrm{Cov}(U_{t+h}, U_t) = \sum_{j,k} c_j c_k \gamma_Z(h - j + k)$, which does not depend on t. This proves (2), and (5.3) follows by linearity and $\nabla_s S_t = 0$. ∎

Fitting an ARMA model to the stationary series $(1-B)^{r-1}\nabla_s X_t - \mu$ gives

$$\phi(B)\{(1-B)^{r-1}\nabla_s X_t - \mu\} = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2). \tag{5.4}$$

Definition 5.2.1 (Seasonal difference model with trend)

The model (5.4) is called the *seasonal difference model with trend*.

For a linear trend ($r = 1$) a single seasonal difference suffices; in practice one often applies ∇_s and then one ordinary difference ∇ to reach stationarity.

5.3 Seasonal ARIMA models

Motivation. Instead of a fixed periodic component, seasonality can be stochastic: an observation depends on the observation one season earlier. ARMA recursions in the seasonal lag B^s model this.

The model

$$X_t = \phi X_{t-s} + \epsilon_t, \qquad \{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2), \tag{5.5}$$

takes seasonality into account. If $|\phi| < 1$, iterating gives the stationary solution

$$X_t = \sum_{i=0}^{\infty} \phi^i \epsilon_{t-si}. \tag{5.6}$$

This $\{X_t\}$ has no seasonal component $\{S_t\}$, yet it shows a seasonal effect of period s: $\gamma_X(h) = 0$ unless h is a multiple of s, so only observations a whole number of seasons apart are correlated. Such data are described by the *seasonal ARMA* (SARMA) model. Since the series may also have a seasonal trend, we allow seasonal differencing and obtain the seasonal ARIMA (SARIMA) model.

Definition 5.3.1 (SARIMA$(P,D,Q)_s$ model)

A time series $\{X_t\}$ follows a *seasonal ARIMA model* SARIMA$(P,D,Q)_s$ if

$$\Phi(B^s)\{(1-B^s)^D X_t - \mu\} = \Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0,\sigma^2), \tag{5.7}$$

where

$$\Phi(z) = 1 - \Phi_1 z - \cdots - \Phi_P z^P, \qquad \Theta(z) = 1 + \Theta_1 z + \cdots + \Theta_Q z^Q$$

satisfy the causality and invertibility conditions. Usually $\mu = 0$ is assumed.

The model of Definition 5.3.1 is appropriate when only observations in the same position of the season are correlated; $D \neq 0$ is appropriate when the seasonal part carries a trend. Note that (5.4) and (5.7) are different models, and which one is preferable depends on the data.

Example 5.3.2 (ACF of seasonal MA and AR models)

(1) For the MA$(2)_s$ model $X_t = \epsilon_t + \theta_1\epsilon_{t-s} + \theta_2\epsilon_{t-2s}$, the autocovariance is that of an MA(2) placed on the lags $0, s, 2s$:

$$\gamma_X(h) = \begin{cases} (1+\theta_1^2+\theta_2^2)\sigma^2, & h = 0, \\ (\theta_1 + \theta_1\theta_2)\sigma^2, & h = s, \\ \theta_2\sigma^2, & h = 2s, \\ 0, & \text{otherwise } (h \geq 0). \end{cases}$$

(2) For the AR$(1)_s$ model $X_t - \phi X_{t-s} = \epsilon_t$, $|\phi| < 1$, (5.6) gives $\gamma_X(sj) = \sigma^2 \sum_{i\geq 0} \phi^i \phi^{i+j} = \phi^j \sigma^2/(1-\phi^2)$ and $\gamma_X(h) = 0$ if $s \nmid h$. Hence

$$\rho(h) = \begin{cases} \phi^{h/s}, & h = 0, s, 2s, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

5.4 Multiplicative seasonal models

Motivation. Seasonal data are correlated not only across seasons but also between neighbouring observations. Multiplying a seasonal ARIMA operator in B^s by an ordinary ARIMA operator in B captures both kinds of dependence with few parameters.

Definition 5.4.1 (Multiplicative seasonal ARIMA model)

A time series $\{X_t\}$ follows a *multiplicative seasonal* ARIMA$(p,d,q)(P,D,Q)_s$ model if

$$\phi(B)\Phi(B^s)(1-B)^d(1-B^s)^D X_t = \theta(B)\Theta(B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0,\sigma^2),$$

with ϕ, θ as in Definition 4.1.1 and Φ, Θ as in Definition 5.3.1.

This model contains those of Definitions 5.1.1, 5.2.1 and 5.3.1 as special cases. The ACF of the stationary part combines the two scales: a short-lag pattern near 0 governed by ϕ, θ, repeated around the seasonal lags $s, 2s, \ldots$ governed by Φ, Θ.

Example 5.4.2 (The ARIMA$(0,1,1)(0,1,1)_s$ model)

The model $\nabla_s(1-B)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ reads

$$X_t - X_{t-1} - X_{t-s} + X_{t-s-1} = \epsilon_t + \theta\epsilon_{t-1} + \Theta\epsilon_{t-s} + \theta\Theta\epsilon_{t-s-1}.$$

Let $W_t = \nabla_s(1-B)X_t$; then $\{W_t\}$ is the stationary MA process ARIMA$(0,0,1)(0,0,1)_s$. Assume $s \geq 3$. Multiplying out the four noise terms and keeping the matching lags,

$$\gamma_W(0) = (1+\theta^2)(1+\Theta^2)\sigma^2, \qquad \gamma_W(1) = \theta(1+\Theta^2)\sigma^2,$$
$$\gamma_W(s) = \Theta(1+\theta^2)\sigma^2, \qquad \gamma_W(s-1) = \gamma_W(s+1) = \theta\Theta\sigma^2,$$

and $\gamma_W(h) = 0$ for $2 \leq h \leq s-2$ and for $h \geq s+2$ (Figure 3).

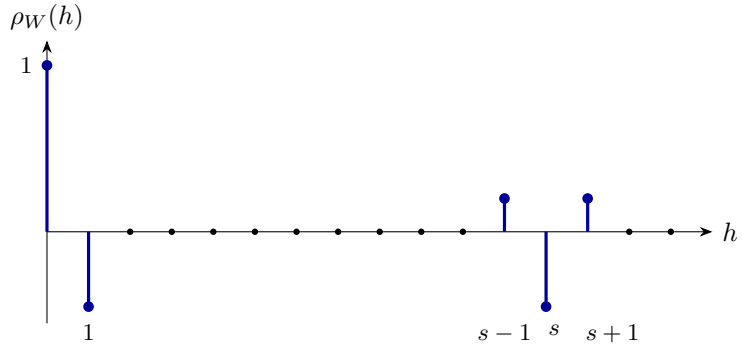


Figure 3: ACF of $W_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ (here $\theta, \Theta < 0$, $s = 12$): nonzero only at lags $0, 1, s-1, s, s+1$.

Example 5.4.3 (The ARIMA$(0,0,1)(1,0,1)_s$ model)

Consider $(1-\Phi B^s)X_t = (1+\theta B)(1+\Theta B^s)\epsilon_t$ with $|\Phi| < 1$ and $s \geq 4$. Write $X_t = (1+\theta B)Y_t$, where $Y_t = \Phi Y_{t-s} + \epsilon_t + \Theta\epsilon_{t-s}$ is a SARMA$(1,1)_s$ process. Since $\mathrm{Cov}(Y_t, \epsilon_t) = \sigma^2$ and $\mathrm{Cov}(Y_t, \epsilon_{t-s}) = (\Phi+\Theta)\sigma^2$,

$$\gamma_Y(0) = \Phi^2\gamma_Y(0) + (1+\Theta^2)\sigma^2 + 2\Phi\Theta\sigma^2 \implies \gamma_Y(0) = \Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_Y(s) = \Phi\gamma_Y(0) + \Theta\sigma^2 = \Big\{\Theta + \Phi + \frac{\Phi(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$

$\gamma_Y(ks) = \Phi^{k-1}\gamma_Y(s)$ for $k \geq 1$, and $\gamma_Y(h) = 0$ if $s \nmid h$. The filter $1+\theta B$ gives $\gamma_X(h) = (1+\theta^2)\gamma_Y(h) + \theta\{\gamma_Y(h-1) + \gamma_Y(h+1)\}$, hence

$$\gamma_X(0) = (1+\theta^2)\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2, \qquad \gamma_X(1) = \theta\Big\{1 + \frac{(\Theta+\Phi)^2}{1-\Phi^2}\Big\}\sigma^2,$$
$$\gamma_X(s) = (1+\theta^2)\gamma_Y(s), \qquad \gamma_X(s-1) = \gamma_X(s+1) = \theta\gamma_Y(s),$$

$\gamma_X(2) = \cdots = \gamma_X(s-2) = 0$, and $\gamma_X(h) = \Phi\gamma_X(h-s)$ for $h \geq s+2$. Compared with Example 5.4.2, the seasonal AR factor makes the pattern around s recur, damped by Φ, around $2s, 3s, \ldots$.

5.5 Selected exercises

Motivation. The exercises below compute the causal representation and the autocovariances of seasonal models directly from the operator identities, and compare a seasonal MA model with an unrestricted MA model of the same order.

Exercise 5.2 (ψ-weights of a seasonal model)

Let $(1-\Phi B^s)X_t = (1+\theta B)\epsilon_t$ with $|\Phi|<1$ and $\{\epsilon_t\}\sim \mathrm{IID}(0,\sigma^2)$. Find the coefficients a_j in the representation $X_t=\sum_{j=0}^{\infty} a_j\epsilon_{t-j}$ of the stationary solution.

Solution.

Step 1. The AR polynomial $\phi(z)=1-\Phi z^s$ vanishes only where $z^s=1/\Phi$, i.e. on the circle $|z|=|\Phi|^{-1/s}>1$ (no zeros at all if $\Phi=0$). By Proposition 2.4.A, the model has a unique stationary solution $X_t=\sum_j a_j\epsilon_{t-j}$, whose coefficients are those of the power series $a(z)=(1+\theta z)/(1-\Phi z^s)$.

Step 2. For $|z|<|\Phi|^{-1/s}$, the geometric series gives $(1-\Phi z^s)^{-1}=\sum_{i\ge 0}\Phi^i z^{si}$, so

$$a(z)=\sum_{i=0}^{\infty}\Phi^i z^{si}+\theta\sum_{i=0}^{\infty}\Phi^i z^{si+1}.$$

Step 3: $s\ge 2$. The exponents si and $si'+1$ never coincide, since $s(i-i')=1$ is impossible for $s\ge 2$. Comparing coefficients,

$$a_{si}=\Phi^i,\qquad a_{si+1}=\theta\Phi^i\quad(i=0,1,2,\dots),\qquad a_j=0 \text{ if } j \bmod s\notin\{0,1\}.$$

Thus $X_t=\sum_{i\ge 0}\Phi^i(\epsilon_{t-si}+\theta\epsilon_{t-si-1})$.

Step 4: $s=1$. Then the model is an ordinary ARMA(1,1) and the two series overlap: $a_0=1$ and $a_j=\Phi^j+\theta\Phi^{j-1}=\Phi^{j-1}(\Phi+\theta)$ for $j\ge 1$.

Step 5: check. In both cases the coefficients satisfy $a_j-\Phi a_{j-s}=\mathbb{1}\{j=0\}+\theta\,\mathbb{1}\{j=1\}$ (with $a_j=0$ for $j<0$), which is the coefficient form of $(1-\Phi z^s)a(z)=1+\theta z$. ■

Exercise 5.3 ($\mathrm{MA}(1)_{12}$ versus $\mathrm{MA}(12)$)

Can the $\mathrm{MA}(1)_{12}$ model $X_t=\epsilon_t+\Theta\epsilon_{t-12}$ be regarded as an MA(12) model? Which of the two formulations is preferable, and why?

Solution.

Step 1: it is an MA(12) model. $X_t=\epsilon_t+\theta_1\epsilon_{t-1}+\cdots+\theta_{12}\epsilon_{t-12}$ with $\theta_1=\cdots=\theta_{11}=0$ and $\theta_{12}=\Theta$. Its autocovariances are those of Example 5.3.2(1) with one seasonal coefficient:

$$\gamma(0)=(1+\Theta^2)\sigma^2,\qquad \gamma(12)=\Theta\sigma^2,\qquad \gamma(h)=0 \text{ for } h\ge 1,\ h\neq 12,$$

because $\mathrm{Cov}(\epsilon_{t+h}+\Theta\epsilon_{t+h-12},\epsilon_t+\Theta\epsilon_{t-12})$ collects a product of equal-time noises only when $h\in\{0,\pm 12\}$.

Step 2: the same invertibility condition. The MA(12) polynomial $\theta(z)=1+\Theta z^{12}$ vanishes exactly when $z^{12}=-1/\Theta$, i.e. on $|z|=|\Theta|^{-1/12}$. Hence all zeros lie outside the unit circle if and only if $|\Theta|<1$, which is the invertibility condition of the seasonal $\mathrm{MA}(1)_{12}$ polynomial $1+\Theta z$. So the two descriptions define the same class of processes.

Step 3: which to prefer. As a model to be *fitted*, the unrestricted MA(12) has twelve free coefficients, eleven of which are zero for this process. Estimating them adds sampling error of order $n^{-1/2}$ to each

fitted coefficient, and hence to the fitted autocovariances and to the forecasts, without improving the fit: the likelihood ratio statistic for the eleven zero restrictions is asymptotically $\chi^2(11)$, whereas AIC charges $2 \cdot 11 = 22$ and BIC $11 \log n$ for the extra parameters. The seasonal MA$(1)_{12}$ model is therefore preferable by the principle of parsimony; it also states the structure directly (dependence only between the same month of consecutive years). The unrestricted MA(12) is preferable only if the sample ACF shows genuine correlation at lags $1, \dots, 11$, in which case a multiplicative model such as ARIMA$(0,0,1)(0,0,1)_{12}$ (Example 5.4.2) is usually tried first. ■

Exercise 5.4 (ACF of ARIMA$(1,0,1)(0,0,1)_s$)

The book asks for the autocovariance functions of four seasonal models; parts (1), (2) and (4) are worked out in Examples 5.4.3 (the process Y_t there), 5.4.2 and 5.4.3. Part (3): let $s \geq 2$, $|\phi| < 1$ and

$$(1 - \phi B)X_t = (1 + \theta B)(1 + \Theta B^s)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2).$$

Find the autocovariance function of the stationary solution.

Solution.

Step 1: reduction to ARMA(1,1). Let Y_t be the causal stationary solution of $(1-\phi B)Y_t = (1+\theta B)\epsilon_t$ and put $X_t = Y_t + \Theta Y_{t-s}$. Since the filters commute,

$$(1 - \phi B)X_t = (1 - \phi B)Y_t + \Theta(1 - \phi B)Y_{t-s} = (1 + \theta B)\epsilon_t + \Theta(1 + \theta B)\epsilon_{t-s} = (1 + \theta B)(1 + \Theta B^s)\epsilon_t.$$

$\{X_t\}$ is stationary as a finite linear filter of a stationary series, and by Proposition 2.4.A (an ARMA$(1, s+1)$ with causal AR part) it is the unique stationary solution.

Step 2: ACVF of Y. By Example 2.4.3, with $c = \gamma_Y(1)$,

$$\gamma_Y(0) = \frac{\sigma^2(1 + 2\phi\theta + \theta^2)}{1 - \phi^2}, \qquad c = \frac{\sigma^2(\phi + \theta)(1 + \phi\theta)}{1 - \phi^2}, \qquad \gamma_Y(h) = c\,\phi^{|h|-1} \ (h \neq 0).$$

Step 3: general formula. By bilinearity of the covariance and stationarity of Y,

$$\begin{aligned} \gamma_X(h) &= \text{Cov}(Y_{t+h} + \Theta Y_{t+h-s},\, Y_t + \Theta Y_{t-s}) \\ &= (1 + \Theta^2)\gamma_Y(h) + \Theta\{\gamma_Y(h - s) + \gamma_Y(h + s)\}. \end{aligned}$$

Step 4: explicit values. Inserting Step 2 for $h \geq 0$ (and $\gamma_X(-h) = \gamma_X(h)$):

$$\begin{aligned} \gamma_X(0) &= (1 + \Theta^2)\gamma_Y(0) + 2\Theta c\,\phi^{s-1}, \\ \gamma_X(h) &= c\{(1 + \Theta^2)\phi^{h-1} + \Theta(\phi^{s-h-1} + \phi^{s+h-1})\}, \quad 1 \leq h \leq s - 1, \\ \gamma_X(s) &= (1 + \Theta^2)c\,\phi^{s-1} + \Theta\{\gamma_Y(0) + c\,\phi^{2s-1}\}, \\ \gamma_X(h) &= c\{(1 + \Theta^2)\phi^{h-1} + \Theta(\phi^{h-s-1} + \phi^{h+s-1})\}, \quad h \geq s + 1. \end{aligned}$$

In the last line every exponent increases by one when h does, so $\gamma_X(h) = \phi\,\gamma_X(h-1)$ for $h \geq s+2$: beyond the seasonal lag the ACF decays geometrically as for the AR(1) factor, while the term $\Theta\gamma_Y(0)$ produces the seasonal spike at $h = s$. The ACF is $\rho_X(h) = \gamma_X(h)/\gamma_X(0)$. ■

Chapter 6. Conditional Heteroscedasticity Models

6.1 Introduction

Motivation. Returns on stocks, interest rates and exchange rates are nearly uncorrelated, yet their size is predictable: large moves cluster together. This chapter models a conditional variance that changes with the past, so that the *volatility* itself becomes the object of inference.

Volatility clustering means that sharp rises and falls come in bursts. Statistically, volatility is a variance: a series is volatile when the conditional variance of the next value, given the present, depends strongly on the current state. Classical models cannot express this. For an AR(1) model $y_t = \phi y_{t-1} + \epsilon_t$ with $\{\epsilon_t\} \sim IID(0, \sigma^2)$,

$$E(y_t \mid y_{t-1}) = \phi y_{t-1}, \qquad \mathrm{Var}(y_t \mid y_{t-1}) = \sigma^2,$$

so the conditional mean moves but the conditional variance is constant. If instead $y_t = 0.1 y_{t-1} + \epsilon_t$ with $\epsilon_t \mid \epsilon_{t-1} \sim N(0,\, 1 + 0.5\epsilon_{t-1}^2)$, a large shock raises the variance of the next one, and the path shows quiet stretches interrupted by turbulent ones (Figure 4).

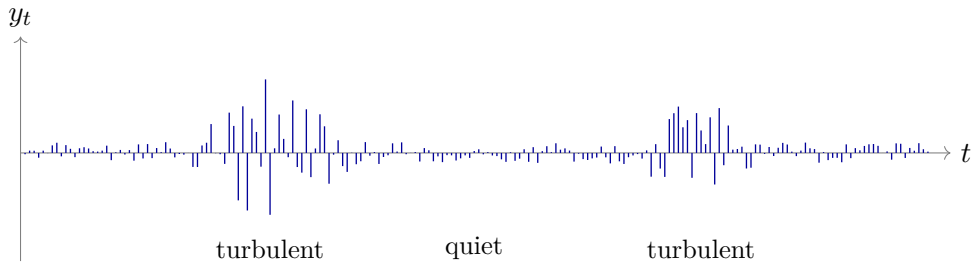


Figure 4: Volatility clustering: the level is unpredictable, but large moves come in bursts.

6.2 ARCH models

Motivation. Engle's ARCH model lets the conditional variance be a function of past squared values. Its simplest version already produces stationary, uncorrelated, heavy-tailed data, and its squares follow an AR model.

Write $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots)$ for the information up to time $t-1$ (the book's I_{t-1}). We wish to model

$$y_t \mid \mathcal{F}_{t-1} \sim N(0, \sigma_t^2), \qquad \sigma_t^2 = \sigma_t^2(y_{t-1}, y_{t-2}, \dots), \tag{6.1}$$

a conditional variance that is itself autoregressive in nature; often only finitely many lags $y_{t-1}, \dots, y_{t-q}$ enter.

6.2.1 ARCH(1)

Let $\{\xi_t\} \sim IID\, N(0,1)$ and consider

$$y_t = \sigma_t \xi_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2, \qquad \alpha_0 > 0,\ 0 \le \alpha_1 < 1, \tag{6.2}$$

where ξ_t is independent of σ_t. Since σ_t is determined by y_{t-1} alone,

$$P(y_t \le x \mid y_{t-1} = y) = P(\xi_t \le x/\sqrt{\alpha_0 + \alpha_1 y^2}) = \Phi(x/\sqrt{\alpha_0 + \alpha_1 y^2}), \tag{6.3}$$

with Φ the standard normal distribution function, so $y_t \mid y_{t-1}, y_{t-2}, \cdots \sim N(0, \sigma_t^2)$ and (6.2) is a concrete instance of (6.1): the *ARCH(1) model*. In practice ξ_t is often only assumed $IID(0,1)$.

Iterating $y_t^2 = \sigma_t^2 \xi_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2$,

$$y_t^2 = (\alpha_0 + \alpha_1(\alpha_0 + \alpha_1 y_{t-2}^2)\xi_{t-1}^2)\xi_t^2 = \cdots = \alpha_0 \sum_{j=0}^{n} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2 + \alpha_1^{n+1} \xi_t^2 \cdots \xi_{t-n}^2 y_{t-n-1}^2, \tag{6.4}$$

and since $\alpha_1^{n+1} \to 0$ this suggests

$$y_t^2 = \alpha_0 \sum_{j=0}^{\infty} \alpha_1^j \xi_t^2 \cdots \xi_{t-j}^2, \qquad \sigma_t^2 = \alpha_0 + \alpha_0 \alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2, \tag{6.5}$$

that is, the candidate solution

$$y_t = \xi_t \Big(\alpha_0 + \alpha_0 \alpha_1 \sum_{j=0}^{\infty} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2 \Big)^{1/2}. \tag{6.6}$$

Being a nonlinear function of $\xi_t, \xi_{t-1}, \ldots$, it cannot be captured by a linear ARMA structure.

Proposition (Properties of ARCH(1), [Supplement])

Let $\alpha_0 > 0$, $0 \le \alpha_1 < 1$ and $\{\xi_t\} \sim IID(0,1)$.
(1) The series in (6.6) converges a.s.; $\{y_t\}$ is a strictly stationary solution of (6.2), σ_t^2 is a function of $\xi_{t-1}, \xi_{t-2}, \ldots$, and (σ_t^2, y_t) is independent of $\xi_{t+1}, \xi_{t+2}, \ldots$.
(2) $E(y_t \mid \mathcal{F}_{t-1}) = 0$, $E(y_t^2 \mid \mathcal{F}_{t-1}) = \mathrm{Var}(y_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$, and

$$E y_t^2 = E\sigma_t^2 = \frac{\alpha_0}{1 - \alpha_1}; \tag{6.7}$$

$\{y_t\}$ is a stationary white noise.
(3) If $\xi_t \sim N(0,1)$, then $E y_t^4 < \infty$ if and only if $3\alpha_1^2 < 1$, and then the kurtosis is

$$\frac{E y_t^4}{(E y_t^2)^2} = 3 \frac{1 - \alpha_1^2}{1 - 3\alpha_1^2} > 3 \quad (\alpha_1 > 0). \tag{6.8}$$

(4) If $E y_t^4 < \infty$, then $\nu_t = y_t^2 - \sigma_t^2$ is a white noise and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t, \tag{6.9}$$

so $\{y_t^2\}$ is an AR(1) process.

Proof.

Step 1. Put $h_t = \alpha_0 + \alpha_0 \alpha_1 \sum_{j \ge 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-j-1}^2$. The terms are nonnegative, and by independence $E(\xi_{t-1}^2 \cdots \xi_{t-j-1}^2) = 1$, so $E h_t = \alpha_0 \sum_{j \ge 0} \alpha_1^j = \alpha_0/(1-\alpha_1) < \infty$; hence the series converges a.s. Moreover

$$\alpha_0 + \alpha_1 \xi_{t-1}^2 h_{t-1} = \alpha_0 + \alpha_0 \alpha_1 \xi_{t-1}^2 + \alpha_0 \sum_{j \ge 0} \alpha_1^{j+2} \xi_{t-1}^2 \xi_{t-2}^2 \cdots \xi_{t-j-2}^2 = h_t,$$

so with $\sigma_t^2 = h_t$ and $y_t = \sigma_t \xi_t$ we get $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$. Since $(y_t)_t = G(\ldots, \xi_{t-1}, \xi_t)$ for one fixed measurable map G applied to the shifted i.i.d. sequence, $\{y_t\}$ is strictly stationary. The independence claims are immediate from the construction.

Step 2. $\mathcal{F}_{t-1} \subset \sigma(\xi_{t-1}, \xi_{t-2}, \ldots)$, which is independent of ξ_t, while σ_t is $\mathcal{F}_{t-1}$-measurable (it equals $(\alpha_0 + \alpha_1 y_{t-1}^2)^{1/2}$). Hence $E(y_t \mid \mathcal{F}_{t-1}) = \sigma_t E\xi_t = 0$ and $E(y_t^2 \mid \mathcal{F}_{t-1}) = \sigma_t^2 E\xi_t^2 = \sigma_t^2$. Taking expectations, $E y_t^2 = E\sigma_t^2 = E h_t = \alpha_0/(1-\alpha_1)$. For $h \ge 1$, $E(y_t y_{t+h}) = E\big(y_t\, E(y_{t+h} \mid \mathcal{F}_{t+h-1})\big) = 0$, so $\{y_t\}$ is white noise with mean 0.

Step 3. Write $y_t^2 = \alpha_0 \sum_{j\ge 0} \alpha_1^j P_j$ with $P_j = \xi_t^2 \cdots \xi_{t-j}^2$. Since $E\xi^4 = 3$, $\|P_j\|_2 = (EP_j^2)^{1/2} = 3^{(j+1)/2}$. If $\sqrt{3}\,\alpha_1 < 1$, Minkowski's inequality gives $\|y_t^2\|_2 \le \alpha_0 \sum_j \alpha_1^j 3^{(j+1)/2} < \infty$, so $Ey_t^4 < \infty$. Conversely, suppose $Ey_t^4 < \infty$; then $m_4 := E\sigma_t^4$ is finite and, by stationarity, does not depend on t. Using $Ey_{t-1}^4 = 3E\sigma_{t-1}^4$,

$$m_4 = E(\alpha_0 + \alpha_1 y_{t-1}^2)^2 = \alpha_0^2 + 2\alpha_0\alpha_1 \frac{\alpha_0}{1-\alpha_1} + 3\alpha_1^2 m_4, \quad \text{so} \quad (1-3\alpha_1^2)m_4 = \alpha_0^2 \frac{1+\alpha_1}{1-\alpha_1} > 0,$$

forcing $3\alpha_1^2 < 1$. Then $Ey_t^4 = 3m_4$ and

$$\frac{Ey_t^4}{(Ey_t^2)^2} = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)} \cdot \frac{(1-\alpha_1)^2}{\alpha_0^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2},$$

which exceeds 3 when $\alpha_1 > 0$ because $1-\alpha_1^2 > 1-3\alpha_1^2$.

Step 4. $\nu_t = \sigma_t^2(\xi_t^2 - 1)$ is $\mathcal{F}_t$-measurable with $E(\nu_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 E(\xi_t^2 - 1) = 0$ and $E\nu_t^2 = E\sigma_t^4\, E(\xi_t^2-1)^2 < \infty$. A square-integrable martingale difference has mean zero, constant variance (by stationarity) and is uncorrelated: for $h \ge 1$, $E(\nu_t\nu_{t+h}) = E(\nu_t E(\nu_{t+h} \mid \mathcal{F}_{t+h-1})) = 0$. Finally $y_t^2 = \sigma_t^2 + \nu_t = \alpha_0 + \alpha_1 y_{t-1}^2 + \nu_t$. ∎

Parts (1)–(2) do not use normality. Part (3) shows that the marginal law of an ARCH(1) series is *leptokurtic*: whatever the distribution of ξ_t, y_t has heavier tails than the normal. For example, 10,000 draws from $\sigma_t^2 = 0.2 + 0.8y_{t-1}^2$ produce a histogram far more peaked and heavy-tailed than the fitted normal density.

Definition 6.2.1 (ARCH(q) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2, \tag{6.10}$$

with $\alpha_0 > 0$ and $\alpha_i \ge 0$ $(1 \le i \le q)$, then $\{y_t\}$ follows an *ARCH(q) model*. If moreover $1 - \alpha_1 z - \cdots - \alpha_q z^q \ne 0$ for all $|z| \le 1$ (equivalently $\sum_{i=1}^q \alpha_i < 1$), then $\{y_t\}$ is strictly stationary and

$$\operatorname{Var}(y_t) = \frac{\alpha_0}{1 - \sum_{i=1}^q \alpha_i}. \tag{6.11}$$

Proof of the stationarity statement. [Supplement] Let $S = \sum_{i=1}^q \alpha_i$ and $A(z) = \sum_i \alpha_i z^i$.

Step 1: the root condition is $S < 1$. If $S < 1$ then $|A(z)| \le S < 1$ for $|z| \le 1$, so $1 - A(z) \ne 0$. If $S \ge 1$, the continuous function A satisfies $A(0) = 0$ and $A(1) = S \ge 1$, so $A(z) = 1$ for some $z \in (0, 1]$.

Step 2: construction. Set $h_t^{(0)} = 0$ and $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^q \alpha_i \xi_{t-i}^2 h_{t-i}^{(m)}$. By induction each $h_t^{(m)} = g_m(\xi_{t-1}, \xi_{t-2}, \dots)$ for a fixed measurable $g_m \ge 0$, and $h_t^{(m)}$ is nondecreasing in m ($h^{(1)} = \alpha_0 \ge h^{(0)}$, and the recursion is monotone because its coefficients are nonnegative). Since $h_{t-i}^{(m)}$ depends only on $\xi_{t-i-1}, \dots$, it is independent of ξ_{t-i}, so $Eh_t^{(m+1)} = \alpha_0 + S\,Eh_t^{(m)}$ and $Eh_t^{(m)} = \alpha_0(1 + S + \cdots + S^{m-1}) \le \alpha_0/(1-S)$.

Step 3: limit. Let $\sigma_t^2 = \lim_m h_t^{(m)}$. By monotone convergence $E\sigma_t^2 = \alpha_0/(1-S) < \infty$, so $\sigma_t^2 < \infty$ a.s., and letting $m \to \infty$ in the recursion gives $\sigma_t^2 = \alpha_0 + \sum_i \alpha_i \xi_{t-i}^2 \sigma_{t-i}^2$. With $y_t = \sigma_t \xi_t$ this is (6.10). As in Proposition , $\{y_t\}$ is a fixed function of the shifted i.i.d. sequence, hence strictly stationary, and $\operatorname{Var}(y_t) = E\sigma_t^2 E\xi_t^2 = \alpha_0/(1-S)$, which is (6.11).

Step 4: uniqueness. Let $\tilde\sigma_t^2$ be another strictly stationary solution that is a function of $\xi_{t-1}, \xi_{t-2}, \dots$ with $E\tilde\sigma_t^2 < \infty$. Then $D_t = \sigma_t^2 - \tilde\sigma_t^2$ satisfies $D_t = \sum_i \alpha_i \xi_{t-i}^2 D_{t-i}$, and by independence of ξ_{t-i} and D_{t-i}, $E|D_t| \le \sum_i \alpha_i E|D_{t-i}| = S\,E|D_0|$. Stationarity gives $(1-S)E|D_0| \le 0$, so $D_t = 0$ a.s. ∎

As for ARCH(1), $\nu_t = y_t^2 - \sigma_t^2$ is a martingale difference (white noise when $Ey_t^4 < \infty$) and

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \cdots + \alpha_q y_{t-q}^2 + \nu_t,$$

so $\{y_t^2\}$ is AR(q). Choosing the order q of an ARCH model therefore reduces to choosing the order of an AR model for $\{y_t^2\}$, using the model-selection tools of Chapter 5.

6.3 GARCH models

Motivation. A large ARCH order q is needed to describe persistent volatility, and then estimation is poor and the constraints are hard to satisfy. Adding lagged conditional variances, as ARMA adds lagged noise to AR, gives the parsimonious GARCH family.

Definition 6.3.1 (GARCH(1,1) model)

If $\{y_t\}$ satisfies

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \alpha_0 > 0,\ \alpha_1, \beta_1 \geq 0, \quad (6.12)$$

then $\{y_t\}$ follows a *GARCH(1,1) model*. If $\alpha_1 + \beta_1 < 1$, $\{y_t\}$ is (second-order) stationary; and if

$$E \log(\beta_1 + \alpha_1 \xi_t^2) < 0,$$

then $\{(y_t, \sigma_t)\}$ is strictly stationary (Nelson, 1990; Bougerol and Picard, 1992).

Proof of the stationarity statements. [Supplement] Put $a(x) = \beta_1 + \alpha_1 x^2$, so $\sigma_t^2 = \alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2$ and, by iteration,

$$\sigma_t^2 = \alpha_0 \Big[1 + \sum_{k=1}^{K} \prod_{i=1}^{k} a(\xi_{t-i})\Big] + \prod_{i=1}^{K+1} a(\xi_{t-i})\, \sigma_{t-K-1}^2. \quad (6.13)$$

Step 1: strict stationarity. Let $\gamma = E \log a(\xi_t) < 0$ (well defined, since $E \log a \leq \log Ea < \infty$; if $\gamma = -\infty$, replace $\log a$ by $\max(\log a, -M)$ with M large, which keeps the mean negative and only enlarges the products below). By the strong law of large numbers, $\frac{1}{k}\sum_{i=1}^{k} \log a(\xi_{t-i}) \to \gamma$ a.s., so $\big(\prod_{i=1}^{k} a(\xi_{t-i})\big)^{1/k} \to e^{\gamma} < 1$, and by the root test

$$\sigma_t^2 := \alpha_0 \Big[1 + \sum_{k=1}^{\infty} \prod_{i=1}^{k} a(\xi_{t-i})\Big]$$

converges a.s. Substituting shows $\alpha_0 + a(\xi_{t-1})\sigma_{t-1}^2 = \sigma_t^2$, so $y_t = \sigma_t \xi_t$ solves (6.12); (y_t, σ_t) is a fixed function of the shifted i.i.d. sequence, hence strictly stationary. It is the only strictly stationary solution with σ_t^2 a function of the past ξ's: for another one, $\tilde{\sigma}_t^2$, (6.13) gives $\tilde{\sigma}_t^2 - \sigma_t^2 = \prod_{i=1}^{K} a(\xi_{t-i})\,(\tilde{\sigma}_{t-K}^2 - \sigma_{t-K}^2)$; the product tends to 0 a.s., while the second factor has a fixed distribution (stationarity) and is therefore tight, so the right side tends to 0 in probability. The left side does not depend on K, hence it is 0.

Step 2: second-order stationarity. If $\alpha_1 + \beta_1 < 1$, Jensen's inequality gives $E \log a(\xi_t) \leq \log Ea(\xi_t) = \log(\alpha_1 + \beta_1) < 0$, so Step 1 applies. By Tonelli and independence, $E\prod_{i=1}^{k} a(\xi_{t-i}) = (\alpha_1 + \beta_1)^k$, hence

$$Ey_t^2 = E\sigma_t^2 = \alpha_0 \sum_{k \geq 0} (\alpha_1 + \beta_1)^k = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} < \infty.$$

Exactly as in Proposition (2), $E(y_t \mid \mathcal{F}_{t-1}) = 0$, so $\{y_t\}$ is a stationary white noise. ∎

The two conditions differ. When $\alpha_1+\beta_1=1$ and ξ_t^2 is nondegenerate, strict Jensen gives $E\log(\beta_1+\alpha_1\xi_t^2)<\log 1=0$: the series is strictly stationary although $Ey_t^2=\infty$.

The parameters of a GARCH(1,1) model fitted to daily returns (for instance $r_t=100\log(P_{t+1}/P_t)$ of a stock index) typically have $\alpha_1+\beta_1$ just below 1. The case $\alpha_1+\beta_1=1$ is the *IGARCH model* (integrated GARCH): a shock at one time keeps influencing all later variances. When $\alpha_1+\beta_1$ is 1 or close to 1 the conditional variance is called *persistent*, for the following reason.

Proposition (Variance forecasts, [Supplement])

In the GARCH(1,1) model let $\phi=\alpha_1+\beta_1$. For $k\geq 0$,

$$E(\sigma_{t+k}^2\mid\mathcal{F}_{t-1})=\alpha_0\sum_{j=0}^{k-1}\phi^j+\phi^k\sigma_t^2, \tag{6.14}$$

which has the form $c+\phi^k\sigma_t^2$. If $\phi\approx 1$, the effect of today's variance on the forecast barely decays as k grows.

Proof. For $s\geq t$, since σ_s^2 is $\mathcal{F}_{s-1}$-measurable and $E(y_s^2\mid\mathcal{F}_{s-1})=\sigma_s^2$,

$$E(\sigma_{s+1}^2\mid\mathcal{F}_{s-1})=\alpha_0+\alpha_1E(y_s^2\mid\mathcal{F}_{s-1})+\beta_1\sigma_s^2=\alpha_0+\phi\sigma_s^2.$$

For $k=0$ the claim is trivial. If it holds for k, the tower property gives $E(\sigma_{t+k+1}^2\mid\mathcal{F}_{t-1})=E(E(\sigma_{t+k+1}^2\mid\mathcal{F}_{t+k-1})\mid\mathcal{F}_{t-1})=\alpha_0+\phi E(\sigma_{t+k}^2\mid\mathcal{F}_{t-1})=\alpha_0\sum_{j=0}^{k}\phi^j+\phi^{k+1}\sigma_t^2$. ∎

6.3.1 Asymmetric models

Returns often react more strongly to bad news (negative returns) than to good news.

Definition (Threshold GARCH(1,1))

Let $x^+=\max\{0,x\}$ and $x^-=\max\{-x,0\}$. If $y_t=\sigma_t\xi_t$ with $\{\xi_t\}\sim IID(0,1)$ and

$$\sigma_t^2=\alpha_0+\alpha_{11}(y_{t-1}^+)^2+\alpha_{12}(y_{t-1}^-)^2+\beta_1\sigma_{t-1}^2,\qquad \alpha_0>0,\ \alpha_{11},\alpha_{12},\beta_1\geq 0,$$

then $\{y_t\}$ follows a *threshold* (or *asymmetric*) *GARCH(1,1) model*. Equivalently, $\sigma_t^2=\alpha_0+\alpha_1y_{t-1}^2+\gamma_1(y_{t-1}^-)^2+\beta_1\sigma_{t-1}^2$ with $\alpha_1=\alpha_{11}$, $\gamma_1=\alpha_{12}-\alpha_{11}$.

With $\alpha_{12}>\alpha_{11}$ a negative return raises the next variance more than a positive one of the same size, which is the typical pattern in stock returns.

6.3.2 Gaussian quasi-maximum likelihood

Assume for the moment $\xi_t\sim N(0,1)$, fix initial values y_0 and σ_0^2 (commonly $y_0^2=y_1^2$, $\sigma_0^2=y_1^2$), and compute recursively from $\theta=(\alpha_0,\alpha_1,\beta_1)^T$

$$\sigma_t^{2*}(\theta)=\alpha_0+\alpha_1y_{t-1}^2+\beta_1\sigma_{t-1}^{2*}(\theta),\qquad t=1,\dots,n.$$

The conditional likelihood is $L(\theta)=\prod_{t=1}^n f(y_t\mid y_{t-1},\dots)$ with

$$f(y_t\mid y_{t-1},\dots)=(2\pi\sigma_t^{2*})^{-1/2}\exp\{-y_t^2/(2\sigma_t^{2*})\},$$

and the *Gaussian MLE* $\hat\theta$ maximizes

$$l(\theta)=\log L(\theta)=-\frac{n}{2}\log 2\pi-\frac{1}{2}\sum_{t=1}^n\Big\{\log\sigma_t^{2*}(\theta)+\frac{y_t^2}{\sigma_t^{2*}(\theta)}\Big\}. \tag{6.15}$$

The same estimator is used when ξ_t is not normal; it is then called the *quasi-MLE* (QMLE), and it remains consistent and asymptotically normal.

Proposition (Asymptotic normality of the QMLE, [Supplement])

Let $\{y_t\}$ be a strictly stationary GARCH(1,1) series with true parameter θ_0 in the interior of a compact parameter set, $\kappa_\xi = E\xi_t^4 < \infty$, and suppose $\hat\theta \to \theta_0$ in probability. Let $D_t = \sigma_t^{-2}\,\partial\sigma_t^2/\partial\theta$ at θ_0 and

$$J = E(D_t D_t^T) = -2\,E\Big[\frac{\partial^2 \log f(y_t \mid y_{t-1},\dots)}{\partial\theta\,\partial\theta^T}\Big],$$

assumed positive definite. Then

$$\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0,\Gamma), \qquad \Gamma = (\kappa_\xi - 1)J^{-1}$$

(Francq and Zakoian, 2004). For normal ξ_t, $\kappa_\xi = 3$ and $\Gamma = 2J^{-1}$.

Proof. [Supplement] We argue at the level of the standard M-estimation scheme, taking for granted two technical facts proved by Francq and Zakoian: the initial values y_0, σ_0^2 have an asymptotically negligible effect (because $\sigma_t^{2*} - \sigma_t^2 = O(\beta_1^t)$), and D_t and the second derivatives of $\log\sigma_t^2$ have finite moments uniformly in a neighbourhood of θ_0. Write $l_t(\theta) = -\tfrac12\{\log\sigma_t^2 + y_t^2/\sigma_t^2\}$.

Step 1: score. $\partial l_t/\partial\theta = \tfrac12(y_t^2/\sigma_t^2 - 1)\sigma_t^{-2}\partial\sigma_t^2/\partial\theta$, which at θ_0 equals $s_t = \tfrac12(\xi_t^2 - 1)D_t$. Since D_t is $\mathcal{F}_{t-1}$-measurable and ξ_t is independent of $\mathcal{F}_{t-1}$, $\{s_t\}$ is a stationary ergodic martingale difference with $E(s_t s_t^T) = \frac{\kappa_\xi - 1}{4}J$. The martingale central limit theorem (Billingsley) gives $n^{-1/2}\sum_{t=1}^n s_t \xrightarrow{d} N\big(0, \frac{\kappa_\xi-1}{4}J\big)$.

Step 2: Hessian. Differentiating again,

$$\frac{\partial^2 l_t}{\partial\theta\,\partial\theta^T} = \frac12\Big(\frac{y_t^2}{\sigma_t^2} - 1\Big)\frac{\partial}{\partial\theta^T}\Big(\sigma_t^{-2}\frac{\partial\sigma_t^2}{\partial\theta}\Big) - \frac12\,\frac{y_t^2}{\sigma_t^2}\,\sigma_t^{-4}\frac{\partial\sigma_t^2}{\partial\theta}\frac{\partial\sigma_t^2}{\partial\theta^T}.$$

At θ_0, $y_t^2/\sigma_t^2 = \xi_t^2$ has conditional mean 1, so the expectation equals $-\tfrac12 J$. By the ergodic theorem and the uniform moment bound, $\frac1n\sum_t \partial^2 l_t(\bar\theta_n)/\partial\theta\partial\theta^T \to -\tfrac12 J$ in probability for every $\bar\theta_n \to \theta_0$.

Step 3: expansion. Since $\hat\theta$ is an interior maximizer for large n, a mean-value expansion of each coordinate of the score gives $0 = n^{-1/2}\sum_t s_t + \big(\frac1n\sum_t \partial^2 l_t(\bar\theta_n)\big)\sqrt{n}(\hat\theta - \theta_0)$ with $\bar\theta_n$ between $\hat\theta$ and θ_0. Hence $\sqrt{n}(\hat\theta - \theta_0) = 2J^{-1}n^{-1/2}\sum_t s_t + o_p(1)$, whose limit is normal with covariance $2J^{-1}\cdot\frac{\kappa_\xi-1}{4}J\cdot 2J^{-1} = (\kappa_\xi - 1)J^{-1}$. ∎

Example 6.3.2 (Gaussian MLE for ARCH(1))

Fix y_0 and let $y_1,\dots,y_n$ follow ARCH(1). The conditional log-likelihood is

$$l(\alpha_0,\alpha_1) = -\frac{n}{2}\log 2\pi - \frac12\sum_{t=1}^n\Big\{\log(\alpha_0 + \alpha_1 y_{t-1}^2) + \frac{y_t^2}{\alpha_0 + \alpha_1 y_{t-1}^2}\Big\},$$

and its maximizer is the MLE. The likelihood equations have no closed-form solution, so the estimate is computed numerically.

6.3.3 GARCH(p, q)

Definition 6.3.3 (GARCH(p, q) model)

If $\{y_t\}$ satisfies

$$y_t = \xi_t \sigma_t, \quad \{\xi_t\} \sim IID(0,1), \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i y_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2, \quad \alpha_0 > 0,\ \alpha_i, \beta_j \geq 0, \tag{6.16}$$

then $\{y_t\}$ follows a *GARCH(p, q) model.*

Proposition (Stationarity and ARMA form of GARCH(p, q), [Supplement])

Let $S = \sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j$.
(1) If $S < 1$, (6.16) has a unique strictly stationary solution with σ_t^2 a function of $\xi_{t-1}, \xi_{t-2}, \ldots$ and $E\sigma_t^2 < \infty$; it satisfies $Ey_t^2 = E\sigma_t^2 = \alpha_0/(1-S)$, and $\{y_t\}$ and $\{\sigma_t^2\}$ are stationary (Bollerslev, 1986). Conversely, a solution with $E\sigma_t^2$ finite and constant in t forces $S < 1$.
(2) With $\nu_t = y_t^2 - \sigma_t^2$, a martingale difference (white noise if $Ey_t^4 < \infty$),

$$\Big(1 - \sum_{i=1}^{q} \alpha_i B^i - \sum_{j=1}^{p} \beta_j B^j\Big) y_t^2 = \alpha_0 + \nu_t - \sum_{j=1}^{p} \beta_j \nu_{t-j}, \tag{6.17}$$

so $\{y_t^2\}$ follows an ARMA$(\max\{p, q\}, p)$ model.

Proof.

Step 1: construction. Set $h_t^{(0)} = 0$ and $h_t^{(m+1)} = \alpha_0 + \sum_{i=1}^{q} \alpha_i \xi_{t-i}^2 h_{t-i}^{(m)} + \sum_{j=1}^{p} \beta_j h_{t-j}^{(m)}$. As in the proof for Definition 6.2.1, $h_t^{(m)} = g_m(\xi_{t-1}, \xi_{t-2}, \ldots) \geq 0$ is nondecreasing in m, ξ_{t-i} is independent of $h_{t-i}^{(m)}$, and $Eh^{(m+1)} = \alpha_0 + S\, Eh^{(m)}$, so $Eh^{(m)} \uparrow \alpha_0/(1-S)$. The monotone limit σ_t^2 has $E\sigma_t^2 = \alpha_0/(1-S)$, satisfies (6.16) with $y_t = \sigma_t \xi_t$, and is strictly stationary as a fixed function of the shifted i.i.d. sequence; $Ey_t^2 = E\sigma_t^2$ and $E(y_t \mid \mathcal{F}_{t-1}) = 0$ make $\{y_t\}$ white noise.

Step 2: uniqueness. If $\tilde{\sigma}_t^2$ is another such solution, $D_t = \sigma_t^2 - \tilde{\sigma}_t^2$ satisfies $D_t = \sum_i \alpha_i \xi_{t-i}^2 D_{t-i} + \sum_j \beta_j D_{t-j}$, so $E|D_0| \leq S\, E|D_0|$ and $D = 0$.

Step 3: necessity. If $E\sigma_t^2 = m < \infty$ for all t, taking expectations in (6.16) with $Ey_{t-i}^2 = E\sigma_{t-i}^2$ gives $m = \alpha_0 + Sm$; as $\alpha_0 > 0$ and $m > 0$, $S < 1$.

Step 4: ARMA form. $E(\nu_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 E(\xi_t^2 - 1) = 0$; white-noise properties follow as in Proposition (4). Substituting $\sigma_{t-k}^2 = y_{t-k}^2 - \nu_{t-k}$ $(k = 0, \ldots, p)$ into (6.16),

$$y_t^2 - \nu_t = \alpha_0 + \sum_{i=1}^{q} \alpha_i y_{t-i}^2 + \sum_{j=1}^{p} \beta_j (y_{t-j}^2 - \nu_{t-j}),$$

which rearranges to (6.17). The AR polynomial has degree $\max\{p, q\}$ and the MA polynomial degree p. ■

Hence p, q can be chosen as in Chapter 5: maximize the Gaussian log-likelihood for each candidate order and minimize AIC or BIC. Since most financial series are well described by GARCH(1,1), higher orders are rarely needed.

Remark 6.3.4 (threshold AR(1)–GARCH(1,1))

In financial applications the following combination of a conditional mean and an asymmetric conditional variance is especially useful:

$$y_t = \phi y_{t-1} + \epsilon_t, \qquad \epsilon_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim IID(0,1),$$

$$\sigma_t^2 = \alpha_0 + \alpha_{11}(\epsilon_{t-1}^+)^2 + \alpha_{12}(\epsilon_{t-1}^-)^2 + \beta_1 \sigma_{t-1}^2, \qquad \alpha_0 > 0,\ \alpha_{11}, \alpha_{12}, \beta_1 \geq 0.$$

Heavy tails of ξ_t can also be built in directly, for example with a standardized t-distribution (*t-GARCH*); the Gaussian QMLE is then less efficient. Deciding whether ξ_t is normal is therefore of practical importance (Section 6.4). Another widely used variant works on the log scale.

Definition 6.3.5 (EGARCH(1,1) model)

If $\{y_t\}$ satisfies $y_t = \sigma_t \xi_t$, $\{\xi_t\} \sim IID(0,1)$, and

$$\log \sigma_t^2 = \alpha_0 + \theta \frac{y_{t-1}}{\sigma_{t-1}} + \gamma\Big\{\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big| - E\Big|\frac{y_{t-1}}{\sigma_{t-1}}\Big|\Big\} + \beta \log \sigma_{t-1}^2,$$

where $\alpha_0, \theta, \gamma \in \mathbb{R}$ and $|\beta| < 1$, then $\{y_t\}$ follows an *EGARCH(1,1) model* (exponential GARCH).

Because the recursion is for $\log \sigma_t^2$, no sign restrictions on the parameters are needed, which removes a drawback of GARCH. When $\theta < 0$, bad news ($\xi_{t-1} < 0$) raises volatility more than good news. Stochastic volatility models, in which $\log \sigma_t^2$ carries its own noise, are a further alternative not treated here.

6.4 Testing normality

Motivation. Errors in ARCH/GARCH models are frequently far from normal, and then the Gaussian MLE loses much of its efficiency. The Jarque–Bera test detects non-normality through skewness and kurtosis.

Given an i.i.d. sample $\xi_1, \dots, \xi_n$, to test H_0: $\xi_1, \dots, \xi_n$ *are normal*, the *Jarque–Bera statistic* is

$$T_{JB} = n\Big(\frac{S_n^2}{6} + \frac{(K_n - 3)^2}{24}\Big), \qquad S_n = \frac{m_3}{m_2^{3/2}}, \quad K_n = \frac{m_4}{m_2^2}, \quad m_k = \frac{1}{n}\sum_{t=1}^{n}(\xi_t - \bar{\xi})^k, \tag{6.18}$$

the sample skewness S_n and sample kurtosis K_n. Under normality these are near 0 and 3, so T_{JB} is small.

Proposition (Null distribution of the Jarque–Bera statistic, [Supplement])

Under H_0, $T_{JB} \xrightarrow{d} \chi^2(2)$.

Proof. All quantities are location-scale invariant, so take $\xi_t \sim N(0,1)$. Let $\hat{\mu}_k = \frac{1}{n}\sum \xi_t^k$; recall $E\xi^2 = 1$, $E\xi^4 = 3$, $E\xi^6 = 15$, $E\xi^8 = 105$, and odd moments vanish.

Step 1: reduction. $\sqrt{n}\,\bar{\xi} = O_p(1)$ and $\bar{\xi} \to 0$, $\hat{\mu}_2 \to 1$, $\hat{\mu}_3 \to 0$ a.s. Expanding,

$$m_2 = \hat{\mu}_2 - \bar{\xi}^2, \quad m_3 = \hat{\mu}_3 - 3\bar{\xi}\hat{\mu}_2 + 2\bar{\xi}^3, \quad m_4 = \hat{\mu}_4 - 4\bar{\xi}\hat{\mu}_3 + 6\bar{\xi}^2\hat{\mu}_2 - 3\bar{\xi}^4.$$

Hence $\sqrt{n}\,m_3 = \sqrt{n}(\hat{\mu}_3 - 3\bar{\xi}) + o_p(1)$, $\sqrt{n}(m_4 - 3) = \sqrt{n}(\hat{\mu}_4 - 3) + o_p(1)$ and $\sqrt{n}(m_2 - 1) = \sqrt{n}(\hat{\mu}_2 - 1) + o_p(1)$.

Step 2: delta method. Since $m_2 \to 1$, $\sqrt{n}S_n = \sqrt{n}\, m_3 + o_p(1)$. The map $(x, y) \mapsto x/y^2$ has gradient $(1, -6)$ at $(3, 1)$, so $\sqrt{n}(K_n - 3) = \sqrt{n}(m_4 - 3) - 6\sqrt{n}(m_2 - 1) + o_p(1)$. Therefore

$$\sqrt{n}\begin{pmatrix} S_n \\ K_n - 3 \end{pmatrix} = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \begin{pmatrix} \xi_t^3 - 3\xi_t \\ \xi_t^4 - 6\xi_t^2 + 3 \end{pmatrix} + o_p(1).$$

Step 3: CLT. The summands are i.i.d. with mean 0 and $\mathrm{Var}(\xi^3 - 3\xi) = 15 - 18 + 9 = 6$, $\mathrm{Var}(\xi^4 - 6\xi^2) = 105 - 180 + 108 - 9 = 24$; their covariance is an odd moment, hence 0. By the multivariate CLT, $\sqrt{n}S_n$ and $\sqrt{n}(K_n - 3)$ are asymptotically independent $N(0, 6)$ and $N(0, 24)$, so $T_{JB} = (\sqrt{n}S_n)^2/6 + (\sqrt{n}(K_n - 3))^2/24 \xrightarrow{d} \chi^2(2)$. ∎

In a GARCH(1,1) model the errors are unobserved, so residuals are used. With the Gaussian MLE $\hat{\alpha}_0, \hat{\alpha}_1, \hat{\beta}_1$, write (6.12) as $(1 - \beta_1 B)\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$, i.e.

$$\sigma_t^2 = (1 - \beta_1 B)^{-1}\alpha_0 + \alpha_1(1 - \beta_1 B)^{-1} y_{t-1}^2 = \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{j=1}^{\infty} \beta_1^{j-1} y_{t-j}^2,$$

and truncate at the first observation:

$$\hat{\xi}_t = \frac{y_t}{\hat{\sigma}_t}, \qquad \hat{\sigma}_t^2 = \frac{\hat{\alpha}_0}{1 - \hat{\beta}_1} + \hat{\alpha}_1 \sum_{j=1}^{t-1} \hat{\beta}_1^{j-1} y_{t-j}^2, \qquad 1 \le t \le n. \tag{6.19}$$

(Alternatively $\hat{\sigma}_t^2$ may be computed by the recursion (6.12) started from $y_0 = \sigma_0 = y_1$; the test result is the same asymptotically.) The residual statistic $\hat{T}_{JB}$, obtained by replacing ξ_t with $\hat{\xi}_t$ in (6.18), has the same $\chi^2(2)$ limit, and normality is rejected at level α if $\hat{T}_{JB} \ge c_\alpha$, where $P(\chi^2(2) \ge c_\alpha) = \alpha$.

6.5 Change-point detection

Motivation. GARCH parameters of real series often shift over time, and ignoring a shift misrepresents the model, typically as a spurious IGARCH with $\alpha_1 + \beta_1 \approx 1$. The CUSUM test compares partial sums of squared residuals with their expected linear growth.

Given $y_1, \dots, y_n$, test

> H_0: $y_1, \dots, y_n$ follow a GARCH(1,1) model with no parameter change;
> H_1: $y_1, \dots, y_n$ follow GARCH(1,1) but the parameters change at some time.

Under H_0 compute the Gaussian MLE, the residuals (6.19), and the *CUSUM statistic* of squared residuals

$$\hat{T}_n = \max_{m \le k \le n} \frac{1}{\sqrt{n}\,\hat{\tau}} \Big| \sum_{t=1}^{k} \hat{\xi}_t^2 - \frac{k}{n} \sum_{t=1}^{n} \hat{\xi}_t^2 \Big|, \qquad \hat{\tau}^2 = \frac{1}{n} \sum_{t=1}^{n} \hat{\xi}_t^4 - \Big(\frac{1}{n} \sum_{t=1}^{n} \hat{\xi}_t^2 \Big)^2, \tag{6.20}$$

where $\hat{\tau}^2$ estimates $\tau^2 = \mathrm{Var}(\xi_1^2)$ and m is a small fixed integer (say between 5 and 10). Under H_0, $\hat{\xi}_t^2 \approx \xi_t^2$, so

$$\hat{T}_n \approx T_n = \max_{m \le k \le n} \frac{1}{\sqrt{n}\,\hat{\tau}} \Big| \sum_{t=1}^{k} \xi_t^2 - \frac{k}{n} \sum_{t=1}^{n} \xi_t^2 \Big|. \tag{6.21}$$

Since $\sum_{t \le k} \xi_t^2 \approx k$ and $\frac{k}{n} \sum_{t \le n} \xi_t^2 \approx k$, T_n is small under H_0. If the parameters change at k^* with k^*/n away from 0 and 1, then $\frac{1}{n} \big| \sum_{t \le k^*} \hat{\xi}_t^2 - \frac{k^*}{n} \sum_{t \le n} \hat{\xi}_t^2 \big|$ converges to a positive constant, so $\hat{T}_n \to \infty$; large values of $\hat{T}_n$ reject H_0.

Proposition (Limit of the CUSUM statistic, [Supplement])

Let ξ_t be i.i.d. with $E\xi_t^2 = 1$, $E\xi_t^4 < \infty$, $\tau^2 = \mathrm{Var}(\xi_t^2) > 0$, and let $\hat\tau \to \tau$ in probability. Then

$$T_n \xrightarrow{d} T = \sup_{0\le u\le 1} |B^0(u)|,$$

where $B^0(u) = W(u) - uW(1)$ is a Brownian bridge and W a standard Brownian motion on $[0,1]$.

Proof.

Step 1. By Donsker's functional CLT, $W_n(u) = \frac{1}{\tau\sqrt{n}}\sum_{t\le \lfloor nu\rfloor}(\xi_t^2 - 1)$ converges in distribution to W in $D[0,1]$.

Step 2. The centring cancels: $\sum_{t\le k}\xi_t^2 - \frac{k}{n}\sum_{t\le n}\xi_t^2 = \sum_{t\le k}(\xi_t^2-1) - \frac{k}{n}\sum_{t\le n}(\xi_t^2-1)$. Hence, with τ in place of $\hat\tau$, the statistic equals $\max_{m\le k\le n}|W_n(k/n) - \frac{k}{n}W_n(1)|$. As W_n is a step function, this differs from $\sup_{0\le u\le 1}|W_n(u) - uW_n(1)|$ by at most $|W_n(1)|/n + \max_{k<m}|W_n(k/n)| + m|W_n(1)|/n$, and $\max_{k<m}|W_n(k/n)| \le \frac{1}{\tau\sqrt{n}}\sum_{t<m}|\xi_t^2 - 1| \to 0$ because m is fixed. So the difference is $o_p(1)$.

Step 3. The map $x \mapsto \sup_u |x(u) - ux(1)|$ is continuous at every continuous x for the sup-norm, and W has continuous paths, so by the continuous mapping theorem $\sup_u|W_n(u) - uW_n(1)| \xrightarrow{d} \sup_u|W(u) - uW(1)| = T$. Finally $\tau/\hat\tau \to 1$ and Slutsky's lemma give $T_n \xrightarrow{d} T$. ∎

The law of T is the Kolmogorov distribution, $P(T \le x) = 1 - 2\sum_{k\ge1}(-1)^{k-1}e^{-2k^2x^2}$, and $P(T \ge 1.358) = 0.05$. Thus H_0 is rejected at the 5% level when $\hat T_n \ge 1.358$, and the change point is estimated by

$$\hat k = \arg\max_{1\le k\le n}\hat T_n(k), \qquad \hat T_n(k) = \frac{1}{\sqrt{n}\,\hat\tau}\Big|\sum_{j=1}^{k}\hat\xi_j^2 - \frac{k}{n}\sum_{j=1}^{n}\hat\xi_j^2\Big|. \tag{6.22}$$

Several change points are found by binary segmentation: repeat the test on $y_1,\dots,y_{\hat k}$ and $y_{\hat k+1},\dots,y_n$ until no further change is detected. The CUSUM test is most powerful for changes near $n/2$ and weaker for changes near either end of the sample. Fitting a single GARCH(1,1) across a change typically yields $\hat\alpha_1 + \hat\beta_1$ very close to 1, so an apparent IGARCH fit should prompt a change-point test; for many stock-price series the test indeed finds changes.

6.6 Value-at-Risk

Motivation. Value-at-Risk is a low conditional quantile of the return distribution. In a GARCH model this conditional quantile is the volatility times a fixed quantile of the error, which can be estimated by quantile regression.

For a random variable X with distribution function F_X and $0 < \tau < 1$, the *τ-quantile* is

$$F_X^{-1}(\tau) = \min\{y : F_X(y) \ge \tau\}$$

(the minimum exists because F_X is right-continuous and nondecreasing).

Proposition (Two properties of quantiles, [Supplement])

(1) For $a > 0$ and $b \in \mathbb{R}$,

$$F_{aX+b}^{-1}(\tau) = aF_X^{-1}(\tau) + b. \tag{6.23}$$

(2) If $E|X| < \infty$ and $\rho_\tau(x) = x(\tau - I(x < 0))$ (the *check function*), then

$$F_X^{-1}(\tau) = \min\Big\{m : \ m \text{ minimizes } E\rho_\tau(X - m)\Big\}. \tag{6.24}$$

Proof.

Step 1. For every x, $F_{aX+b}(ax+b) = P(aX+b \le ax+b) = F_X(x)$. As $x \mapsto ax+b$ is an increasing bijection of $\mathbb{R}$,

$$\min\{y : F_{aX+b}(y) \ge \tau\} = \min\{ax+b : F_X(x) \ge \tau\} = a\min\{x : F_X(x) \ge \tau\} + b.$$

Step 2. For fixed x, $u \mapsto \rho_\tau(x-u)$ is Lipschitz with derivative $I(x < u) - \tau$ except at $u = x$, so $\rho_\tau(x-m') - \rho_\tau(x-m) = \int_m^{m'} (I(x<u) - \tau)du$ for $m < m'$. The integrand is bounded, so Fubini gives, with $g(m) = E\rho_\tau(X-m)$ (finite since $\rho_\tau(x) \le |x|$),

$$g(m') - g(m) = \int_m^{m'} (P(X < u) - \tau)du = \int_m^{m'} (F_X(u) - \tau)du,$$

the two integrands differing only at the countably many atoms of X. Let $q = F_X^{-1}(\tau)$. For $m < q$, $F_X(u) < \tau$ on $[m, q)$, so $g(q) - g(m) < 0$; for $m > q$, $F_X(u) \ge \tau$ on $[q, m]$, so $g(m) - g(q) \ge 0$. Thus q minimizes g and no smaller m does. ∎

By (6.24), from a sample $X_1, \dots, X_n$ the quantile is estimated by the minimizer of

$$\frac{1}{n}\sum_{t=1}^{n} \rho_\tau(X_t - m). \tag{6.25}$$

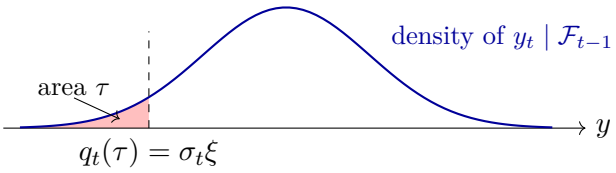


Figure 5: Conditional VaR at level τ: the τ-quantile of the conditional law of y_t, which scales with σ_t.

Now consider GARCH(1,1) in the reparametrized form

$$y_t = \sigma_t \xi_t, \quad \{\xi_t\} \sim iid(0, \omega), \ \omega > 0, \qquad \sigma_t^2 = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \alpha, \beta \ge 0, \ \alpha + \beta < 1,$$

where the scale has been moved from the constant term into the error variance ω. Since σ_t is $\mathcal{F}_{t-1}$-measurable and ξ_t is independent of $\mathcal{F}_{t-1}$, (6.23) with $a = \sigma_t$, $b = 0$ gives the *conditional quantile*

$$q_t(\tau) = \sigma_t F_\xi^{-1}(\tau) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2)^{1/2}\, \xi, \qquad \xi = F_\xi^{-1}(\tau).$$

Thus $q_t(\tau) = q_t(\xi, \alpha, \beta)$ depends on three parameters and, by the conditional version of (6.24),

$$q_t(\tau) = \min\{m : \ m \text{ minimizes } E(\rho_\tau(y_t - m) \mid \mathcal{F}_{t-1})\}.$$

Given $y_1, \dots, y_n$, the analogue of (6.25) is: choose $(\hat\xi, \hat\alpha, \hat\beta)$ to minimize

$$\frac{1}{n}\sum_{t=1}^{n} \rho_\tau(y_t - q_t^*(\xi, \alpha, \beta)), \qquad q_t^*(\xi, \alpha, \beta) = (1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*})^{1/2}\xi,$$

where $\sigma_t^{2*} = 1 + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^{2*}$ is computed recursively from arbitrary initial values y_0, σ_0^{2*}. The estimate

$$\hat q_t(\tau) = q_t^*(\hat\xi, \hat\alpha, \hat\beta) =: \mathrm{VaR}_{Y_t \mid \mathcal{F}_{t-1}}(\tau)$$

is the *conditional Value-at-Risk* at level τ (Figure 5); in risk management τ is small, such as 0.01 or 0.05.

6.7 Selected exercises

Motivation. The exercises below fill in three steps that the ARCH(1) construction uses without proof: the conditional normal law, the uniqueness of the series solution, and the strict stationarity of functions of an i.i.d. sequence.

Exercise 6.1 (Conditional law of ARCH(1))

Let X, Y be independent random elements (with values in $\mathbb{R}^k$ or $\mathbb{R}^\infty$) and f a measurable real function. It is known that

$$P(f(X,Y) \le z \mid X = x) = P(f(x,Y) \le z).$$

Prove this identity, and use it to prove (6.3).

Solution.

Step 1: the identity. Fix z and put $g(x) = P(f(x,Y) \le z) = \int \mathbb{1}\{f(x,u) \le z\}\, P_Y(du)$. By independence the joint law of (X,Y) is the product $P_X \otimes P_Y$, and the integrand $\mathbb{1}\{f(x,u) \le z\}$ is jointly measurable; by Tonelli's theorem g is measurable and, for every Borel set A,

$$P(f(X,Y) \le z,\ X \in A) = \int_A \int \mathbb{1}\{f(x,u) \le z\}\, P_Y(du)\, P_X(dx) = E\big[g(X)\mathbb{1}\{X \in A\}\big].$$

This is the defining property of the conditional probability, so $g(x)$ is (a version of) $P(f(X,Y) \le z \mid X = x)$.

Step 2: application. By Proposition (1), y_{t-1} is a function of $\xi_{t-1}, \xi_{t-2}, \dots$ and hence independent of ξ_t. Take $X = y_{t-1}$, $Y = \xi_t$ and $f(x,u) = u\,(\alpha_0 + \alpha_1 x^2)^{1/2}$, so that $f(X,Y) = \sigma_t \xi_t = y_t$. Step 1 and $(\alpha_0 + \alpha_1 y^2)^{1/2} > 0$ give

$$P(y_t \le z \mid y_{t-1} = y) = P\big(\xi_t(\alpha_0 + \alpha_1 y^2)^{1/2} \le z\big) = \Phi\big(z/\sqrt{\alpha_0 + \alpha_1 y^2}\big),$$

which is (6.3).

Step 3: the whole past. The vector $X = (y_{t-1}, y_{t-2}, \dots)$ is also a function of $\xi_{t-1}, \xi_{t-2}, \dots$, hence independent of ξ_t. Applying Step 1 with this X and $f(x,u) = u(\alpha_0 + \alpha_1 x_1^2)^{1/2}$ shows that the conditional law of y_t given $\mathcal{F}_{t-1}$ is $N(0, \alpha_0 + \alpha_1 y_{t-1}^2) = N(0, \sigma_t^2)$. ■

Exercise 6.2 (Iterating the ARCH(1) recursion)

Consider the ARCH(1) model (6.2).
(a) Prove (6.4) for every $n \ge 0$.
(b) Let $\{y_t\}$ be a solution with $Ey_t^2 = c < \infty$ for all t, where each y_s is a function of $\xi_s, \xi_{s-1}, \dots$. Prove that y_t^2 and σ_t^2 are given by the series (6.5); consequently y_t equals (6.6).

Solution.

(a) Write $P_{t,j} = \xi_t^2 \xi_{t-1}^2 \cdots \xi_{t-j}^2$. For $n = 0$, $y_t^2 = \sigma_t^2 \xi_t^2 = (\alpha_0 + \alpha_1 y_{t-1}^2)\xi_t^2 = \alpha_0 P_{t,0} + \alpha_1 P_{t,0}\, y_{t-1}^2$. If (6.4) holds for n, substitute $y_{t-n-1}^2 = (\alpha_0 + \alpha_1 y_{t-n-2}^2)\xi_{t-n-1}^2$ into the remainder:

$$\alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2 = \alpha_0 \alpha_1^{n+1} P_{t,n+1} + \alpha_1^{n+2} P_{t,n+1}\, y_{t-n-2}^2,$$

which turns (6.4) for n into (6.4) for $n+1$.

(b)

Step 1. Let $S_n = \alpha_0 \sum_{j=0}^{n} \alpha_1^j P_{t,j}$ and $R_n = \alpha_1^{n+1} P_{t,n}\, y_{t-n-1}^2$, so $y_t^2 = S_n + R_n$ by (a). All terms are nonnegative, so S_n is nondecreasing and $R_n = y_t^2 - S_n$ is nonincreasing in n; hence $R_n \downarrow R$ a.s. for some $R \geq 0$.

Step 2. y_{t-n-1} is a function of $\xi_{t-n-1}, \xi_{t-n-2}, \ldots$, hence independent of $(\xi_t, \ldots, \xi_{t-n})$. Therefore

$$ER_n = \alpha_1^{n+1} E(\xi_t^2) \cdots E(\xi_{t-n}^2)\, E y_{t-n-1}^2 = \alpha_1^{n+1} c \to 0,$$

using $0 \leq \alpha_1 < 1$. Since $0 \leq R \leq R_n$, $ER = 0$, so $R = 0$ a.s. and $y_t^2 = \lim_n S_n = \alpha_0 \sum_{j \geq 0} \alpha_1^j P_{t,j}$, the first identity in (6.5).

Step 3. Applying Step 2 at time $t-1$ and using $\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2$,

$$\sigma_t^2 = \alpha_0 + \alpha_1 \alpha_0 \sum_{j \geq 0} \alpha_1^j \xi_{t-1}^2 \cdots \xi_{t-1-j}^2,$$

the second identity in (6.5). As $\sigma_t \geq 0$ and $y_t = \sigma_t \xi_t$, y_t is given by (6.6). Together with Proposition (1) this shows that (6.6) is the *only* causal solution with finite variance. ■

Exercise 6.4 (Functions of an i.i.d. sequence)

Let $\{\xi_t\}$ be an i.i.d. sequence.
(a) For measurable $f_n : \mathbb{R}^n \to \mathbb{R}$, show that $Y_t^{(n)} = f_n(\xi_t, \ldots, \xi_{t-n+1})$ is strictly stationary (in particular for $n = 2$).
(b) Let $f : \mathbb{R}^\infty \to \mathbb{R}$ be measurable and suppose $f_n(\xi_t, \ldots, \xi_{t-n+1}) \to f(\xi_t, \xi_{t-1}, \ldots)$ a.s. for every t. Show that $Y_t = f(\xi_t, \xi_{t-1}, \ldots)$ is strictly stationary.

Solution.

(a) Fix times $t_1, \ldots, t_m$ and a shift h; let $L = \min_i t_i - n + 1$ and $U = \max_i t_i$. Each $Y_{t_i}^{(n)}$ is f_n applied to a fixed block of coordinates of $\boldsymbol{\xi} = (\xi_L, \ldots, \xi_U)$, so

$$(Y_{t_1}^{(n)}, \ldots, Y_{t_m}^{(n)}) = g(\boldsymbol{\xi}), \qquad (Y_{t_1+h}^{(n)}, \ldots, Y_{t_m+h}^{(n)}) = g(\boldsymbol{\xi}^h),$$

with the same measurable $g : \mathbb{R}^{U-L+1} \to \mathbb{R}^m$ and $\boldsymbol{\xi}^h = (\xi_{L+h}, \ldots, \xi_{U+h})$. By independence and identical distribution, both $\boldsymbol{\xi}$ and $\boldsymbol{\xi}^h$ have the product law $\mu^{\otimes(U-L+1)}$, where μ is the law of ξ_1. Measurable images of equally distributed vectors are equally distributed, so the two vectors of Y's have the same joint law, which is Definition 1.2.2. For $n = 2$, g consists of the coordinates $f_2(\xi_{t_i}, \xi_{t_i-1})$.

(b) Fix $t_1, \ldots, t_m$ and h. By assumption, $\mathbf{V}_n = (Y_{t_1}^{(n)}, \ldots, Y_{t_m}^{(n)}) \to \mathbf{V} = (Y_{t_1}, \ldots, Y_{t_m})$ a.s., and likewise $\mathbf{V}_n^h \to \mathbf{V}^h$ for the times shifted by h. Almost sure convergence implies convergence in distribution. By (a), $\mathbf{V}_n$ and $\mathbf{V}_n^h$ have the same law for every n, so this single sequence of laws converges weakly both to the law of $\mathbf{V}$ and to that of $\mathbf{V}^h$. Weak limits are unique, hence $\mathbf{V} \overset{d}{=} \mathbf{V}^h$, and $\{Y_t\}$ is strictly stationary. For example, with f_n the truncation of the series in (6.6) after its first $n-1$ terms (a function of $\xi_t, \ldots, \xi_{t-n+1}$), this proves the strict stationarity claimed in Proposition (1). ■

Chapter 7. Multivariate Time Series

Many applications involve several series observed together. A portfolio, for instance, depends on each asset and on how the assets move jointly, and risk measures such as Value-at-Risk for a portfolio require a multivariate model. This chapter introduces multivariate time series, their stationarity, and the basic models: the vector ARMA (in particular VAR) model, cointegration, multivariate GARCH and copula-based models.

7.1 Multivariate time series models

Motivation. Stacking k univariate series into a vector, the notions of stationarity, autocovariance and ARMA modelling carry over almost verbatim, with matrices in place of scalars; the new phenomena are cross-dependence and cointegration.

Given k univariate series $Y_{1t}, \dots, Y_{kt}$, the vector $\mathbf{Y}_t = (Y_{1t}, \dots, Y_{kt})^T$ is a *k-dimensional multivariate time series*. For vectors $\mathbf{x} = (x_1, \dots, x_k)^T$ and $\mathbf{y} = (y_1, \dots, y_k)^T$, $\mathbf{x} \le \mathbf{y}$ means $x_i \le y_i$ for all i.

Definition 7.1.1 (Strict stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *strictly stationary* if for all integers $t_1, \dots, t_m$ and h ($m \ge 1$) and all $\mathbf{y}_1, \dots, \mathbf{y}_m \in \mathbb{R}^k$,

$$P(\mathbf{Y}_{t_1} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m} \le \mathbf{y}_m) = P(\mathbf{Y}_{t_1+h} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m+h} \le \mathbf{y}_m).$$

Definition 7.1.2 (Stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *stationary* if
(1) $E\mathbf{Y}_t = \boldsymbol{\mu}$ for all t;
(2) $EY_{it}^2 < \infty$ for all $1 \le i \le k$ and all t;
(3) $\operatorname{Cov}(Y_{it}, Y_{j,t+h}) = \operatorname{Cov}(Y_{i0}, Y_{jh})$ for all $1 \le i, j \le k$ and all t, h.

Definition 7.1.3 (Autocovariance function)

For a stationary $\{\mathbf{Y}_t\}$, the $k \times k$ matrix-valued function $\Gamma(\cdot) = (\Gamma_{ij}(\cdot))_{i,j=1}^k$ with

$$\Gamma_{ij}(h) = \operatorname{Cov}(Y_{it}, Y_{j,t+h}), \qquad h = 0, \pm 1, \pm 2, \dots,$$

is the *autocovariance function*; that is, $\Gamma(h) = E(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T$ with $\boldsymbol{\mu} = E\mathbf{Y}_1$.

Theorem 7.1.4 (Properties of the autocovariance function)

The autocovariance function of a stationary series satisfies:
(1) $\Gamma(-h) = \Gamma(h)^T$;
(2) $|\Gamma_{ij}(h)| \le \sqrt{\Gamma_{ii}(0)}\sqrt{\Gamma_{jj}(0)}$;
(3) for all $n \ge 1$ and $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^k$, $\sum_{i,j=1}^n \mathbf{a}_i^T \Gamma(i-j)\mathbf{a}_j \ge 0$.

Proof. [Supplement] (The book leaves this as an exercise.)

(1) By stationarity, $\Gamma(-h) = E(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t-h} - \boldsymbol{\mu})^T = E(\mathbf{Y}_{s+h} - \boldsymbol{\mu})(\mathbf{Y}_s - \boldsymbol{\mu})^T$ with $s = t - h$, and this is the transpose of $E(\mathbf{Y}_s - \boldsymbol{\mu})(\mathbf{Y}_{s+h} - \boldsymbol{\mu})^T = \Gamma(h)$.

(2) By the Cauchy–Schwarz inequality, $|\operatorname{Cov}(Y_{it}, Y_{j,t+h})| \le \sqrt{\operatorname{Var} Y_{it}}\sqrt{\operatorname{Var} Y_{j,t+h}} = \sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}$.

(3) Since $\operatorname{Cov}(\mathbf{a}_i^T\mathbf{Y}_i, \mathbf{a}_j^T\mathbf{Y}_j) = \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j$,

$$0 \le \operatorname{Var}\Big(\sum_{i=1}^{n} \mathbf{a}_i^T\mathbf{Y}_i\Big) = \sum_{i,j=1}^{n} \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j = \sum_{i,j=1}^{n} \mathbf{a}_j^T\Gamma(i-j)\mathbf{a}_i,$$

where the last step transposes each scalar term and uses (1). Relabelling $i \leftrightarrow j$ gives the claim. ∎

Definition 7.1.5 (Vector ARMA model)

If $\{\mathbf{Y}_t\}$ satisfies

$$\mathbf{Y}_t - \Phi_1\mathbf{Y}_{t-1} - \cdots - \Phi_p\mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t + \Theta_1\boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q\boldsymbol{\epsilon}_{t-q}, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma), \tag{7.1}$$

where $\Phi_1, \ldots, \Phi_p, \Theta_1, \ldots, \Theta_q, \Sigma$ are $k \times k$ matrices and Σ is positive definite, then $\{\mathbf{Y}_t\}$ follows a *vector ARMA(p, q) model* (VARMA(p, q)). With $q = 0$ it is a *VAR(p) model*.

Write $\Phi(z) = I - \Phi_1 z - \cdots - \Phi_p z^p$ and $\Theta(z) = I + \Theta_1 z + \cdots + \Theta_q z^q$, so (7.1) reads $\Phi(B)\mathbf{Y}_t = \Theta(B)\boldsymbol{\epsilon}_t$. The next theorem is the analogue of the univariate causality result: the condition on the zeros of $\phi(z)$ becomes a condition on the zeros of $\det \Phi(z)$.

Theorem 7.1.6 (Causal VARMA)

If

$$\det(I - \Phi_1 z - \cdots - \Phi_p z^p) \neq 0 \qquad \text{for all complex } |z| \le 1,$$

then the VARMA model (7.1) has a stationary solution $\{\mathbf{Y}_t\}$, it is unique, and it has the representation

$$\mathbf{Y}_t = \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}, \tag{7.2}$$

where $C_0 = I_k$ and the $k \times k$ matrices C_i are determined by

$$(I - \Phi_1 z - \cdots - \Phi_p z^p) \sum_{i=0}^{\infty} C_i z^i = I + \Theta_1 z + \cdots + \Theta_q z^q, \qquad |z| \le 1. \tag{7.3}$$

With mean $\boldsymbol{\mu}$ the model reads $\mathbf{Y}_t - \boldsymbol{\mu} - \Phi_1(\mathbf{Y}_{t-1} - \boldsymbol{\mu}) - \cdots - \Phi_p(\mathbf{Y}_{t-p} - \boldsymbol{\mu}) = \boldsymbol{\epsilon}_t + \Theta_1\boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q\boldsymbol{\epsilon}_{t-q}$.

Proof. [Supplement] (The book omits the proof.) Matrix products do not commute, so the order of factors in (7.3) matters: $\Phi(z)$ stands on the left.

Step 1: the coefficients. $\det \Phi(z)$ is a polynomial with finitely many zeros, none in the closed unit disc, so it has no zeros in $|z| < 1 + \delta$ for some $\delta > 0$. Hence $\Phi(z)^{-1} = \operatorname{adj}\Phi(z)/\det\Phi(z)$ has entries that are rational functions without poles in $|z| < 1 + \delta$, i.e. analytic there, and so is $C(z) := \Phi(z)^{-1}\Theta(z) = \sum_{i \ge 0} C_i z^i$. The power series converges at $z = 1 + \delta/2$, so its terms are bounded: $\|C_i\| \le K(1 + \delta/2)^{-i}$ for a constant K. In particular $\sum_i \|C_i\| < \infty$, $C_0 = C(0) = I$, and $\Phi(z)C(z) = \Theta(z)$, which is (7.3). Similarly $\Phi(z)^{-1} = \sum_i D_i z^i$ with $\sum_i \|D_i\| < \infty$.

Step 2: existence. Since $E\|\boldsymbol{\epsilon}_t\| \le (\operatorname{tr}\Sigma)^{1/2}$, $E\sum_i \|C_i\|\|\boldsymbol{\epsilon}_{t-i}\| < \infty$, so the series (7.2) converges absolutely a.s.; it also converges in mean square because $\big(E\|\sum_{i=m}^{n} C_i\boldsymbol{\epsilon}_{t-i}\|^2\big)^{1/2} \le (\operatorname{tr}\Sigma)^{1/2}\sum_{i=m}^{n}\|C_i\| \to 0$. Then $E\mathbf{Y}_t = \mathbf{0}$ and, as $E\boldsymbol{\epsilon}_s\boldsymbol{\epsilon}_r^T = \Sigma\,\mathbb{1}\{s = r\}$,

$$E\mathbf{Y}_t\mathbf{Y}_{t+h}^T = \sum_{i,j} C_i E(\boldsymbol{\epsilon}_{t-i}\boldsymbol{\epsilon}_{t+h-j}^T)C_j^T = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T \qquad (h \ge 0),$$

which does not depend on t; so $\{\mathbf{Y}_t\}$ is stationary. Writing $\Phi(z)=\sum_{j=0}^{p}A_j z^j$ ($A_0=I$, $A_j=-\Phi_j$) and rearranging the absolutely convergent double series,

$$\Phi(B)\mathbf{Y}_t=\sum_{j=0}^{p}A_j\sum_{i\ge 0}C_i\boldsymbol{\epsilon}_{t-j-i}=\sum_{m\ge 0}\Big(\sum_{j=0}^{\min(p,m)}A_jC_{m-j}\Big)\boldsymbol{\epsilon}_{t-m}=\sum_{m=0}^{q}\Theta_m\boldsymbol{\epsilon}_{t-m},$$

because the inner sum is the coefficient of z^m in $\Phi(z)C(z)=\Theta(z)$ ($\Theta_0=I$). Thus (7.2) solves (7.1).

Step 3: uniqueness. Let $\{\mathbf{Y}_t\}$ be any stationary solution. Applying the filter $\sum_i D_iB^i$ (with summable coefficients, so it acts in mean square on stationary sequences and composes like multiplication of power series) to $\Phi(B)\mathbf{Y}_t=\Theta(B)\boldsymbol{\epsilon}_t$ gives $\mathbf{Y}_t=\Phi(B)^{-1}\Theta(B)\boldsymbol{\epsilon}_t=\sum_i C_i\boldsymbol{\epsilon}_{t-i}$. ∎

Example 7.1.7 (Coefficients of a bivariate ARMA(1,1))

Let $k=2$, $p=q=1$, $\Phi_1=\begin{pmatrix}0.5 & 0.5\\ 0 & 0.5\end{pmatrix}$ and $\Theta_1=\Phi_1^T$. Here $\det\Phi(z)=(1-0.5z)^2\neq 0$ for $|z|\le 1$. From (7.3),

$$\begin{aligned}\sum_{i=0}^{\infty}C_iz^i&=\begin{pmatrix}1-0.5z & -0.5z\\ 0 & 1-0.5z\end{pmatrix}^{-1}\begin{pmatrix}1+0.5z & 0\\ 0.5z & 1+0.5z\end{pmatrix}\\&=(1-0.5z)^{-2}\begin{pmatrix}1 & 0.5z(1+0.5z)\\ 0.5z(1-0.5z) & 1-0.25z^2\end{pmatrix}.\end{aligned}$$

Expanding $(1-0.5z)^{-2}=\sum_{m\ge 0}(m+1)2^{-m}z^m$ and collecting the coefficient of z^i entrywise,

$$C_0=I,\qquad C_i=2^{-i}\begin{pmatrix}i+1 & 2i-1\\ 1 & 2\end{pmatrix},\quad i=1,2,\dots$$

For instance the $(1,2)$ entry is $\frac{1}{2}\,i\,2^{-(i-1)}+\frac{1}{4}(i-1)2^{-(i-2)}=(2i-1)2^{-i}$.

7.1.1 Cointegration

Definition 7.1.8 (Integration order and cointegration)

If $(1-B)^d\mathbf{Y}_t=\big((1-B)^dY_{1t},\dots,(1-B)^dY_{kt}\big)^T$ is stationary while $\mathbf{Y}_t,(1-B)\mathbf{Y}_t,\dots,(1-B)^{d-1}\mathbf{Y}_t$ are nonstationary, we write $\mathbf{Y}_t\sim I(d)$. If $\mathbf{Y}_t\sim I(d)$ and, for some positive integer $\delta<d$ and some vector $\boldsymbol{\alpha}\in\mathbb{R}^k$, $\boldsymbol{\alpha}^T\mathbf{Y}_t\sim I(\delta)$, then $\mathbf{Y}_t$ is *cointegrated* with *cointegrating vector* $\boldsymbol{\alpha}$. (The most common case is $d=1$, $\delta=0$: a stationary combination of $I(1)$ series.)

Cointegration must be distinguished from *spurious regression*. If X_t and Y_t are two *independent* random walks, there is no relation between them, yet regressing Y_t on X_t typically produces a large R^2 and a highly significant slope, because both series wander with apparent trends. Cointegration, in contrast, is a genuine long-run equilibrium relation.

Example 7.1.10 (Consumption and income)

Let C_t be real per-capita consumption and I_t real per-capita disposable income, with $C_t,I_t\sim I(1)$, and suppose $C_t=a+bI_t+u_t$ for some $a,b\in(0,1)$ with $Eu_t=0$, $u_t\sim I(0)$. Then $\boldsymbol{\alpha}^T\mathbf{Y}_t=C_t-bI_t=a+u_t\sim I(0)$ for $\mathbf{Y}_t=(C_t,I_t)^T$ and $\boldsymbol{\alpha}=(1,-b)^T$: the pair is cointegrated.

When the components are cointegrated, the *vector error correction model* (VECM) is the natural form.

Definition 7.1.11 (VECM)

The VECM(p) with *cointegration rank* r ($\le k$) is

$$\Delta \mathbf{Y}_t = \boldsymbol{\delta} + \Pi \mathbf{Y}_{t-1} + \sum_{i=1}^{p-1} \Phi_i^* \Delta \mathbf{Y}_{t-i} + \boldsymbol{\epsilon}_t, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma), \tag{7.4}$$

where $\Delta = 1 - B$ is the differencing operator, $\boldsymbol{\delta}$ is a $k \times 1$ constant vector, $\Pi = \boldsymbol{\alpha}\boldsymbol{\beta}^T$ with $\boldsymbol{\alpha}, \boldsymbol{\beta}$ $k \times r$ matrices, and the Φ_i^* are $k \times k$ matrices.

If $\mathbf{Y}_t \sim I(1)$, then $\Delta \mathbf{Y}_t$ and $\boldsymbol{\epsilon}_t$ are stationary, so (7.4) forces $\boldsymbol{\alpha}\boldsymbol{\beta}^T \mathbf{Y}_{t-1}$, and hence (when $\boldsymbol{\alpha}$ has full column rank) $\boldsymbol{\beta}^T \mathbf{Y}_{t-1}$, to be stationary: the columns of $\boldsymbol{\beta}$ are r cointegrating vectors, and $\boldsymbol{\alpha}$ measures how fast deviations from equilibrium are corrected.

Johansen (1988) proposed a test of $H_0 : r \le r_0$ against $H_1 : r > r_0$ based on the *trace statistic*

$$\lambda_{\text{trace}}(r_0) = -n \sum_{i=r_0+1}^{k} \log(1 - \hat{\lambda}_i), \tag{7.5}$$

where n is the sample size and $\hat{\lambda}_1 \ge \cdots \ge \hat{\lambda}_k$ are the estimated eigenvalues associated with Π. If $r \le r_0$, the true eigenvalues beyond r_0 vanish and the statistic stays small; large values reject H_0. Its null distribution is nonstandard and depends on the constant term, r_0, trend terms and the dimension (see Johansen, 1988).

7.2 Inference for stationary multivariate series

Motivation. The sample mean and sample autocovariance matrices are consistent estimators, exactly as in the univariate case, and a VAR(p) model is fitted by least squares one equation at a time.

Theorem 7.2.1 (Sample mean)

Let $\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}$ with $\{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma)$ and $k \times k$ matrices C_i satisfying $\sum_{i=0}^{\infty} \|C_i\| < \infty$, where $\|C\| = \sup\{\|C\mathbf{y}\| : \|\mathbf{y}\| \le 1\}$. Let $\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{t=1}^{n} \mathbf{Y}_t$. Then:
(1) $P(\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu},\ n \to \infty) = 1$;
(2) if $\Gamma_{ii}(n) \to 0$ for all $1 \le i \le k$, then $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to 0$;
(3) if $\sum_{h=0}^{\infty} |\Gamma_{ii}(h)| < \infty$ for all $1 \le i \le k$, then

$$n E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to \sum_{i=1}^{k} \sum_{h=-\infty}^{\infty} \Gamma_{ii}(h).$$

Proof. [Supplement] (The book omits the proof.) By Theorem 7.1.6's Step 2 argument, the series converges a.s. and in mean square and $\{\mathbf{Y}_t\}$ is stationary.

(1) $\mathbf{Y}_t = G(\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_{t-1}, \dots)$ for one fixed measurable G, so $\{\mathbf{Y}_t\}$ is a strictly stationary ergodic sequence (a function of an i.i.d. shift), and $E\|\mathbf{Y}_t\| < \infty$. Birkhoff's ergodic theorem gives $\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu}$ a.s.

(2) $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) = \sum_{i=1}^{k} \mathrm{Var}(\bar{Y}_i)$ and

$$\mathrm{Var}(\bar{Y}_i) = \frac{1}{n^2} \sum_{s,t=1}^{n} \Gamma_{ii}(t-s) = \frac{1}{n} \sum_{|h|<n} \Big(1 - \frac{|h|}{n}\Big) \Gamma_{ii}(h) \le \frac{1}{n} \sum_{|h|<n} |\Gamma_{ii}(h)|.$$

If $\Gamma_{ii}(h) \to 0$ then its Cesàro means $\frac{1}{n}\sum_{|h|<n} |\Gamma_{ii}(h)|$ tend to 0.

(3) $n\,\mathrm{Var}(\bar{Y}_i) = \sum_{|h|<n}(1-|h|/n)\Gamma_{ii}(h) \to \sum_h \Gamma_{ii}(h)$ by dominated convergence (dominating sequence $|\Gamma_{ii}(h)|$, summable by assumption, using $\Gamma_{ii}(-h)=\Gamma_{ii}(h)$). Summing over i gives the claim. ■

Theorem 7.2.2 (Sample autocovariance)

Let $\{\mathbf{Y}_t\}$ be as in Theorem 7.2.1. Then $\Gamma(h) = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T$ $(h \ge 0)$, and the sample autocovariance function

$$\hat{\Gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-|h|}(\mathbf{Y}_t - \hat{\boldsymbol{\mu}})(\mathbf{Y}_{t+|h|} - \hat{\boldsymbol{\mu}})^T$$

(for $h<0$ take the transpose) satisfies $P(\hat{\Gamma}(h) \to \Gamma(h),\ n\to\infty) = 1$; that is, $\hat{\Gamma}(\cdot)$ is a consistent estimator of $\Gamma(\cdot)$.

Proof. [Supplement] The formula for $\Gamma(h)$ was derived in Step 2 of the proof of Theorem 7.1.6. Fix $h \ge 0$ and write $\mathbf{Y}_t - \hat{\boldsymbol{\mu}} = (\mathbf{Y}_t - \boldsymbol{\mu}) - (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})$:

$$\begin{aligned}\hat{\Gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t-\boldsymbol{\mu})(\mathbf{Y}_{t+h}-\boldsymbol{\mu})^T - \Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t-\boldsymbol{\mu})\Big)(\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})^T \\ - (\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})\Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_{t+h}-\boldsymbol{\mu})\Big)^T + \frac{n-h}{n}(\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})(\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})^T.\end{aligned}$$

The matrices $Z_t = (\mathbf{Y}_t-\boldsymbol{\mu})(\mathbf{Y}_{t+h}-\boldsymbol{\mu})^T$ are a fixed function of the shifted i.i.d. sequence and integrable (Cauchy–Schwarz), so the ergodic theorem gives $\frac{1}{n}\sum_{t\le n-h} Z_t \to EZ_t = \Gamma(h)$ a.s. (dropping h terms changes nothing in the limit). The averages in the middle terms tend to $\mathbf{0}$ a.s. and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \to \mathbf{0}$ a.s. by Theorem 7.2.1(1), so the other three terms vanish in the limit. ■

7.2.1 Inference in VAR(p) models

In the VAR(p) model $\mathbf{Y}_t - \Phi_1\mathbf{Y}_{t-1} - \cdots - \Phi_p\mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t$, $\{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0},\Sigma)$, the coefficients are estimated by least squares, given initial values $\mathbf{Y}_{-p+1},\dots,\mathbf{Y}_0$. Define

$$\mathbf{y}_t = \begin{pmatrix}\mathbf{Y}_t\\ \vdots \\ \mathbf{Y}_{t-p+1}\end{pmatrix}\ (kp\times 1), \qquad \begin{aligned} Y &= (\mathbf{Y}_1,\dots,\mathbf{Y}_n)\ (k\times n), & \Phi &= (\Phi_1,\dots,\Phi_p)\ (k\times kp),\\ X &= (\mathbf{y}_0,\dots,\mathbf{y}_{n-1})\ (kp\times n), & \epsilon &= (\boldsymbol{\epsilon}_1,\dots,\boldsymbol{\epsilon}_n)\ (k\times n).\end{aligned}$$

Then the model is $Y = \Phi X + \epsilon$.

Proposition (Least squares for VAR(p), [Supplement])

If XX^T is invertible, then for every Φ

$$(Y-\Phi X)(Y-\Phi X)^T = (Y-\hat{\Phi}X)(Y-\hat{\Phi}X)^T + (\hat{\Phi}-\Phi)XX^T(\hat{\Phi}-\Phi)^T, \qquad \hat{\Phi} = YX^T(XX^T)^{-1}.$$

Hence $\hat{\Phi}$ minimizes $\sum_{t=1}^n \|\mathbf{Y}_t - \Phi\mathbf{y}_{t-1}\|^2 = \mathrm{tr}\{(Y-\Phi X)(Y-\Phi X)^T\}$ (and the determinant of the residual cross-product matrix), uniquely.

Proof. Write $Y-\Phi X = (Y-\hat{\Phi}X) + (\hat{\Phi}-\Phi)X$ and expand. The cross terms vanish because $(Y-\hat{\Phi}X)X^T = YX^T - YX^T(XX^T)^{-1}XX^T = 0$. The second term on the right is positive semidefinite, and it is zero only if $\Phi = \hat{\Phi}$ since XX^T is positive definite. Taking traces (or determinants, using monotonicity of det on positive semidefinite matrices) gives the minimization claims. ■

The least squares estimators are therefore

$$\hat{\Phi} = YX^T(XX^T)^{-1}, \qquad \hat{\Sigma} = \frac{1}{n-kp}(Y-\hat{\Phi}X)(Y-\hat{\Phi}X)^T;$$

note that $\hat{\Phi}$ is the same as ordinary least squares applied separately to each of the k equations.

The order p can be chosen as follows. For a sufficiently large p_0, fit VAR(i), $i = 1, \dots, p_0$, compute the residuals $\hat{\boldsymbol{\epsilon}}_t^{(i)} = \mathbf{Y}_t - \hat{\Phi}_1^{(i)}\mathbf{Y}_{t-1} - \cdots - \hat{\Phi}_i^{(i)}\mathbf{Y}_{t-i}$ and their covariance matrix

$$\hat{\Sigma}_i = \frac{1}{n-2i}\sum_{t=i+1}^{n} \hat{\boldsymbol{\epsilon}}_t^{(i)}\hat{\boldsymbol{\epsilon}}_t^{(i)T},$$

and select the i minimizing

$$AIC(i) = \log|\hat{\Sigma}_i| + 2k^2 i/n \qquad \text{or} \qquad BIC(i) = \log|\hat{\Sigma}_i| + k^2 i\log(n)/n.$$

The penalty counts the $k^2 i$ autoregressive parameters.

7.3 Multivariate GARCH models

Motivation. The volatility of a portfolio depends on the joint, time-varying covariance of its assets, so the univariate GARCH idea must be extended to a conditional covariance matrix.

Let y_{it}, $i = 1, \dots, d$, be d series forming $\mathbf{Y}_t$, and let $\mathcal{F}_{t-1}$ be the information up to $t-1$. With $\sigma_{iit} = \mathrm{Var}(y_{it} \mid \mathcal{F}_{t-1})$ and $\sigma_{ijt} = \mathrm{Cov}(y_{it}, y_{jt} \mid \mathcal{F}_{t-1})$, the matrix $\Sigma_t = (\sigma_{ijt})$ is the conditional covariance matrix of $\mathbf{Y}_t$. The simplest multivariate GARCH model is

$$\mathbf{Y}_t = \Sigma_t^{1/2}\boldsymbol{\epsilon}_t, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, I_d), \qquad E(\mathbf{Y}_t\mathbf{Y}_t^T \mid \mathcal{F}_{t-1}) = \Sigma_t, \tag{7.6}$$

with Σ_t a function of the past. Since Σ_t has $d(d+1)/2$ distinct entries, an unrestricted recursion for Σ_t has a number of parameters growing like d^4, and estimation quickly becomes infeasible. Practical models therefore restrict the form of Σ_t; well-known examples are the BEKK (Baba–Engle–Kraft–Kroner) model, which keeps Σ_t positive definite by a quadratic-form recursion, and the CCC (constant conditional correlation) GARCH model, $\Sigma_t = D_t R D_t$ with $D_t = \mathrm{diag}(\sigma_{11t}^{1/2}, \dots, \sigma_{ddt}^{1/2})$, univariate GARCH volatilities and a constant correlation matrix R. See Bollerslev (1990) and Engle and Kroner (1995).

7.4 SCOMDY models

Motivation. A copula separates a joint distribution into its marginals and a dependence function. Combining univariate GARCH volatilities with a copula for the standardized errors gives a flexible, semiparametric multivariate model.

A simple instance of (7.6) consists of d GARCH(1,1) series whose errors are jointly normal:

$$y_{it} = \sigma_{it}\xi_{it}, \quad i = 1, \dots, d, \qquad \boldsymbol{\xi}_t = (\xi_{1t}, \dots, \xi_{dt})^T \text{ i.i.d. multivariate normal.} \tag{7.7}$$

If $\boldsymbol{\xi}_t$ is not normal, its distribution must be modelled. The *semiparametric copula-based multivariate dynamic* (SCOMDY) model does this with a copula.

Theorem (Sklar, continuous case, [Supplement])

Let $\mathbf{Y} = (Y_1, \dots, Y_d)^T$ have joint distribution function F and continuous marginals $F_1, \dots, F_d$. Then there is a function C on $[0,1]^d$, namely the distribution function of $(F_1(Y_1), \dots, F_d(Y_d))$, such that

$$F(\mathbf{y}) = C(F_1(y_1), \dots, F_d(y_d)), \qquad \mathbf{y} = (y_1, \dots, y_d)^T,$$

and C has uniform $(0,1)$ marginals (a *copula*).

Proof. If F_i is continuous then $U_i = F_i(Y_i)$ is uniform: for $u \in (0,1)$ let $x_u = \max\{x : F_i(x) = u\}$ (exists by continuity, as $F_i(x) \to 0, 1$ at $\mp\infty$); then $\{F_i(Y_i) \le u\} = \{Y_i \le x_u\}$ up to a null set, so $P(U_i \le u) = F_i(x_u) = u$. Moreover $\{Y_i \le y_i\} \subset \{F_i(Y_i) \le F_i(y_i)\}$, and the difference $\{Y_i > y_i,\ F_i(Y_i) = F_i(y_i)\}$ has probability 0 because F_i is flat on the corresponding interval. Hence

$$F(\mathbf{y}) = P(Y_i \le y_i\ \forall i) = P(U_i \le F_i(y_i)\ \forall i) = C(F_1(y_1), \dots, F_d(y_d)).$$

■

Thus the copula C carries all the dependence. Since parametric multivariate families other than the normal are scarce, one models C instead of F. Common choices are the Gaussian, Archimedean, Frank and Clayton copulas. For $d = 2$ the *Gaussian copula* is

$$C(u, v; \rho) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v)),$$

where Φ is the standard normal distribution function and Φ_ρ the bivariate normal distribution function with means 0, variances 1 and correlation ρ. An *Archimedean copula* has the form

$$C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$$

for a convex generator φ on $[0,1]$ (decreasing, $\varphi(1) = 0$); $\varphi(t) = \log(e^{-\theta} - 1) - \log(e^{-\theta t} - 1)$, $\theta \neq 0$, gives the *Frank copula* and $\varphi(t) = (t^{-\theta} - 1)/\theta$, $\theta > 0$, the *Clayton copula*. As these examples show, copulas usually involve unknown parameters (see Joe, 1997).

Marginal distribution functions are estimated by empirical distribution functions. Given $\mathbf{Y}_j = (Y_{1j}, \dots, Y_{dj})^T$, $j = 1, \dots, n$, choose a copula family C and estimate F by $\hat{F}(\mathbf{y}) = C(\hat{F}_1(y_1), \dots, \hat{F}_d(y_d))$, where

$$\hat{F}_i(y) = \frac{1}{n} \sum_{j=1}^{n} I(Y_{ij} \le y), \qquad i = 1, \dots, d.$$

For a series following (7.7) with copula-distributed errors, this yields a conditional likelihood for $(y_{1t}, \dots, y_{dt})^T$, $t = 1, \dots, n$, on which inference is based. The SCOMDY model is very general: GARCH, VAR and Markov-switching models are special cases. Chen and Fan (2006) estimated the copula parameters by a Kullback–Leibler criterion, derived their asymptotic distribution, and proposed model-selection tests.

7.5 Selected exercises

Motivation. The first exercise turns the geometric decay of the causal coefficients into geometric decay of the autocovariance matrices; the second computes the autocovariance function of a bivariate ARMA(1,1) by the matrix analogue of the univariate recursions.

Exercise 7.2 (Geometric decay of the autocovariances)

Let $\{\mathbf{Y}_t\}$ be the stationary solution of a VARMA model satisfying the condition of Theorem 7.1.6. Show that there are $\rho \in (0,1)$ and a constant K such that $|\Gamma_{ij}(h)| \le K\rho^{|h|}$ for all i, j and h.

Solution.

Step 1: decay of C_i. Step 1 of the proof of Theorem 7.1.6 gives $\delta > 0$ and K_0 with $\|C_i\| \le K_0\rho^i$ for all $i \ge 0$, where $\rho = (1+\delta/2)^{-1} \in (0,1)$.

Step 2: bound on $\Gamma(h)$. The mean does not affect Γ, and by Theorem 7.2.2, $\Gamma(h) = \sum_{i\ge 0} C_i \Sigma C_{i+h}^T$ for $h \ge 0$. Using $\|AB\| \le \|A\|\|B\|$ and $\|C^T\| = \|C\|$,

$$\|\Gamma(h)\| \le \sum_{i\ge 0} \|C_i\|\,\|\Sigma\|\,\|C_{i+h}\| \le \|\Sigma\| K_0^2 \sum_{i\ge 0} \rho^i \rho^{i+h} = \frac{\|\Sigma\| K_0^2}{1-\rho^2}\,\rho^h.$$

Step 3: entries. With unit vectors $\mathbf{e}_i$, $|\Gamma_{ij}(h)| = |\mathbf{e}_i^T \Gamma(h) \mathbf{e}_j| \le \|\Gamma(h)\|$. For $h < 0$, Theorem 7.1.4(1) gives $\Gamma(h) = \Gamma(-h)^T$, which has the same norm. Hence the claim holds with $K = \|\Sigma\| K_0^2/(1-\rho^2)$. ■

Exercise 7.3 (Autocovariances of a bivariate ARMA(1,1))

Find the autocovariance matrix function of the bivariate ARMA(1,1) model $\mathbf{Y}_t = \Phi \mathbf{Y}_{t-1} + \boldsymbol{\epsilon}_t + \Theta \boldsymbol{\epsilon}_{t-1}$ with

$$\Phi = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.6 \end{pmatrix}, \qquad \Theta = \begin{pmatrix} 0.5 & 0.6 \\ -0.7 & 0.8 \end{pmatrix}, \qquad \{\boldsymbol{\epsilon}_t\} \sim \text{IID } N\Big(\mathbf{0}, \begin{pmatrix} 1 & 0.71 \\ 0.71 & 2 \end{pmatrix}\Big).$$

Solution.

Step 1: causality. $\det(I - \Phi z) = (1 - 0.7z)(1 - 0.6z)$ has zeros $1/0.7$ and $1/0.6$, outside the unit disc. By Theorem 7.1.6 the stationary solution is $\mathbf{Y}_t = \sum_{i\ge 0} C_i \boldsymbol{\epsilon}_{t-i}$ with $C_0 = I$; hence $E\mathbf{Y}_t = \mathbf{0}$, $E\mathbf{Y}_t \boldsymbol{\epsilon}_{t+j}^T = 0$ for $j \ge 1$, and $E\mathbf{Y}_t \boldsymbol{\epsilon}_t^T = C_0 \Sigma = \Sigma$.

Step 2: lag 0. Write $\mathbf{Y}_t = \Phi \mathbf{Y}_{t-1} + \mathbf{U}_t$ with $\mathbf{U}_t = \boldsymbol{\epsilon}_t + \Theta \boldsymbol{\epsilon}_{t-1}$. By Step 1, $E\mathbf{Y}_{t-1}\mathbf{U}_t^T = \Sigma\Theta^T$ and $E\mathbf{U}_t\mathbf{U}_t^T = \Sigma + \Theta\Sigma\Theta^T$, so

$$\Gamma(0) = \Phi\Gamma(0)\Phi^T + Q, \qquad Q = \Phi\Sigma\Theta^T + \Theta\Sigma\Phi^T + \Sigma + \Theta\Sigma\Theta^T.$$

Numerically $\Sigma\Theta^T = \begin{pmatrix} 0.926 & -0.132 \\ 1.555 & 1.103 \end{pmatrix}$, $\Phi\Sigma\Theta^T = \begin{pmatrix} 0.6482 & -0.0924 \\ 0.9330 & 0.6618 \end{pmatrix}$ (and $\Theta\Sigma\Phi^T$ is its transpose), $\Theta\Sigma\Theta^T = \begin{pmatrix} 1.3960 & 0.5958 \\ 0.5958 & 0.9748 \end{pmatrix}$, hence

$$Q = \begin{pmatrix} 3.6924 & 2.1464 \\ 2.1464 & 4.2984 \end{pmatrix}.$$

Since $\Phi = \operatorname{diag}(\phi_1, \phi_2)$, $(\Phi\Gamma(0)\Phi^T)_{ij} = \phi_i\phi_j\Gamma_{ij}(0)$, so $\Gamma_{ij}(0) = Q_{ij}/(1 - \phi_i\phi_j)$:

$$\Gamma(0) = \begin{pmatrix} 3.6924/0.51 & 2.1464/0.58 \\ 2.1464/0.58 & 4.2984/0.64 \end{pmatrix} = \begin{pmatrix} 7.2400 & 3.7007 \\ 3.7007 & 6.7163 \end{pmatrix}.$$

Step 3: lag 1 and beyond. For $h \ge 1$,

$$\Gamma(h) = E\mathbf{Y}_t\big(\Phi\mathbf{Y}_{t+h-1} + \boldsymbol{\epsilon}_{t+h} + \Theta\boldsymbol{\epsilon}_{t+h-1}\big)^T = \Gamma(h-1)\Phi^T + E(\mathbf{Y}_t\boldsymbol{\epsilon}_{t+h-1}^T)\Theta^T,$$

and by Step 1 the last term is $\Sigma\Theta^T$ for $h = 1$ and 0 for $h \geq 2$. Therefore

$$\Gamma(1) = \Gamma(0)\Phi^T + \Sigma\Theta^T = \begin{pmatrix} 5.9940 & 2.0884 \\ 4.1455 & 5.1328 \end{pmatrix}, \qquad \Gamma(h) = \Gamma(1)(\Phi^T)^{h-1} \ (h \geq 1),$$

that is, $\Gamma_{i1}(h) = 0.7^{h-1}\Gamma_{i1}(1)$ and $\Gamma_{i2}(h) = 0.6^{h-1}\Gamma_{i2}(1)$; for negative lags $\Gamma(-h) = \Gamma(h)^T$. For instance the contemporaneous correlation is $3.7007/\sqrt{7.24 \cdot 6.7163} \approx 0.531$, and $\Gamma_{21}(1) = \mathrm{Cov}(Y_{2t}, Y_{1,t+1}) = 4.1455$ differs from $\Gamma_{12}(1) = 2.0884$: unlike the univariate case, $\Gamma(1)$ is not symmetric. The asymmetry comes mainly from the MA term, $(\Sigma\Theta^T)_{21} = 1.555$ against $(\Sigma\Theta^T)_{12} = -0.132$. ■

Chapter 8. Integer-Valued Time Series Models

Time series of *counts* take values in $\{0, 1, 2, \dots\}$: the number of trades of a stock per minute, monthly traffic accidents in a region, daily hospital admissions, monthly earthquakes. For such *count time series*, ARMA and GARCH models are inappropriate because $\phi y_{t-1} + \epsilon_t$ is not integer-valued. Two standard alternatives are the INAR (integer-valued autoregressive) model of McKenzie (1985) and Al-Osh and Alzaid (1987), and the INGARCH (integer-valued GARCH) model of Ferland et al. (2006), whose conditional law is Poisson with a mean depending on the past; inference for INGARCH was developed by Davis and Liu (2016), Lee and Lee (2019) and others (see Weiss, 2018). Throughout, $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \dots)$ denotes the past (the book's I_t).

8.1 The INAR(1) model

Motivation. Multiplying an integer by $\alpha \in (0, 1)$ does not give an integer, but keeping each of Y units independently with probability α does. This binomial thinning replaces scalar multiplication and yields an integer-valued analogue of AR(1) with the same autocorrelations.

Definition 8.1.1 (Binomial thinning)

For a nonnegative integer-valued random variable Y and $\alpha \in (0, 1)$, the *thinning operator* $\circ$ is

$$\alpha \circ Y = \sum_{i=1}^{Y} w_i,$$

where the w_i are i.i.d. Bernoulli variables with success probability α, independent of Y (the *counting sequence*); $\alpha \circ Y = 0$ when $Y = 0$.

Proposition (Properties of thinning, [Supplement])

Let Y be nonnegative integer-valued with $EY^2 < \infty$, with probability generating function $G_Y(s) = Es^Y$, and let $\alpha, \beta \in (0, 1)$.
(1) $E(\alpha \circ Y \mid Y) = \alpha Y$ and $\operatorname{Var}(\alpha \circ Y \mid Y) = \alpha(1-\alpha)Y$; hence $E(\alpha \circ Y) = \alpha EY$ and $\operatorname{Var}(\alpha \circ Y) = \alpha(1-\alpha)EY + \alpha^2 \operatorname{Var} Y$.
(2) $G_{\alpha \circ Y}(s) = G_Y(1 - \alpha + \alpha s)$.
(3) With independent counting sequences, $\alpha \circ (\beta \circ Y) \overset{d}{=} (\alpha\beta) \circ Y$.
(4) If $Y \sim \text{Poisson}(\lambda)$ then $\alpha \circ Y \sim \text{Poisson}(\alpha\lambda)$.

Proof.

(1) Given $Y = y$, $\alpha \circ Y$ is Binomial(y, α), with mean αy and variance $\alpha(1-\alpha)y$. The unconditional formulas follow from $E(\cdot) = E(E(\cdot \mid Y))$ and $\operatorname{Var}(\cdot) = E \operatorname{Var}(\cdot \mid Y) + \operatorname{Var} E(\cdot \mid Y)$.

(2) $E(s^{\alpha \circ Y} \mid Y) = (Es^{w_1})^Y = (1 - \alpha + \alpha s)^Y$; take expectations.

(3) By (2) twice, $G_{\alpha \circ (\beta \circ Y)}(s) = G_{\beta \circ Y}(1-\alpha+\alpha s) = G_Y\big(1-\beta+\beta(1-\alpha+\alpha s)\big) = G_Y(1-\alpha\beta+\alpha\beta s)$, the generating function of $(\alpha\beta) \circ Y$.

(4) $G_Y(s) = e^{\lambda(s-1)}$, so $G_{\alpha \circ Y}(s) = e^{\lambda(1-\alpha+\alpha s-1)} = e^{\alpha\lambda(s-1)}$. ■

Definition 8.1.2 (INAR(1) model)

A series y_t, $t = 1, 2, \dots$, with values in $\{0, 1, 2, \dots\}$ follows the *INAR(1) model* if

$$y_t = \alpha \circ y_{t-1} + \epsilon_t, \tag{8.1}$$

where $\alpha \in (0,1)$, the ϵ_t are i.i.d. nonnegative integer-valued random variables, and ϵ_t and the counting sequence of $\alpha \circ y_{t-1}$ are independent of each other and of $\mathcal{F}_{t-1}$. The term $\alpha \circ y_{t-1}$ is closely related to a branching process (each unit present at $t-1$ survives to t with probability α), and ϵ_t can be viewed as the number of immigrants.

Write $\mu = E\epsilon_t$ and $\sigma_\epsilon^2 = \operatorname{Var}\epsilon_t$. Usually ϵ_t is taken Poisson or geometric.

Proposition (Stationarity, moments and ACF of INAR(1), [Supplement])

Let $0 < \alpha < 1$ and $\sigma_\epsilon^2 < \infty$.
(1) (8.1) has a strictly stationary solution; every stationary solution has the marginal law of $\sum_{j=0}^{\infty} \alpha^j \circ \epsilon_{t-j}$ (independent thinnings), which is finite a.s.
(2) $E(y_t \mid \mathcal{F}_{t-1}) = \alpha y_{t-1} + \mu$ and $\operatorname{Var}(y_t \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha)y_{t-1} + \sigma_\epsilon^2$.
(3) $Ey_t = \dfrac{\mu}{1-\alpha}$ and $\operatorname{Var} y_t = \dfrac{\sigma_\epsilon^2 + \alpha\mu}{1-\alpha^2}$.
(4) $\gamma(h) = \operatorname{Cov}(y_t, y_{t+h}) = \alpha^{|h|}\gamma(0)$, so the ACF is $\rho(h) = \alpha^{|h|}$, as for an AR(1) process.

Proof.

Step 1: construction. Take i.i.d. ϵ_s ($s \in \mathbb{Z}$) and, independently, i.i.d. Bernoulli(α) indicators $B_{s,i,r}$ for $s < r$ and $i \geq 1$. Immigrant i of time s ($1 \leq i \leq \epsilon_s$) is *alive at time* $t \geq s$ if $B_{s,i,r} = 1$ for all $s < r \leq t$, and let y_t be the number of individuals alive at time t. Since $Ey_t = \sum_{j\geq 0} \alpha^j \mu < \infty$, y_t is finite a.s. Each individual alive at $t-1$ survives to t independently with probability α, via indicators $B_{\cdot,\cdot,t}$ independent of $\mathcal{F}_{t-1}$ and of ϵ_t; so the survivors form $\alpha \circ y_{t-1}$ and $y_t = \alpha \circ y_{t-1} + \epsilon_t$. The construction is shift-invariant, so $\{y_t\}$ is strictly stationary. The contributions of the cohorts $s = t-j$ are independent, and each is distributed as $\alpha^j \circ \epsilon_{t-j}$ (survival over j steps has probability α^j); hence $y_t \overset{d}{=} \sum_{j\geq 0} \alpha^j \circ \epsilon_{t-j}$.

Step 2: uniqueness of the law. For any solution, iterating (8.1) and using Proposition (3) gives $y_t \overset{d}{=} \alpha^t \circ y_0 + \sum_{j=0}^{t-1} \alpha^j \circ \epsilon_{t-j}$ with all thinnings independent. As $P(\alpha^t \circ y_0 \geq 1) = E\left[1-(1-\alpha^t)^{y_0}\right] \to 0$ by dominated convergence, the law of y_t converges to that of $\sum_{j\geq 0} \alpha^j \circ \epsilon_{t-j}$. If the solution is stationary its law does not change with t, so it equals this limit.

Step 3: conditional moments. Given $\mathcal{F}_{t-1}$, $\alpha \circ y_{t-1}$ is Binomial(y_{t-1}, α) and independent of ϵ_t, so Proposition (1) gives (2); in particular

$$E(y_t \mid y_{t-1}) = \alpha y_{t-1} + \mu. \tag{8.2}$$

Step 4: moments. The representation of Step 1 has finite variance, since $\operatorname{Var}(\alpha^j \circ \epsilon) = \alpha^j(1-\alpha^j)\mu + \alpha^{2j}\sigma_\epsilon^2$ is summable. Taking expectations in (8.2), $m = Ey_t$ solves $m = \alpha m + \mu$. By the variance decomposition and (2),

$$v = \operatorname{Var} y_t = E(\alpha(1-\alpha)y_{t-1} + \sigma_\epsilon^2) + \operatorname{Var}(\alpha y_{t-1} + \mu) = \alpha(1-\alpha)m + \sigma_\epsilon^2 + \alpha^2 v,$$

and $(1-\alpha)m = \mu$ gives $v(1-\alpha^2) = \alpha\mu + \sigma_\epsilon^2$.

Step 5: ACF. For $h \geq 1$, y_t is $\mathcal{F}_{t+h-1}$-measurable, so $\gamma(h) = \operatorname{Cov}(E(y_{t+h} \mid \mathcal{F}_{t+h-1}), y_t) = \operatorname{Cov}(\alpha y_{t+h-1} + \mu, y_t) = \alpha\gamma(h-1)$, whence $\gamma(h) = \alpha^h\gamma(0)$; and $\gamma(-h) = \gamma(h)$. ∎

Example 8.1.3 (Poisson innovations)

Let $\epsilon_t \sim \text{Poisson}(\mu)$. By Proposition (4), $\alpha^j \circ \epsilon_{t-j} \sim \text{Poisson}(\alpha^j\mu)$, and a sum of independent Poisson variables is Poisson, so the stationary law is

$$y_t \sim \text{Poisson}\Big(\frac{\mu}{1-\alpha}\Big),$$

with mean equal to variance, consistent with Proposition (3) for $\sigma_\epsilon^2 = \mu$. Larger α gives a higher level and longer runs of similar counts, the autocorrelation at lag h being α^h.

8.1.1 Conditional least squares

By (8.2), (α, μ) can be estimated by least squares: given $y_0, y_1, \ldots, y_n$, $(\hat\alpha, \hat\mu)$ minimizes $\sum_{t=1}^n (y_t - \alpha y_{t-1} - \mu)^2$. Solving the normal equations,

$$\begin{pmatrix} \hat\alpha \\ \hat\mu \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^n y_{t-1}^2 & \sum_{t=1}^n y_{t-1} \\ \sum_{t=1}^n y_{t-1} & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{t=1}^n y_{t-1} y_t \\ \sum_{t=1}^n y_t \end{pmatrix}. \tag{8.3}$$

Proposition (Asymptotics of the least squares estimator, [Supplement])

Let $\{y_t\}$ be the stationary INAR(1) series with $\operatorname{Var} y_t > 0$ and $Ey_t^3 < \infty$. Put $\theta = (\alpha, \mu)^T$, $\mathbf{x}_t = (y_{t-1}, 1)^T$, $u_t = y_t - \alpha y_{t-1} - \mu$, $A = E(\mathbf{x}_t \mathbf{x}_t^T)$ and $V = E(u_t^2 \mathbf{x}_t \mathbf{x}_t^T)$. Then $\hat\theta \to \theta$ a.s. and

$$\sqrt{n}(\hat\theta - \theta) \xrightarrow{d} N(0, A^{-1} V A^{-1}).$$

Proof.

Step 1. $\{y_t\}$ is a fixed function of an i.i.d. sequence (Step 1 above), hence stationary and ergodic. A is positive definite because $\det A = \operatorname{Var} y_{t-1} > 0$. Since $y_t = \theta^T \mathbf{x}_t + u_t$, (8.3) gives

$$\hat\theta - \theta = \Big(\frac{1}{n} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t^T \Big)^{-1} \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t u_t.$$

Step 2. By the ergodic theorem $\frac{1}{n}\sum \mathbf{x}_t \mathbf{x}_t^T \to A$ and $\frac{1}{n}\sum \mathbf{x}_t u_t \to E(\mathbf{x}_t u_t) = E\big(\mathbf{x}_t E(u_t \mid \mathcal{F}_{t-1})\big) = 0$ a.s., so $\hat\theta \to \theta$ a.s.

Step 3. $\{\mathbf{x}_t u_t\}$ is a stationary ergodic martingale difference with $E\|\mathbf{x}_t u_t\|^2 < \infty$ (because $E(u_t^2 \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha) y_{t-1} + \sigma_\epsilon^2$ and $Ey^3 < \infty$). The martingale CLT gives $n^{-1/2} \sum \mathbf{x}_t u_t \xrightarrow{d} N(0, V)$, and Slutsky's lemma completes the proof. ∎

8.2 The INGARCH(1,1) model

Motivation. Whereas INAR models the count itself, INGARCH models its conditional mean: the count is Poisson given the past, with an intensity following a GARCH-type recursion. Because the Poisson variance equals the mean, this is simultaneously a model for the conditional variance.

Suppose the conditional mean of y_t given the past follows the recursion

$$\mu_t = \omega + \alpha y_{t-1} + \beta \mu_{t-1}, \qquad t = 1, \ldots, n, \tag{8.4}$$

where $\omega > 0$, $\alpha, \beta > 0$ with $\alpha + \beta < 1$, and the initial values y_0 (a nonnegative integer) and μ_0 (a positive real) are arbitrary, and that y_t is generated from a Poisson distribution with parameter μ_t:

$$P(y_t = k \mid \mathcal{F}_{t-1}) = \frac{e^{-\mu_t} \mu_t^k}{k!}, \qquad k = 0, 1, \ldots \tag{8.5}$$

Definition 8.2.1 (INGARCH(1,1) model)

Combining (8.4) and (8.5), the *INGARCH(1,1) model* is

$$y_t \mid \mathcal{F}_{t-1} \sim \text{Poisson}(\mu_t), \qquad \mu_t = \omega + \alpha y_{t-1} + \beta \mu_{t-1}. \tag{8.6}$$

By the Poisson property, the conditional variance of y_t also equals μ_t.

Writing (8.6) with the backshift operator, $(1-\beta B)\mu_t = \omega + \alpha y_{t-1}$, so

$$\mu_t = \frac{\omega}{1-\beta} + \alpha \sum_{j=1}^{\infty} \beta^{j-1} y_{t-j}, \tag{8.7}$$

and μ_t is a function of (ω, α, β) and the past counts. The analogy with GARCH(1,1) is exact: $\sigma_t^2 \leftrightarrow \mu_t$, $y_{t-1}^2 \leftrightarrow y_{t-1}$.

Proposition (Moments of INGARCH(1,1), [Supplement])

Let $\{y_t\}$ be a second-order stationary solution of (8.6) (Ferland et al., 2006, show that one exists when $\alpha + \beta < 1$). Let $m = Ey_t$. Then
(1) $m = E\mu_t = \dfrac{\omega}{1-\alpha-\beta}$;
(2) $\operatorname{Var} \mu_t = \dfrac{\alpha^2 m}{1-(\alpha+\beta)^2}$ and $\operatorname{Var} y_t = m \dfrac{1-(\alpha+\beta)^2+\alpha^2}{1-(\alpha+\beta)^2} \geq m$, so the counts are overdispersed;
(3) for $h \geq 1$, $\operatorname{Cov}(y_t, y_{t-h}) = (\alpha+\beta)^{h-1}(\alpha \operatorname{Var} y_t + \beta \operatorname{Var} \mu_t)$: the ACF decays geometrically at rate $\alpha+\beta$, like that of an ARMA(1,1).

Proof. Let $\nu_t = y_t - \mu_t$. Since μ_t is $\mathcal{F}_{t-1}$-measurable, $E(\nu_t \mid \mathcal{F}_{t-1}) = 0$ and $E(\nu_t^2 \mid \mathcal{F}_{t-1}) = \mu_t$; hence ν_t is uncorrelated with every square-integrable $\mathcal{F}_{t-1}$-measurable variable, and $E\nu_t^2 = m$.

Step 1. Taking expectations in (8.4), $m = \omega + (\alpha+\beta)m$ (note $E\mu_t = Ey_t$).

Step 2. Substituting $y_{t-1} = \mu_{t-1} + \nu_{t-1}$, $\mu_t = \omega + (\alpha+\beta)\mu_{t-1} + \alpha\nu_{t-1}$, where μ_{t-1} and ν_{t-1} are uncorrelated. Hence $\operatorname{Var} \mu_t = (\alpha+\beta)^2 \operatorname{Var} \mu_{t-1} + \alpha^2 m$, which by stationarity gives the formula for $\operatorname{Var} \mu_t$. Then $\operatorname{Var} y_t = \operatorname{Var} \mu_t + E\nu_t^2 = \operatorname{Var} \mu_t + m$.

Step 3. For $h \geq 1$, y_{t-h} is $\mathcal{F}_{t-1}$-measurable, so $\operatorname{Cov}(y_t, y_{t-h}) = \operatorname{Cov}(\mu_t, y_{t-h})$. For $h = 1$: $\operatorname{Cov}(\omega + \alpha y_{t-1} + \beta\mu_{t-1}, y_{t-1}) = \alpha \operatorname{Var} y_{t-1} + \beta \operatorname{Cov}(\mu_{t-1}, \mu_{t-1} + \nu_{t-1}) = \alpha \operatorname{Var} y + \beta \operatorname{Var} \mu$. For $h \geq 2$: $\operatorname{Cov}(\mu_t, y_{t-h}) = \operatorname{Cov}((\alpha+\beta)\mu_{t-1} + \alpha\nu_{t-1}, y_{t-h}) = (\alpha+\beta)\operatorname{Cov}(\mu_{t-1}, y_{t-h}) = (\alpha+\beta)\operatorname{Cov}(y_{t-1}, y_{t-h})$, using that ν_{t-1} is uncorrelated with y_{t-h}. Induction on h finishes the proof. ■

Given $y_1, \ldots, y_n$, the conditional maximum likelihood estimator of (ω, α, β) maximizes the conditional likelihood or, equivalently, the log-likelihood (up to a constant)

$$L(\omega, \alpha, \beta) = \prod_{t=1}^{n} \frac{e^{-\mu_t} \mu_t^{y_t}}{y_t!}, \qquad l(\omega, \alpha, \beta) = \sum_{t=1}^{n} (y_t \log \mu_t - \mu_t),$$

with μ_t computed recursively from (8.4). The estimator $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$ is consistent and asymptotically normal (Lee and Lee, 2019). Besides the Poisson, the negative binomial distribution and, more generally, one-parameter exponential families are used as conditional distributions.

8.3 Change-point tests

Motivation. As for GARCH, the parameters of count models may shift; a CUSUM test on the residuals, which form a martingale difference sequence under the null, detects the change.

The test applies the CUSUM idea of Section 6.5 to residuals. For INAR(1),

$$\hat{\epsilon}_t = y_t - \hat{\alpha} y_{t-1} - \hat{\mu}, \tag{8.8}$$

and for INGARCH(1,1),

$$\hat{\epsilon}_t = y_t - \hat{\mu}_t, \qquad \hat{\mu}_t = \hat{\omega} + \hat{\alpha} y_{t-1} + \hat{\beta}\hat{\mu}_{t-1}, \tag{8.9}$$

with arbitrary initial values $y_0, \hat{\mu}_0$. The residual-based CUSUM statistic is

$$\hat{T}_n = \max_{1 \le k \le n} \frac{1}{\sqrt{n}\,\hat{\tau}} \Big| \sum_{t=1}^{k} \hat{\epsilon}_t - \frac{k}{n} \sum_{t=1}^{n} \hat{\epsilon}_t \Big|, \qquad \hat{\tau}^2 = \frac{1}{n}\sum_{t=1}^{n} \hat{\epsilon}_t^2 - \Big(\frac{1}{n}\sum_{t=1}^{n} \hat{\epsilon}_t\Big)^2. \tag{8.10}$$

At the 5% level a change is declared if $\hat{T}_n > 1.358$, and the maximizing k is taken as the change point.

Proposition (Null limit of the residual CUSUM test, [Supplement])

Let $\{y_t\}$ be a stationary INAR(1) series satisfying the assumptions of Proposition , with no parameter change, and let $\hat{\epsilon}_t$ be the residuals (8.8) from the least squares estimator. Then $\hat{T}_n \xrightarrow{d} \sup_{0 \le u \le 1} |B^0(u)|$, the Brownian-bridge limit of Proposition .

Proof.

Step 1: true innovations. The true residuals $u_t = y_t - \alpha y_{t-1} - \mu = (\alpha \circ y_{t-1} - \alpha y_{t-1}) + (\epsilon_t - \mu)$ form a stationary ergodic martingale difference sequence with variance $\tau^2 = E u_t^2 = \alpha(1-\alpha)Ey_t + \sigma_\epsilon^2 > 0$. By the functional CLT for such martingale differences (Billingsley), $W_n(s) = \frac{1}{\tau\sqrt{n}} \sum_{t \le \lfloor ns \rfloor} u_t \xrightarrow{d} W$. Exactly as in the proof of Proposition (Steps 2–3), $\max_k \frac{1}{\tau\sqrt{n}} \big| \sum_{t \le k} u_t - \frac{k}{n} \sum_{t \le n} u_t \big| \xrightarrow{d} \sup_s |B^0(s)|$.

Step 2: estimation effect. $\hat{\epsilon}_t - u_t = (\alpha - \hat{\alpha}) y_{t-1} + (\mu - \hat{\mu})$. Constants cancel in $\sum_{t \le k} c - \frac{k}{n}\sum_{t \le n} c = 0$, so

$$\frac{1}{\sqrt{n}} \Big| \sum_{t \le k} (\hat{\epsilon}_t - u_t) - \frac{k}{n} \sum_{t \le n} (\hat{\epsilon}_t - u_t) \Big| = \sqrt{n}|\hat{\alpha} - \alpha| \cdot \frac{1}{n} \Big| \sum_{t \le k} (y_{t-1} - m) - \frac{k}{n} \sum_{t \le n} (y_{t-1} - m) \Big|,$$

with $m = Ey_t$. Here $\sqrt{n}(\hat{\alpha} - \alpha) = O_p(1)$ by Proposition , and $\max_{k \le n} \frac{1}{n} |\sum_{t \le k} (y_{t-1} - m)| \to 0$ a.s. because $\frac{1}{k}\sum_{t \le k}(y_{t-1} - m) \to 0$ a.s. (ergodic theorem). So the maximum over k of this difference is $o_p(1)$.

Step 3: variance estimate. Similarly $\hat{\epsilon}_t = u_t + O(|\hat{\alpha} - \alpha| y_{t-1} + |\hat{\mu} - \mu|)$ and the ergodic theorem give $\hat{\tau}^2 \to \tau^2$ in probability. Slutsky's lemma combines Steps 1–3. ∎

For INGARCH(1,1) the residuals $y_t - \mu_t$ are again martingale differences, and the same limit holds for the residuals (8.9) under the regularity conditions of Lee and Lee (2019); the critical value 1.358 is thus used in both models (cf. Section 6.5).

8.4 Selected exercises

Motivation. Two structural facts about INGARCH(1,1) are proved here: the conditional mean is a geometrically weighted sum of past counts, and the counts themselves follow an ARMA(1,1) model driven by a martingale-difference noise.

Exercise 8.3 (The intensity as a weighted sum of past counts)

Let $\{y_t\}_{t\in\mathbb{Z}}$ be a second-order stationary solution of (8.6) with $\omega > 0$, $\alpha, \beta > 0$, $\alpha + \beta < 1$. Prove (8.7), the series converging a.s. and in mean.

Solution.

Step 1: finite iteration. For every $K \geq 1$,

$$\mu_t = \omega \sum_{j=0}^{K-1} \beta^j + \alpha \sum_{j=1}^{K} \beta^{j-1} y_{t-j} + \beta^K \mu_{t-K}.$$

For $K = 1$ this is (8.4). If it holds for K, inserting $\mu_{t-K} = \omega + \alpha y_{t-K-1} + \beta\mu_{t-K-1}$ into the last term gives $\beta^K \mu_{t-K} = \omega\beta^K + \alpha\beta^K y_{t-K-1} + \beta^{K+1}\mu_{t-K-1}$, which is the identity for $K + 1$.

Step 2: the remainder. $\mu_t > 0$ and, by Proposition (1), $E\mu_t = m < \infty$ for all t. Hence $E|\beta^K \mu_{t-K}| = \beta^K m \to 0$ as $K \to \infty$, because $0 < \beta < 1$.

Step 3: the series. The terms $\beta^{j-1} y_{t-j}$ are nonnegative and $\sum_{j\geq 1} \beta^{j-1} E y_{t-j} = m/(1-\beta) < \infty$. By monotone convergence $S = \sum_{j\geq 1} \beta^{j-1} y_{t-j}$ has finite mean, so it is finite a.s., the partial sums S_K converge to S a.s., and $E|S - S_K| = \sum_{j>K} \beta^{j-1} m \to 0$.

Step 4: conclusion. By Step 1, $D_K = \mu_t - \omega(1-\beta^K)/(1-\beta) - \alpha S_K = \beta^K \mu_{t-K}$. By Step 3, $D_K \to \mu_t - \omega/(1-\beta) - \alpha S$ a.s., while by Step 2 $D_K \to 0$ in mean, hence a.s. along a subsequence. The two limits agree, so $\mu_t = \omega/(1-\beta) + \alpha S$ a.s., which is (8.7).

Step 5: remark on initial values. If the recursion is started at $t = 1$ from arbitrary y_0, μ_0, Step 1 with $K = t$ gives

$$\mu_t = \omega \frac{1-\beta^t}{1-\beta} + \alpha \sum_{j=1}^{t} \beta^{j-1} y_{t-j} + \beta^t \mu_0,$$

so the effect of μ_0 on μ_t decays like β^t; this is why the initial values do not matter asymptotically in the conditional likelihood. ■

Exercise 8.4 (INGARCH(1,1) as an ARMA(1,1) model)

Under the assumptions of Exercise 8.3, show that $\{y_t\}$ follows an ARMA(1,1) model. (Hint: consider $\nu_t = y_t - \mu_t$.)

Solution.

Step 1: the noise. μ_t is $\mathcal{F}_{t-1}$-measurable and $y_t \mid \mathcal{F}_{t-1} \sim \text{Poisson}(\mu_t)$, so $E(\nu_t \mid \mathcal{F}_{t-1}) = 0$ and $E(\nu_t^2 \mid \mathcal{F}_{t-1}) = \mu_t$. Hence $E\nu_t = 0$ and $E\nu_t^2 = E\mu_t = m = \omega/(1-\alpha-\beta)$, the same for all t. For $s < t$, ν_s is $\mathcal{F}_{t-1}$-measurable and $\nu_s\nu_t$ is integrable (Cauchy–Schwarz), so $E\nu_s\nu_t = E(\nu_s E(\nu_t \mid \mathcal{F}_{t-1})) = 0$. Thus $\{\nu_t\}$ is a white noise with variance m (a martingale difference, but not i.i.d.: its conditional variance μ_t moves with the past).

Step 2: the ARMA equation. Substituting $\mu_{t-1} = y_{t-1} - \nu_{t-1}$ in $y_t = \mu_t + \nu_t = \omega + \alpha y_{t-1} + \beta\mu_{t-1} + \nu_t$,

$$y_t = \omega + (\alpha+\beta) y_{t-1} + \nu_t - \beta\nu_{t-1}, \qquad \text{i.e.} \qquad (1 - (\alpha+\beta)B)(y_t - m) = (1-\beta B)\nu_t,$$

using $\omega = (1-\alpha-\beta)m$. This is an ARMA(1,1) model with AR coefficient $\phi = \alpha + \beta \in (0,1)$ and MA coefficient $\theta = -\beta \in (-1,0)$; it is causal and invertible.

Step 3: causality and a check. Iterating Step 2, $y_t - m = \sum_{j=0}^{K-1} \phi^j(\nu_{t-j} - \beta\nu_{t-j-1}) + \phi^K(y_{t-K} - m)$, and the remainder has second moment $\phi^{2K} \operatorname{Var} y_t \to 0$. Hence $y_t - m$ is a mean-square limit of combinations of ν_s, $s \leq t$, and is uncorrelated with $\nu_{t+1}, \nu_{t+2}, \ldots$. These are the only facts used in Example 2.4.3, which therefore gives

$$\operatorname{Var} y_t = \frac{m(1 + 2\phi\theta + \theta^2)}{1-\phi^2} = m\,\frac{1 - 2\alpha\beta - \beta^2}{1-(\alpha+\beta)^2} = m\,\frac{1-(\alpha+\beta)^2 + \alpha^2}{1-(\alpha+\beta)^2},$$

in agreement with Proposition (2), and $\gamma(h) = (\alpha + \beta)^{h-1}\gamma(1)$ as in Proposition (3). ■